



# LIGHTING PRACTICE: DOCUMENTING CONTROL INTENT NARRATIVES AND SEQUENCES OF OPERATIONS

AN AMERICAN NATIONAL STANDARD





**ANSI/IES LP-16-22**

**LIGHTING PRACTICE:  
DOCUMENTING CONTROL INTENT NARRATIVES AND  
SEQUENCES OF OPERATIONS  
AN AMERICAN NATIONAL STANDARD**

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has been approved by IES.  
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should be directed to IES.

**Prepared for IES by the  
Lighting Control Systems Committee**



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## 1.0 Introduction and Scope

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### 1.1 Introduction

In order for lighting control systems to work, the designer needs to create a vision for the design and provide clear instructions for how the lighting controls within the project boundaries, as well as their connections to other systems, are to operate. The design vision is documented in a Control Intent Narrative (CIN), a plain text document that can be easily understood by a layperson. The CIN is a springboard for the Sequence of Operation (SOO) document, the instructions for how to program the lighting control system. The SOO has the contractually enforceable language for informing the installer how to program the system to meet the design intent. This Lighting Practice document will describe how to create a project CIN and SOO that:

- 1) are detailed enough to ensure that the delivered system provides the functionality required;
- 2) provide enough information that the construction team can provide exactly the programming required; and
- 3) are contractually enforceable if something goes wrong.

### 1.2 Scope

This document provides guidance on the documentation of Control Intent Narratives and Sequences of Operation (CIN and SOO). It is not intended to be a design guide, but rather a reference manual of best practices on how the design, once formulated, is included in the project documentation and communicated to the construction and commissioning teams.

This document is intended for people in the lighting community, including building owners, managers, architects, lighting practitioners, electrical engineers, control systems practitioners, luminaire and controls manufacturers, systems integrators, and installers.

Equipment, sensors, software, and network infrastructure are necessary components of a complete lighting control system, but these are not covered here (refer to *ANSI/IES LP-6-20, Lighting Practice: Lighting Control Systems – Products, Selection, and Specification*<sup>1</sup> for information on these aspects of the lighting control system).

There are different ways to convey intent and sequences of operation. This document presents two of the most common formats: a matrix and a narrative. Each has its strengths and weaknesses. Information is provided to help the reader make a decision that will benefit the project. Furthermore, it is imperative that the method chosen be in harmony with the capabilities of the project team so that the SOOs will be updated as they evolve through the project.

It is not necessary to read this document cover to cover. The reader may skip to the sections of interest and the example narratives that will be useful for a given project. While the examples given are real world examples used on various projects, the example sequences should *not* be used as is; they will almost certainly not meet the requirements for a specific project. Instead, the examples should be used as a starting point and then adapted to specify the control system the project requires. The examples given may be used for their structure and level of detail, but the requirements and setpoints should be changed to meet the specific project's needs.

In addition, each specific project may require unique sequences that have never been created before. A questionnaire is provided in **Annex C** to help with creating a CIN and SOO for a specific project.

Finally, a List of **Abbreviations** and a **Glossary** can be found just before the annexes.

## 2.0 Control Intent Narratives and Sequences of Operation

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The CIN and SOO do not hold equal positions in the work products for a project. The CIN is part of the Basis of Design (BOD) and is used to convey the overall intentions for the specified system(s). The CIN may be included in the contract documents (often the specifications or drawings); however, it should not be relied on because it deals in generalities. There is usually a lot of room for interpretation and enforcement; the language is often so vague that it is unenforceable. However, this

allows it to be a more easily understandable document for an owner or end user. The CIN can be made enforceable by the addition of details and setpoints. When measurable outcomes are added they become contractually enforceable. A side-by-side example of a CIN and an SOO is provided in **Table 2-1**. A clear CIN and SOO included with the specifications can help clarify approval or rejection of product submittals to meet project requirements.

It should be noted that the terms *Basis of Design* (BOD) and *Owner's Project Requirements* (OPR) primarily belong to the practice of systems commissioning. Within the context of this discussion, the term *OPR* refers to the specific control related requirements as articulated by the owner for the project under development. These may be extremely simple (for example, "save as much energy as possible") or more complex ("provide a seamless experience within the finished project in which the facility seems to operate independently of human operational commands"). The BOD takes the information from the OPR and combines it with the designer's vision to present a plain-language description of the design. The plain-language description of the lighting control function within the BOD creates the CIN. Readers are directed to ANSI/IES LP-8-20<sup>2</sup> for a detailed exploration of the commissioning process as it relates to both

lighting and lighting control systems. In particular, Table 4-1 in that document provides a thorough discussion of the process of writing a complete BOD document. **Section 2.1** of this document focuses on the portion of a BOD that describes the function of the lighting control system, the Control Intent Narrative. It is important to note that the entire BOD is a living document and will be revised as the project progresses.

## 2.1 Control Intent Narrative (CIN)

The CIN serves as the high-level descriptor of what the lighting control system is, what it will to do, and how the end users will interact with it. It is the primary means of ensuring that the ownership and user teams understand and agree with the lighting control design as it is being developed. The CIN includes:

- A general description of the project goals
- Control strategies to satisfy these goals
- A description of the control system
- A basic preliminary sequence of operation for each typical space type

It is important for the Control Intent Narrative to be written in clear, simple terms to allow an owner to review, understand, and approve the functionality of the lighting control system. Industry jargon should be avoided. This will help to ensure that the CIN describes

**Table 2-1. Example: A CIN and SOO Comparison**

| Control Intent Narrative*                                                                                                                                                                                                                                                                             | Sequence of Operation*                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Provides a <i>general</i> description of the design vision.                                                                                                                                                                                                                                           | Contains highly <i>specific</i> descriptions of what the lighting will do. It will include setpoints, will represent all the lighting of the project (interior, exterior, and emergency), and will address integration with other systems (e.g., HVAC).                                                                                                                   |
| <i>Example:</i> In <i>all areas</i> where lighting power within primary side-lit or top-lit daylighting zones exceeds the wattage threshold of the relevant energy code, such lighting shall <i>automatically dim</i> based on daylight contribution to maintain design illuminance and distribution. | <i>Example:</i> Adjust general lighting in <i>private offices</i> to maintain <i>300 lux</i> within the side-lit daylighting zone. When light level changes, delay change for <i>1 min. (adj.**)</i> . If daylight contribution decreases, increase lighting by <i>5% (adj.)</i> each second. If daylight increases, ramp down lighting by <i>20% (adj.)</i> each second. |

Table notes:

\* Italics have been added to emphasize the differences between CIN and SOO.

\*\* adj.: Setpoints usually need to be adjusted in the field. Denoting adjustable setpoints is standard practice by mechanical engineers when writing SOOs for HVAC control systems. Adding "adjustable" (or "adj.") is an indicator to the controls contractor that the setpoint can and will be adjusted during startup and testing at no additional cost to the owner. Best practice is to reinforce setpoint adjustability by including this work in the control system specification. Note: Setpoints that do not need to be adjustable do not have "adj." after the setpoint value.

a system that meets the fundamental project needs. In addition, if there are significant departures from the OPR, they should be clearly described in the CIN and noted as changes with an accompanying justification. The owner or client should be able to understand the recommendations and specific causes and effects described in the document without assistance, in order to be confident that the system will be operationally and functionally appropriate.

## 2.2 Sequence of Operation (SOO)

The SOO is the contractually enforceable outgrowth of the CIN. The move from design narrative (how the system *might* or *will* behave) to how it is required to (*shall*) operate requires changing the tone of the document from a descriptive and purely informative document to a prescriptive one with specific goals, setpoints, and other attributes. Although the steps and elements of the SOO may seem similar to those described in the CIN, it is important to remember that the goal of CIN documentation is different from that of the SOO. In the former case, the documentation provides

a description of the various attributes of the system. In the latter, the focus is on the programming and functional requirements of the equipment necessary to achieve the stated goals. It may be helpful to think of the CIN as the “what” and the SOO as the “how.”

## 2.3 Key Elements of the CIN and SOO

Every project is different, and any list of control elements will need to be geared to the specific requirements outlined in the OPR. **Table 2-2** provides an example of the types of information addressed within a CIN and an SOO. It is not intended to be exhaustive.

It is critical that the CIN consider the scale of the system it is describing. A whole-building system will have different attributes than a single-room solution. In a whole-building project, building-level controls may originate from a master control station that propagates commands to spaces inside and outside the building. Such building-wide events usually require a master-level input or an input from an external system like a utility demand-response signal or integration with fire

**Table 2-2. Key Elements for a CIN and an SOO**

| Element                                   | CIN                                                                               | SOO*                                                 |
|-------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------|
| Scale of the system or project            | Describe in general                                                               | N.A.                                                 |
| System topology                           | Wired or wireless?<br>Centralized server or distributed devices?<br>Combinations? | Provide network diagram in the contract documents    |
| Special infrastructure                    | Existing wire type or special wire type needed?                                   | N.A.                                                 |
| Integration with other system(s)          | What are the systems?                                                             | How will they be integrated?                         |
| Automatic or manual control               | Where are automatic controls used?                                                | Specific devices                                     |
| User interface                            | What kinds? (e.g., keypads, touchscreen)                                          | Specific engraving, annunciation, or graphic details |
| Sensors for occupancy or vacancy response | On/off?<br>Partial dimming?                                                       | Specific setpoints, timing, and methods              |
| Daylight harvesting and response          | Describe general approach                                                         | Specific setpoints and locations                     |
| Scene controls                            | Describe preliminary approach                                                     | Specific zones, output, schedule, and how to trigger |
| Setpoints                                 | Preliminary and general                                                           | Specifics for all                                    |
| Specialized lighting                      | Preliminary approach                                                              | Specifics with details                               |
| Emergency controls                        | General approach                                                                  | Specific setpoints and how to trigger                |
| Night lighting                            | General approach                                                                  | Specific zones, output, and schedule                 |

\* Table note: N.A. = not applicable.

and life-safety systems. When part of the building or only a few rooms are designed, it is important to clearly describe whether the area is controlled by a new control system that interfaces with the existing control system in the building, or is an extension of the existing control system. The word “interface” can be a verb or a noun.

Most of the features of a building-level system may be available within the context of a smaller-scale system as well. As an example, a networked lighting control system that is designed appropriately can interface with other building systems if necessary, via a BACNet gateway (refer to ANSI/IES LP-6-20<sup>1</sup> for further information), and the presence of a processor in a networked system will provide an integral clock that can be utilized for time-based controls.

The specifier should clearly delineate the key parameters of the system, so that as the project progresses through bid, construction, and turnover the intended outcomes are achieved. This should include detailed information concerning both the system functionality and the system topology. Information about user interfaces and sensors should include details about wired or wireless methodologies and equipment. This information should also describe whether system commands may be originated by a centralized server, via devices distributed throughout the system, or a combination of the two.

The clearest way to document system topology is in a network diagram (see **Figure B-3** in **Annex B** for an example). Failure to include a network diagram within the contract documents will often result in the networking being incomplete or left out entirely because vendors and contractors often fail to connect the requirements for networked command in the SOO with system topology. Areas not connected to the central server will not be able to enact sequences that require networked commands (e.g., time switch or demand-response commands).

It is critical that any other key infrastructure or lighting network requirements be clearly defined within the CIN so that the document can function as a touchstone that can be referenced throughout the project to measure the success of the execution of the design. If there

are specific requirements, infrastructure needs, or integration requirements that could lead the project down a sole-sourcing path, those requirements need to be defined here as well. Conversely, if there are no such requirements, or no reason to choose one topology over another that needs to be specifically documented, it is important to state that all design avenues have been thoroughly considered and that the design team has made a conscious choice to allow the system to be open to a variety of proposed solutions. This serves to indicate that the design team has provided as much insight as possible.

When properly incorporated into the contract documents, the CIN and SOO work together to fully present the control design. Although the SOO gives details on how the controls operate and interact in a space, in many cases, a wide variety of lighting controls may meet the SOO needs but still not achieve the owner’s desires as articulated in the CIN. It is critical that the CIN elements be incorporated into the specifications jointly with the SOO in the contract documents so that both the control function is met and the owner’s project requirements for controls are fulfilled by the lighting control equipment.

Emergency controls are generally separate from building-level controls, as their operation may be determined by wiring rather than by control signals (in some jurisdictions, this may be a code requirement). However, they are an important layered part of the overall lighting control design that should be described in the CIN and SOO.

## 2.4 Creation Steps for the CIN and SOO

The steps described in this section comprise one suggested method by which a project can be broken down into smaller components. As each phase of project development progresses (see **Section 3.0 Creation Timeline**), the level of detail with which each room, action, or response is considered increases. As an example, a document created during the Schematic Design (SD) phase may identify rooms with daylight involvement, while the same document during the Design Development (DD) phase may indicate the quantity of daylight responsive zones and the types of responses each zone should exhibit. The version of

the same narrative in the Construction Documentation (CD) phase will include specific information about the sensor being deployed and its mounting (e.g., within or external to the luminaire), the illuminance to be maintained at the work surface, and the fade rate of the lighting between states.

It may be helpful to think of the various steps in this process as an opportunity for each of the several “voices” present within the general BOD documentation to be heard and captured in the CIN—specifically, the building, the code or legal statutes governing its design and construction, the owner or users, and the design team. This is what is meant by the “voice” indicators in **Section 2.4.1**.

### 2.4.1 CIN Steps.

**Step 1:** What does the owner want and consider important for end-user needs? [Owner’s “voice”]

Central question: What is the design team being asked to provide?

- Read the OPR to find out the Client’s goals and wishes.

**Step 2:** What are the spaces involved? [Building’s “voice”]

Central question: How will each type of space be addressed by the control system?

- List each space and function.
- List which part of the building needs what type of system, such as a standalone system or a networked system.

**Step 3:** Interaction (Building’s “voice”)

Central question: How do various devices interact to provide control?

- List other building systems that need to interface with the lighting control system and how they should interact.

**Step 4:** List the code-required and certification-required functions. [Code’s “voice,” such as energy codes, building codes, and certification and rating systems like LEED or WELL Building]

Central question: What equipment will be necessary to address each of the specific code or rating-certification requirements?

- List code and standard requirements that apply to the project.
- List spaces that have specific control requirements by code or standard, and list the device required; this may include occupancy or daylight sensors, for example.

**Step 5:** What is the purpose of the system? [Design team’s “voice”]

Central question: How does the control system function as an aspect of the project design?

- Define building schedule information as provided by owner.
- List spaces that require specialized control (e.g., color-changing or color tuning) to satisfy the client’s requirements (cite the OPR).
- List spaces that require automatic or other specific control—such as occupancy sensors, daylight sensors, or video teleconferencing—as a function of the design rather than code compliance.
- List spaces or space types that require multi-scene programming.
- List spaces that require integration with other systems, such as building automation, HVAC, security, and fire control.

**Step 6:** List the actions and functional responses.

Central question: Does the visual environment change during the use cycle?

- Think about or coordinate emergency lighting strategy.
- Think about what type of system is needed to satisfy requirements and goals.

**Step 7:** Identify opportunities for setpoints.

Central question: What are the quantifiable conditions that will cause the system to react?

- When is the facility operational? Are there different requirements for after-hours occupancy, for example?

- Does the space (or spaces) have different operational needs for daytime and nighttime?

#### Step 8: Review.

Central question: Have all space types, control types and known scenarios been considered?

- Think about how the system described in the document will ultimately be configured. Some portions of the building or project may require a networked system, while other aspects may operate more efficiently on a standalone system.

#### SIDEBAR: Why Describe CIN Early?

Although turning all the lights off at night can lead to significant energy savings, owners may object to the loss of visibility represented by a dark building. This is a key example of why the operation of the lighting control system should be described early. It is much easier to propose alternative solutions (e.g., night lights, occupancy sensors) that satisfy the owner's expectations during the Schematic Design and Design Development phases than it is to redesign the system in the construction documentation phase (i.e., re-zone the system to provide circuit-level nighttime lighting) or, worse yet, with a change order during Construction Administration (CA).

**2.4.2 SOO Steps.** Creating an SOO is its own multistep process that specifies how to achieve the CIN. As is the case with the CIN, a number of constituents or “voices” have been identified to which the finished project needs to answer. (*Note:* Additional information may be found in **Annex A – Lighting Controller Sequence of Operation.**)

#### Step 1: Consider code-required functions or attributes.

Primary question: What are the lighting control functions that are necessary in order to meet the statutory requirements for this project, and what equipment will be used to meet them?

#### Step 2: List the spaces.

Primary question: How will the various types of spaces within the project respond to both the code requirements

and the aesthetic principles outlined in the CIN? Space types include not only individual rooms, but also the building as a whole and exterior spaces. Including these large-space concepts aids in the inclusion of sequences that involve networked commands.

#### Step 3: Consider what the owner wants.

Primary question: What specific aspects of the OPR and BOD documents have been integrated into the LCS design, and how will each of those functions be achieved?

#### Step 4: Decide on actions and functional responses.

Primary question: What happens in typical use cases?

#### Step 5: Identify setpoints.

Primary question: What are the start and end points of each operation?

- What control parameters need to be established to allow the installation, functional testing, and commissioning teams to configure and program the system to meet the goals presented in the CIN?

#### Step 6: Review the SOO.

Primary question: Have all system integration and functional requirements been addressed?

- *Note:* The designer should make sure that the scene labels match the details for the wall control station. If there is no detail, the contractor may provide a wall station that is in conflict with the design vision or the Owner's needs.

## 3.0 Creation Timeline

The design process is iterative in that each aspect of a project is considered in increasing detail. The creation of a CIN and SOO benefits from this iterative approach, with continued development of detail and clarity through the design and construction process. Most North American projects follow a standardized progression



from Schematic Design (SD) through Construction Administration (CA). More detailed information on the lighting design and construction process can be found in ANSI/IES LP-7-20.<sup>3</sup> While there may be variations in timing, terminology, or sequence, depending on project type or region of the world, CIN and SOO development follows a fairly set structure through the design and construction process. Post-occupancy services, while not part of the creation timeline, are often specified. **Table 3-1** shows an example timeline.

### 3.1 Schematic Design (SD)

The project Schematic Design, or pre-design, phase is the time to start developing the Owner Project Requirements (OPR), and from this create the project Basis of Design (BOD). It may be helpful to think of this back-and-forth as a dialog that documents the control goals, needs and requirements. For example, the OPR may identify a need to “automatically turn off luminaires in empty rooms.” The BOD may cite “All classrooms should be provided occupancy sensors to automatically turn lighting off when the room is no longer occupied.”

This high-level documentation of the intent should inform the project team about how the overall system should function. The Schematic Design phase should conclude with conceptual design approval of the OPR and BOD.

### 3.2 Design Development (DD)

During Design Development, the specific attributes described in the OPR and established in the BOD should be identified, and a means of achieving each should be proposed and described in the CIN. Once the Client (or the Owner’s rep) has reviewed and provided feedback, the Control Intent Narrative should be updated so that it identifies a means of achieving the goals of the BOD. As an example, where the BOD cites the relevant code for lighting shut-off control, the CIN describes in detail what needs to be accomplished by the lighting controls: “All classrooms should be provided with ceiling mounted occupancy sensors to automatically turn on the light level to 50% when someone enters the room. Lighting will be turned off when the room is no longer occupied.”

**Table 3-1. Example of a CIN and SOO Timeline**

| Phase                            | Lighting Control Scope and Deliverables                                                                                                                                                                                                                       | CIN and SOO Development                                                                                                                                                                                     |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Schematic Design (SD)            | <ul style="list-style-type: none"> <li>Owner Project Requirements (OPR)</li> <li>Basis of Design (BOD)</li> <li>CIN to be included in BOD</li> </ul>                                                                                                          | <ul style="list-style-type: none"> <li>Document goals, criteria, relevant lighting control requirements, and needs</li> <li>Conceptual development of control</li> <li>Owner review and feedback</li> </ul> |
| Design Development (DD)          | <ul style="list-style-type: none"> <li>Control Intent Narrative</li> <li>Preliminary SOO</li> </ul>                                                                                                                                                           | <ul style="list-style-type: none"> <li>CIN further developed and completed</li> <li>Preliminary SOO documentation</li> <li>Owner review, feedback</li> </ul>                                                |
| Construction Documentation (CD)  | <ul style="list-style-type: none"> <li>Luminaire specification with specific dimming protocol</li> <li>Lighting control zoning</li> <li>Controls specification</li> <li>SOO</li> <li>One-line Diagrams</li> <li>Lighting and relay panel schedules</li> </ul> | <ul style="list-style-type: none"> <li>SOO further developed and completed</li> <li>Contractual construction documents approved</li> </ul>                                                                  |
| Construction Administration (CA) | <ul style="list-style-type: none"> <li>Shop drawing submittals</li> <li>RFI (Request for Information)</li> <li>Installation, programming, testing</li> <li>Site observation</li> <li>Functional testing</li> <li>Training and manuals</li> </ul>              | <ul style="list-style-type: none"> <li>Review submittals against CIN and SOO</li> <li>Adjust CIN and SOO as field conditions require</li> <li>Programming and functional testing, based on SOO</li> </ul>   |

The conceptual design is translated into formal plans. General concepts and strategies will become increasingly detailed, until a finished design is produced that satisfies the OPR and BOD. When the 100% DD documentation is presented to the owner for review, and prior to commencing with the next phase, the lighting control-system designer should have considered the system as a whole; this may include how lighting loads are zoned and to which control points, specific equipment and/or control settings, schedules, and required programming.

### 3.3 Construction Documentation (CD)

The lighting control system described in the project drawings and within the project manual should be identifiable as having grown directly from the earlier phases of project development. If the development process has been iterative and collaborative, there will not be major changes in approach between the system described at the end of Design Development and the fully articulated lighting specifications and control specifications that grow out of the CD process. As with other aspects of the lighting design, the move from DD to CD is one of refinement and clarification, and includes the creation of the SOO.

Although the CIN is finalized during this phase, the CIN is not contractually enforceable unless it is included in the contract documents. Project teams may accomplish this by appending the CIN to the specification as described and defined by the Control Specifications Institute (CSI). If the CIN is to be considered a contractually enforceable document, it will need to be reformatted to replace suggestive terms (should or will or could) with imperative terms (shall). The SOO is formulated and finalized during this phase and is part of the contractually enforceable documents. In the example of lighting shut-off control, the SOO would state how the lighting controls presented in the CIN operate: “Ceiling mounted occupancy sensors in a classroom shall automatically turn on all the lights to 50% light level when an occupant enters a classroom. All lights will be turned off after the classroom is vacant for 15 minutes.”

### 3.4 Construction Administration (CA)

During the CA phase of the project, the lighting control system submittals and shop drawings are created

and reviewed to ensure that they meet the goals and requirements of the specifications and the SOO. Manufacturers’ submittal documents should clearly indicate that sufficient hardware and software will be provided to enable installation and programming to accomplish the specified SOO. The submittal documents should clearly indicate what they are proposing to provide, as well as what they are not providing. Manufacturer-specific issues are identified and resolved in this process in order to maintain compliance with the contractually enforceable construction documents. A more complete explanation of the submittal review process can be found in ANSI/IES LP-7-20.<sup>3</sup>

Field situations arising during the lighting control installation, programming, and testing process are addressed, and the SOO is adjusted to align with any field updates and changes. Document updates resulting from field situations are critical in order to support the accuracy and efficiency of the startup and commissioning process. Through the lighting control installation process, the system is configured and programmed to operate as documented in the SOO, followed by functional testing where adopted energy codes require testing for certification of compliance.

## 4.0 Roles and Responsibilities

Each professional on the team has a specific role to play. These roles and responsibilities are not always clear, and it is best at the outset of a project to learn how the project will come to fruition, as well as who is responsible and at what phase. The descriptions below are specific to the various team members’ roles relative to the lighting control system.

- **Owner:** The Owner should review the BOD, including the CIN, and provide input. This basic overview of the lighting control system should be reviewed with the Project Architect as well as with the Client and, if possible, the end users, to ensure alignment with basic project requirements.
- **Architect:** The Architect is the professional with the most direct contact with the Client and therefore has the most knowledge of the requirements of the

Client, Owner, or end users, as well as their technical sophistication. Thus, the Architect can communicate those requirements to the design team, enabling them to develop the CIN and SOO and specify an appropriate control system. Additionally, when the Architect is the prime consultant and is gathering a group of sub-consultants, it is important to make sure that there is no scope gap among the sub-consultants. The Architect is responsible for ensuring that the scopes of all sub-consultants are clearly documented in detail so that it is clear which consultant is responsible for each task, such as developing zoning, writing the CIN, and developing the SOO.

- **Lighting Designer:** The lighting designer is the primary design professional responsible for the illumination on the project. Various design team professionals may serve this role, such as the Architect, Interior Designer, Electrical Engineer, or professionals that specialize in lighting design. Lighting control design responsibilities include understanding the requirements of the Client, Owner, and end users, and communicating specific needs, such as zoning, scene setting if any, or special control needs such as color tuning or tunable white control, to the Lighting Control Specifier (LCS).
- **Interior Designer:** Interior designers work intimately with clients, owners, and end users, and often know the specific needs for each space. Additionally, they are a key player in developing the programming of interior spaces, and therefore may have additional insight into the function of a space, and how lighting controls can support that function. It is important that the Interior Designer coordinate overall concepts with both Lighting Designer and Lighting Control Specifier, such as if the space needs multiple light settings or a specific user interface, so that the desired intent is captured in the CIN and SOO, and therefore delivered on the project. It is important to discuss where the control stations will go and what they look like with the LCS, so that the delivered system meets the design needs and expectations of the Interior Designer.
- **Electrical Engineer:** The Engineer may be tasked with identifying the specific components necessary to bring the control concept to fruition. It is critical that the Engineer be involved in the process and ensure that the specified system will comply with all applicable codes.
- **Lighting Control Specifier (LCS):** Various design team professionals may serve this role. These can include the project Lighting Designer, the Architect, and the Electrical Engineer. Less commonly, the Systems Integrator or Contractor may be tasked with specifying the control system. Regardless of the other roles the LCS may fulfill within the project, in this role they are responsible for specifying the lighting control system.
- **Manufacturer's Sales Representative:** The manufacturer's sales representative is a third-party representative, or agent, hired by a manufacturer to represent their brand and products in each market. They act as a "go-between" amongst the customer, distributor, and manufacturer. They help fill in whatever details are left out when orders are placed, and generally help the design team and sales partners as needed to enable a smooth flow of orders to manufacturers, and products onto job sites. They provide many services, as needed, from project to project. On any given project they might:
  - Assist with product selection based on the CIN and SOO
  - Assemble cut sheets for submittal approvals
  - Manage delivery dates
  - Provide onsite verification of product application and installation
  - Prepare lighting control plans using guidance from the manufacturer
  - Provide installation instruction to the installing contractor
  - Provide startup service and product programming
  - Assist with the activities of the commissioning provider
  - Manage the flow of money between the customer and the manufacturer
  - Manage the price margins and discounts
- **Lighting Controls System Vendor:** This person or company is responsible for translating the information put forth in the CIN, the SOO, and the material specification into a specific Bill of Materials (BOM), which can be represented by equipment cut sheets and/or catalog numbers or stock keeping units (SKUs).

- **Lighting Control System Integrator (LCSI):** This person or company trains and oversees contractors on startup, programming, and commissioning of the lighting control system, and trains the Owner or owner's representative. The LCSI may also have scope to verify controls design concepts prior to procurement, review specifications, and ensure that the products being delivered will meet the project's control design specifications. The LCSI is responsible for identifying the pieces of equipment or aspects of programming that will allow the various parts of the control system to work together. These interfaces may be required to achieve interoperability among systems from different manufacturers or across platforms.
- **Construction Manager (CM):** The CM can serve as the Owner's agent or employee, or can be a separate contractor. The role varies from project to project and may also be referred to as Project Manager.
- **Installing Contractor:** Generally, the Contractor's responsibility is part of the Construction Administration phase of the project and therefore may not start until after the CIN and SOO have been created and accepted by the design team. However, the Contractor and the rest of the construction team are typically responsible for realizing the project vision as communicated by the design team, so it is important that all terms and topics covered in both the CIN and the BOD documentation (if separate) be clearly understood and agreed to in advance of the Contractor beginning work. The typical bid process provides a mechanism for the Contractor to ask questions in advance, especially those which might create conflict or result in added costs or schedule delays. For this reason, it is extremely important that all members of both the design and construction teams take the Request for Information (RFI) process seriously and respond quickly and accurately when questions arise.
- **System Startup Provider:** This is the entity responsible for realizing the lighting controls SOO, utilizing the hardware that is delivered and

installed on the job site. The more advanced the system, or more complex the controls, the more instrumental the startup provider is in the success of the project. On simple systems, there may be no system startup provider, and the system may simply work as designed out of the box. On more complex projects, and because hardware may have no functionality out of the box, the startup provider is an instrumental part of the installation team; they rely on the work of the system startup provider to combine all inputs using the system logic as outlined in the SOO to deliver the desired outcomes.

- **Commissioning Provider:** Within the specific context of the CIN, the commissioning provider should have the opportunity to comment on the feasibility of the goals as expressed in both the CIN and the proposed SOO, and should be provided with final copies of both for use in determination of the building's successful completion of the design goals as outlined in the BOD. (*Note: Occasionally, the Commissioning Provider title is used to describe the System Startup Provider. In the context of this document, the Startup Provider is the contractor or agency executing the SOO and programming the hardware, and the Commissioning Provider is the individual or agency that ensures all energy savings and design goals are met as outlined in the BOD, and that the contractors and the Startup Provider have delivered on their promises.*) Additional information may be found in *ANSI/IES LP-8-20, Lighting Practice: The Commissioning Process Applied to Lighting and Control Systems*.<sup>2</sup>
- **Facilities Engineer:** This person is responsible for operation and maintenance of the lighting control system over its lifespan. It is important to make sure that the Facilities Engineer's input is sought whenever possible to ensure that the specified system will be able to operate as intended. Therefore, their acceptance of and education about the system are critical to the long-term success of the lighting controls. A facility engineer is an appropriate person to train end users.

## 5.0 Lighting Control Strategies

### 5.1 Strategies Overview

There are many strategies used in lighting controls. **Table 5-1** lists a number of commonly used strategies a lighting controls specifier may use to provide desired outcomes. It is critical, therefore, that they provide functional, plain-language expectations (the Control Intent Narrative), as well as enforceable contract language (the Sequence of Operations) to communicate not only the inclusion of a given strategy, but also how each strategy shall be incorporated, such as, setpoints, timing values, and various other details. The following section provides examples. (See Section 4.0 in ANSI/IES LP-6-20<sup>1</sup> for a more comprehensive list, as well as definitions and details on each control strategy.) **Table 5-1** shows the most basic of manual control strategies, followed by some automatic control strategies, and ending with the control results.

The information in **Sections 5.2** and **5.3** provide a detailed investigation into two different ways of looking at lighting controls specifications—the lighting CIN and the SOO. The material in **Section 5.2** is broken down by strategy type, with some examples of how a strategy may be documented for a given project. The material in **Section 5.3** illustrates some examples by space type and helps to show how these various strategies can be implemented together to provide the full desired controls scheme.

### 5.2 Lighting Control Strategies – By Discrete Strategy Type

In the subsections that follow, CIN and SOO examples for discrete lighting control strategies are presented to demonstrate the elements needed to adequately describe a specific type of control. These examples are intended to be generic and do not contain specific setpoints, settings, or other information necessary for specific project implementation. The designer is responsible for providing this specific information to fully complete the project CIN or SOO for a given application.

Although these are discrete lighting control strategy examples, in a project, lighting control functionality for a space is made up of multiple discrete controls interacting together to achieve the lighting control objective for the space. A complete CIN and SOO for a specific space would combine individual discrete control descriptions to create a complete intent for how the lighting control is to be installed, configured, and operated in the space.

The information presented in these examples is for designers' reference. The examples provide sample content of CIN and SOO components needed to specify discrete lighting control strategies in contractual documents. These examples should not be copied and used for a real project; instead, they should be adapted and modified to suit the project's budget and specific control requirements.

**5.2.1 Manual Controls.** Manual controls provide a user interface to enable control of the lighting level in a space. Manual controls typically consist of a lighting

**Table 5-1. Lighting Control Strategies**

| Control Inputs                           |                      |                         | Control Results    |                           |
|------------------------------------------|----------------------|-------------------------|--------------------|---------------------------|
| Manual                                   | Automatic            |                         |                    |                           |
| Manual User Interface (switch or dimmer) | Occupancy Sensing    | Vacancy Sensing         | Switching          | Lumen Maintenance Dimming |
| Scene Control                            | Daylight Harvesting  | Demand Response         | Continuous Dimming | Color Tuning              |
| Room Partitioning                        | Institutional Tuning | Circadian Control       | Step Dimming       | Tunable White             |
| Personal Tuning                          | Time Scheduling      | Adaptation Compensation | Load Shedding      | Fading: On, Off, Cross    |
|                                          | High End Trim        | Scene Control           | Shade Control      |                           |



zone switch, a step-dimming control switch, or a dimmer installed in an electrical switch box that is conveniently located for occupant access and control. More information on the application of user interfaces and manual controls can be found in Sections 3.7.2, 4.1, and 7.1 of ANSI/IES LP-6-20.<sup>1</sup> **Table 5-2** shows an example of manual on/off switches used to control both general and decorative lighting. These switches also provide occupants the ability to override the current lighting state until the occurrence of the next

automatic event. **Table 5-3** shows a similar example of controlling general and decorative lighting but uses manual dimming controls instead of on/off switches. This example also demonstrates utilization of low- and high-end trim settings with the dimming controls.

**5.2.2 High-End Trim.** High-end trim is a lighting control strategy that sets the maximum control level in a space. This strategy is often used to reduce

**Table 5-2. Manual On/Off Switching Control Example**

| Control Intent Narrative                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Sequence of Operation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| General and decorative lighting in <room names or designations> is turned on and off by occupant manual operation of switches installed in electrical boxes at each entrance or exit door. Switches will be located such that the controlled lighting is within the line of sight from each switch location. Three switches will be installed providing on/off control of: uniform bi-level control for general lighting as required by code; remaining <x%> bi-level control of general lighting; and decorative wall sconce lighting. Except for automatic daylight harvesting control, the manual control switches will temporarily override any automatic control such as time-switch scheduling and occupancy sensors, until the next automatic event occurs. The manual control switches will not allow lighting to be increased above the automatic daylight-harvesting control lighting level. | <ul style="list-style-type: none"> <li>• Wall switch “a” shall provide manual on/off control of “a” luminaires, uniformly operating &lt;x%&gt; of the general lighting.</li> <li>• Wall switch “b” shall provide manual on/off control of “b” luminaires, uniformly operating &lt;y%&gt; of the general lighting.</li> <li>• Wall switch “c” shall provide manual on/off control of “c” luminaires, operating the decorative wall sconce lighting.</li> <li>• Wall switches shall be capable of turning lighting on and off at any time regardless of automatic control by time-switch scheduling or occupancy sensors. Wall switches shall not increase the lighting level above the automatic daylight-harvesting control lighting level.</li> </ul> |

**Table 5-3. Manual Dimming Control Example**

| Control Intent Narrative                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Sequence of Operation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| General and decorative lighting in <room names or designations> are turned on and off and dimmed by occupant manual operation of dimmers installed in electrical boxes at each entrance or exit door. Dimmers will be located such that the controlled lighting is within line of sight from each dimmer location. Dimmers shall operate across the range of minimum lighting power of <x%> to a maximum trimmed level of <y%> lighting power. Two dimmers will be installed providing on/off and dimming control of general lighting and decorative wall sconce lighting. Except for automatic daylight-harvesting control, the manual dimming controls will temporarily override any automatic control such as time-switch scheduling and occupancy sensors, until the next automatic event occurs. The manual dimming controls will not allow lighting to be increased above the automatic daylight-harvesting control lighting level. | <ul style="list-style-type: none"> <li>• Wall dimmer “a” shall provide manual on/off and dimming control of “a” luminaires, uniformly operating the general lighting. Low-end-trim shall be set to &lt;x%&gt; of full lighting power, and high-end trim shall be &lt;y%&gt;.</li> <li>• Wall dimmer “b” shall provide manual on/off and dimming control of “b” luminaires, uniformly operating the decorative wall sconce lighting. Minimum allowed lighting power shall be &lt;x%&gt;, and high-end trim shall be &lt;y%&gt;.</li> <li>• Wall switches shall be capable of turning lighting on and off at any time regardless of automatic control by time-switch scheduling or occupancy sensors. Wall dimmers shall not increase the lighting level above the automatic daylight-harvesting control lighting level.</li> </ul> |

energy consumption, deliver task-illumination tuning, or compensate for light loss factors over the life of a luminaire. More information on the related topics of institutional tuning and lumen maintenance can be found in Sections 4.3 and 4.8 of ANSI/IES LP-6-20.<sup>1</sup> **Table 5-4** shows an example for a high-end trim scenario.

**5.2.3 Low-End Trim.** Low-end trim is a lighting control strategy that sets the minimum control level in a space. This strategy is used to ensure that lights are never dimmed below a specific lighting level. Also, low-end trim may be used to mitigate the light flicker of some lamps that can occur in the very low dimming range. While low-end trim is sometimes used to provide a minimum walk-through illuminance, it does not replace the function of emergency lighting. **Table 5-5** shows an example for a low-end trim scenario.

**5.2.4 Time Scheduling Control.** Time scheduling is a lighting control strategy used to automate lighting levels where there is a repeatable time-of-use space occupancy pattern. Time scheduling may turn lighting on or off, set lighting to a specific level, or change lighting parameters based on space utilization at different times and days. In many applications, overrides are needed, or required, giving occupants manual control of the space lighting at all times. Occupant overrides are typically temporary until the next scheduled event. A common scheduled event is multiple shutoff times, which turn lighting off to meet energy code and energy efficiency requirements. It is

common to refer to repeated scheduled shutoff events as a “lighting sweep off” schedule or “lighting sweeps,” as in cleaning up or sweeping through lighting left on. The time schedule may be activated by a hardwired discreet component or driven by programmable software. More information on the features and application of time scheduling control can be found in Section 4.6 of ANSI/IES LP-6-20.<sup>1</sup>

Time scheduling control can be applied at the circuit level or via the lighting control system network. In either case, it is important that additional details be provided to ensure that the sequence is enacted properly. (See **Figure B-3** in **Annex B** for an example of a network diagram.)

**Table 5-6** shows the use of time scheduled control to repeatedly turn off the lighting in a room based on the maximum amount of time the energy code allows for lighting override outside of scheduled business hours. The example also incorporates a lighting “blink” warning prior to turning the lighting off, notifying occupants of a coming shutoff event, thereby allowing them time to further override the lighting by actuating the lighting switch.

**5.2.5 Occupancy and Vacancy Sensing.** *Occupancy sensors* detect human occupancy in the controlled space and turn the lighting on or off based on the presence or absence of people. *Vacancy sensors* combine a manual switch for on and an occupancy sensor for automatic off operation. More information on the features and application of occupancy and vacancy sensing control can be found in

**Table 5-4. High-End Trim Example**

| Control Intent Narrative                                                                                                                                                            | Sequence of Operation                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Light levels will be limited to uniform levels during system startup, to allow for periodic adjustment over the life of the luminaires to compensate for normal light loss factors. | <ul style="list-style-type: none"> <li>The maximum operating level shall be set to &lt;x%&gt; of maximum for all general lighting.</li> </ul> |

**Table 5-5. Low-End Trim Example**

| Control Intent Narrative                                                                                                                                                    | Sequence of Operation                                                                                                                                                                                                                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Low-end trim will be set in all common and circulation areas to provide occupants a certain minimum amount of light regardless of energy-management dimming or time of day. | <ul style="list-style-type: none"> <li>The minimum operating low-end trim level shall be set for all lighting in all common and circulation areas to the minimum level specified on the luminaire schedule for each luminaire type.</li> </ul> |

**Table 5-6. Time Scheduling: Off Control Example**

| Control Intent Narrative                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Sequence of Operation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lighting in <room names or designations> use time scheduling from a centralized system time switch to turn lighting off after business hours. User control switches override lighting after business hours. User overrides allow lighting to remain on for the maximum duration allowed by the energy code. Lighting will warn occupants <x> minutes before scheduled shutoff events by momentarily turning lighting off and back on within <y> seconds. After the shutoff warning, user control switches may cancel the pending scheduled shutoff and reset the override to the maximum time allowed or until the system enters operating business hours, whichever occurs sooner. | <ul style="list-style-type: none"> <li>• All space lighting, except required egress lighting, shall be scheduled by centralized time switch to be turned off at the end of business operating hours and remain off until the beginning business operating hours the next business day.</li> <li>• Wall control switches or dimmers shall be configured to provide override-on control of lighting after business hours. Actuation of a wall control after business hours shall turn the lighting on for &lt;w&gt; minutes.</li> <li>• Lighting in the space shall momentarily turn off and back on within &lt;y&gt; seconds, to warn occupants &lt;x&gt; minutes before shut-off. Manual actuation of the wall control after this warning signal shall cancel the pending shutoff.</li> </ul> |

Sections 4.5 and 7.4 of ANSI/IES LP-6-20.<sup>1</sup> **Table 5-7** shows an example of an occupancy sensor used to automatically turn on the general lighting to the code maximum-allowed level upon detection of occupancy. All lighting in the room is automatically turned off when no occupancy is detected. **Table 5-8** demonstrates the use of a vacancy sensor control, where occupants need to manually turn on the lighting, as there is no automatic-on function. Similarly, when no occupancy is detected in the room, and following the shutoff time delay, all lighting is turned off.

**5.2.6 Daylight Harvesting.** Daylight harvesting entails reducing electric light output to maintain a

near-constant illuminance in a space receiving daylight. Energy codes requiring daylight harvesting (also called “daylight responsive control”) often only require daylight harvesting for general illumination, and not for decorative, accent, or supplemental task lighting. More information on the features and application of daylight harvesting control can be found in Section 4.7 of ANSI/IES LP-6-20.<sup>1</sup> **Table 5-9** shows an example of a daylight harvesting CIN for a single daylight zone using a closed-loop photosensor that monitors the lighting level in the room and modulates the lighting output to maintain a specified lighting level.

**Table 5-7. Occupancy Sensor Example**

| Control Intent Narrative                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Sequence of Operation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| General lighting in <room names or designations> will automatically turn on to the code maximum-allowed lighting level when an occupant enters. Lighting remains on while the room is occupied. When the room is no longer occupied, all room lighting will automatically turn off after the maximum time delay allowed by the code. Controls at the entrance to the space provide users the ability to turn on, increase, decrease, or turn off the general lighting. To save energy, lighting should quickly turn off when the room is only briefly occupied. | <ul style="list-style-type: none"> <li>• Upon detection of occupants, general lighting shall automatically turn on to &lt;w%&gt; of full power.</li> <li>• A wall switch at the entrance shall provide the user control to turn on, increase, decrease, or turn off the general lighting.</li> <li>• Occupancy sensor mounting location and sensitivity shall be set for major-motion full-room coverage and minor-motion coverage in the workstation area.</li> <li>• All room lighting shall be automatically turned off &lt;x&gt; minutes after occupancy was last detected.</li> <li>• The occupancy sensor walk-through mode shall be enabled, automatically turning lighting off &lt;y&gt; minutes after initial detection, when occupancy is detected for a duration of less than &lt;z&gt; seconds.</li> </ul> |



**Table 5-8. Vacancy Sensor Example**

| Control Intent Narrative                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Sequence of Operation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| General lighting in <room names or designations> is turned on by user manual operation of the vacancy sensor switch (vacancy mode) at the entrance to the space. Lighting remains on while the room is occupied. When the room is no longer occupied, all room lighting automatically turns off after the maximum time delay allowed by the code. The vacancy sensor switch provides the ability for users to turn on, increase, decrease, or turn off the general lighting. To save energy, the lighting should quickly turn off when the room is only briefly occupied. | <ul style="list-style-type: none"> <li>• A vacancy sensor switch shall be installed at the entrance to the space, providing user's manual control to turn on, increase, decrease, or turn off the general lighting.</li> <li>• Vacancy sensor mounting location and sensitivity shall be set for major-motion full-room coverage and minor-motion coverage in the workstation area.</li> <li>• All lighting shall be automatically turned off &lt;v&gt; minutes after occupancy was last detected. Lighting shall automatically turn back on if occupancy is detected within &lt;w&gt; seconds of automatically turning off. After &lt;x&gt; seconds after an automatic shutoff, the lighting shall only be turned on from the vacancy sensor switch.</li> <li>• The vacancy sensor's walk-through mode shall be enabled, automatically turning lighting off &lt;y&gt; minutes after initial detection, when occupancy is detected for a duration of less than &lt;z&gt; seconds.</li> </ul> |

**Table 5-9. Daylight Harvesting Example**

| Control Intent Narrative                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Sequence of Operation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| General lighting is continuously dimmed in the daylit areas of <room names or designations> as daylight is contributed. A closed-loop photosensor monitors the light level at the task plane and modulates the general lighting to maintain near-constant illuminance. General lighting will be automatically turned off when the daylit area exceeds <x%> of the design illuminance target. Manual switch control of the general lighting may not increase or override the lighting level above the daylight-controlled level. Accent-lighting luminaires are not daylight harvesting controlled. A portable handheld device or app, accessible to authorized personnel, will be used for programming and setting daylight control parameters while standing on the floor. | <ul style="list-style-type: none"> <li>• The daylighting control photosensor shall be set to continuously increase or decrease the electric lighting luminous intensity to maintain the target illuminance of the daylight area.</li> <li>• When combined electric and daylight illuminance exceeds &lt;a%&gt; of target levels for &lt;b&gt; seconds, electric light shall be reduced by &lt;c%&gt; per second.</li> <li>• When combined electric and daylight illuminance falls below &lt;d%&gt; of target levels for &lt;e&gt; seconds, electric light output shall be increased by &lt;f%&gt; per second.</li> <li>• When the combined electric and daylight illuminance exceeds &lt;g%&gt; of target levels for &lt;h&gt; seconds, the electric lighting shall be automatically turned off.</li> <li>• The interface shall not allow the user to override the electric lighting controls in a way that will provide greater illuminance than allowed by the daylight control photosensor.</li> <li>• The daylight control photosensor shall not require physical adjustment at the sensor. All settings shall be accessible by authorized personnel when standing on the floor of the space for which the sensor has control.</li> </ul> |

**5.2.7 Demand Response Control.** Demand response control is a lighting control strategy that reduces building lighting power by dimming or turning off selected lighting in response to a need or a request

to reduce building power demand. Demand response events can be manually or automatically initiated within a building or can be signaled from external actions such as electrical grid events or demand response signals

from utilities and grid operators. In the U.S., demand response is becoming integrated into distributed energy resource (DER) systems, operating in conjunction with onsite power generation and storage within grid-enabled efficient buildings (GEB). More information on the features and application of demand response control can be found in Section 4.9 of ANSI/IES LP-6-20.<sup>1</sup>

Demand response requires a networked control system because the command originates from a central server. It is crucial to include a network diagram because vendors and contractors often fail to realize that the strategy requires a networked topology. (See **Figure B-3** in **Annex B** for an example of a network diagram.)

An example of a CIN and SOO that will reduce the lighting power when a demand-response request signal is received through the commonly used Open ADR 2.0b interface (required interface standard for some energy codes) is shown in **Table 5-10**.

**Table 5-10. Demand Response Control Example**

| Control Intent Narrative                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Sequence of Operation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Building general lighting will respond to Open ADR 2.0b utility demand-response request signals from the utility or demand-response aggregator by automatically shedding or reducing lighting power by at least <x%> of the building's total lighting load. Light reduction for demand response will be accomplished through dimming the general lighting output over <y> seconds in all spaces except restrooms, stairwells, storage rooms, or any enclosed spaces under <z> square <meters or feet>. At the completion of a demand response request, the general lighting will be restored to the last commanded illuminance. If the general lighting level was adjusted during the demand response event, the lighting will remain at its current level. | <ul style="list-style-type: none"> <li>When a demand-response request signal is received, general lighting in spaces designated for demand response control shall automatically shed or reduce lighting power by at least &lt;x%&gt; over &lt;y&gt; seconds.</li> <li>When a demand-response signal request concludes, general lighting in spaces designated for demand response control shall automatically be restored to the last commanded illuminance or, if the lighting level was changed during the event, shall remain at the changed lighting level.</li> <li>Spaces not designated for demand response control shall not change the lighting level during a demand response event.</li> </ul> |

**Table 5-11. Emergency Lighting Control Example**

| Control Intent Narrative                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Sequence of Operation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| General-lighting luminaires will automatically provide life-safety code level of emergency illumination upon loss of normal power, using battery-backup units in luminaires that will power full light output from the emergency-designated luminaires. Any lighting control (manual switch, occupancy sensor, daylighting control, or otherwise) will be circumvented so luminaires will be turned on to full light output. When normal power returns, the luminaires will revert back to normal operation and provide the last commanded illuminance. | <ul style="list-style-type: none"> <li>Upon loss of normal power, emergency-lighting designated luminaires shall be automatically turned on to full light output operating on battery-backup power. No other control shall be capable of operating or overriding the full light-output state during loss of normal power.</li> <li>Upon restoration of normal power, emergency-lighting designated luminaires shall revert back to normal operation and provide the last commanded illuminance.</li> </ul> |

**5.2.8 Emergency Lighting.** Emergency lighting provides a required level of illumination during the loss of normal power. Requirements for where and at what illuminance emergency lighting is to be provided are found in the life safety codes enforced by the local authority having jurisdiction. More information on the application of emergency lighting can be found in Section 4.12 of ANSI/IES LP-6-20.<sup>1</sup> **Table 5-11** shows a CIN and SOO example that utilizes luminaires with emergency battery-backup equipment (also called “unit equipment”) serving both as general lighting during normal operation, and as emergency lighting via power from the battery during emergency events. In this example, the emergency luminaires turn to full output either when there is a loss of normal power.

**5.2.9 Room Partitioning.** Room partitioning is a lighting control strategy that allows joined or independent operation of lighting and controls for rooms that have movable partitions between them. This is typically found in classroom, training room, ballroom, conference room,

and convention room settings. When partitions are closed between adjoining rooms, the lighting controls within individual rooms operate just to serve the lighting in the individual partitioned room. When partitions or curtains between rooms are opened, all lighting controls and lighting serving the open space are “combined” to operate together for the now-single controlled room. **Table 5-11** shows an example of a flexible, multi-use space where movable partition walls have sensors indicating when a partition is open or closed. This allows the control system to automatically combine the wall station lighting controls in individual rooms to function together when the partitions are opened to create a larger room, or to configure the individual rooms’ lighting controls to operate separately when the partitions are closed.

**5.2.10 Color or Spectrum Adjustment.** For luminaires that offer color or spectrum adjustment, there are three primary strategies by which the luminous flux and/or spectral power distribution of a lamp or luminaire is adjusted: warm dim, white tuning, and color tuning.

Which strategy is used is determined by the needs of the space, and is implemented by selection of lamps or luminaires that permit the necessary control of spectral power distribution. Manufacturers use different names for these strategies for branding purposes, but any strategy of color adjustment of lamps and luminaires fits within just one of these three categories.

The specifier determines which is being used, ensures that all equipment is compatible with this feature, and documents how or by what metric the color is changed.

Some end users may prefer to have the ability to manually adjust the light spectrum to coordinate with the space utilization at a given time. Conversely, some industries or owners may choose to preprogram CCT for various reasons—for example, to match daylight, to intentionally mimic daylight at night, or to provide a warm comfortable glow at certain times of day.

**5.2.10.1 Warm Dim.** Warm dim, also known as “dim to warm” or “warm glow” or, less often, “amber shift,” is the simplest of the three. As the control—whether line voltage dimming, 0-10V, or a digital signal—is reduced from the maximum level, the CCT shifts toward a lower value (appears warmer), and the luminous intensity is reduced. In a warm-dim lamp or luminaire, the relationship between luminous intensity and CCT has a fixed curve. An appropriately designed LED lighting system can take advantage of this characteristic to mimic the dimming behavior of incandescent lamps without allowing the user to adjust CCT separately from luminous intensity.

**5.2.10.2 White Tuning.** White tuning, also known as “tunable white,” “variable white,” or “dynamic white,” is the adjustment of light source CCT through mixing of two or more independently controlled white emitters, which typically have a CCT difference of 1000 K or higher. When the control values of the luminaire or lamp are proportionally mastered, the luminous intensity may be adjusted largely independent of the CCT. White-tuning luminaires only offer adjustments nominally along the blackbody curve, and therefore the output at any setting can reasonably described as white.

**Table 5-11. Room Partitioning Control Example**

| Control Intent Narrative                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Sequence of Operation                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Flexible rooms with movable partitions have lighting controls that operate only the lighting within the individual partitioned room. When partitions are opened between adjoining rooms, creating a larger room, the controls will automatically be configured for any lighting control location or interface within the combined room, to control all lighting in the combined room together. This provides convenient, uniform control from any control location or interface. The control system receives a signal from sensors detecting the partition’s open or closed status. The signal combines the operation of the user controls when the partitions between rooms are opened more than <x%>. | <ul style="list-style-type: none"> <li>When partitions between rooms are closed, only the lighting control devices within each individual room shall operate the lighting within that room.</li> <li>When partitions between rooms are opened at least &lt;x%&gt;, a sensor signal shall combine operation of all lighting control devices within the combined room to operate all lighting within the combined room together.</li> </ul> |

Most white-tuning luminaires allow independent control of the lumen output of multiple light sources of different CCTs; varying the output of the individual sources allows the creation of varying CCTs for the luminaire as a whole. This is referred to as direct emitter control, which is preferred for most applications where there are multiple manufacturers. Occasionally, luminaires have one control channel for CCT and one for luminous intensity, which can make it difficult to match the color of their light to that of other luminaires in a space unless they are all calibrated together.

**Table 5-12** shares an example of white tuning in a lobby area. In this case, the white tuning follows the CCT reading from a sensor on the building roof. If readings from the CCT sensor should cease, the white tuning system uses a preprogrammed CCT setting based on time of day from sunrise to sunset.

**5.2.10.3 Color Tuning.** Color tuning, also called spectral tuning, uses controls to adjust lighting color(s) in spaces. Color-tuning luminaires with three or more emitters will

**Table 5-12. White Tuning Example**

| Control Intent Narrative                                                                                                                                                                                                                                                      | Sequence of Operation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| <p>The lobby downlights, cove lights, and entry vestibule luminaires employ white tuning to track the correlated color temperature of daylight via a CCT sensor. In this way, the CCT of the lighting in these spaces will match that of the daylight entering the space.</p> | <p>The CCT of lighting zones &lt;zone names or designations&gt; shall be white tunable. The luminaires in these zones shall be programmed to track the CCT of daylight as presented to the CCT sensor installed on a north facing wall on the roof.</p> <p>This CCT sensor shall be calibrated onsite at no fewer than five settings ranging from 2300 K to 6000 K, and shall interpolate all CCTs in between. For all CCTs measured below 2300 K, the system shall command the use of 2300 K for the relevant interior zones. For all CCTs read above 6000 K, the system shall command the use of 6000 K for the zones.</p> <p>When no light is sensed, the system shall command the use of 2300 K in the zones.</p> <p>All luminaires shall be calibrated onsite at no fewer than five settings ranging from the luminaires' maximum CCT to their minimum CCT, such that their CCTs match those commanded by the system. Given variations in LED chips, minor adjustments to this calibration may be required with the lighting designer onsite.</p> <p>CCT changes shall be applied at a rate of &lt;x%&gt; per second, and there shall be a CCT deadband of &lt;y&gt; K.</p> <p>If the CCT sensor input is lost, the system shall provide notification via the touchscreen and revert to a schedule of:</p> <ul style="list-style-type: none"> <li>- 2300 K: from 1 hour before sunset to 1 hour before sunrise.</li> <li>- 3500 K: from 1 hour before sunrise to sunrise</li> <li>- 4500 K: from sunrise to 1 hour after sunrise</li> <li>- 6000 K: from 1 hour after sunrise to 3 hours before sunset</li> <li>- 4500 K: from 3 hours before sunset to 2 hours before sunset</li> <li>- 3500 K: from 2 hours before sunset to 1 hour before sunset</li> </ul> <p>This backup CCT schedule shall be applied at a rate of &lt;x%&gt; per second.</p> |

usually have a gamut that includes the blackbody curve, allowing them to also be used for white tuning. When manufactured with emitters of the necessary spectral power distribution, color-tuning lights may be used in conjunction with daylight to match lighting CCT, or used to

provide circadian stimulation when daylight is not available. Color tuning can change the feel of a space over time.

The example in **Table 5-13** shows a tunable lighting system that will provide color on a lobby wall. The color-

**Table 5-13. Color Tuning Example**

| Control Intent Narrative                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Sequence of Operation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>The lobby wall's grazing luminaires will provide full color adjustment. This adjustment will be made manually from a touchscreen located &lt;location&gt; and via a wireless interface device provided by owner. Additionally, color settings can be scheduled in the processor to reflect the following list of holidays. Initial looks (i.e., presets) and schedule are to be set with the Lighting Designer and Programmer onsite. Future adjustments to be end-user adjustable via the Owner's wireless interface device.</p> <p>&lt;List of holidays and a brief explanation of what the lights should do on each holiday.&gt;</p> | <p>The color of lighting zones &lt;names or designations of zones&gt; shall be controlled via full-spectrum color tuning. The lighting in the zones shall be preprogrammed per a schedule, and over-ridable by an end user.</p> <p>Provide the following list of color presets.</p> <p>Valentine's Day<br/>St Patrick's Day<br/>Easter<br/>[...]<br/>Spare 1<br/>Spare 2<br/>Spare 3</p> <p>These shall be programmed into the processor and scheduled as shown below. All dates begin and end at 4 a.m., and if the luminaires are off at 4 a.m., this color shall take effect the next time the lights are turned on. This color schedule shall be asserted every morning at 4 a.m. during the programmed time.</p> <p>Feb. 7 thru Feb. 14: Valentine's Day<br/>Mar. 10 thru Mar. 19: St Patrick's Day<br/>[...]</p> <p>These presets and schedules shall be stored in the system by the Programmer, as directed onsite by the Lighting Designer. The Owner shall be able to adjust both the content and the scheduled days via a user interface displayed on the Owner's wireless interface device. The spare color presets are intended to represent sports teams, local academic institutions, or local holidays of importance to the ownership team. These will be defined and programmed on the user interface display by ownership. Prepare to adjust labels accordingly.</p> <p>When no programmed color is set for a given day, the lights shall &lt;define action&gt;.</p> <p>Additionally, users shall be able to override color manually via a color picker presented at the lobby touchscreen as well as on the Owner's wireless interface device. These changes shall take effect immediately and shall remain in effect until 4 a.m. the following morning, when the above specified schedule shall be reasserted.</p> <p>A button shall be located on the touchscreen next to the color picker to reassert the schedule, overriding the manual entry and returning the lighting to the prescheduled color(s).</p> |

tuning system provides prescheduled control of specific lighting colors on holidays, user-scheduled or manually set lighting color via a user interface display, and both manual control and override control of lighting color via a touch screen located in the lobby space.

**5.2.10.4 Control Considerations for Tuning.** Complex control is required for all color adjustment, with greater complexity required for color tuning than for warm dim. Designers should select or specify the appropriate luminaires and control interface for each space, while being aware that occupants will generally come from diverse backgrounds and experiences.

Warm-dim tuning essentially replicates incandescent dimming, and often uses dimmers that might also dim incandescent lamps. It is important to confirm the compatibility of warm-dim lamps and luminaires with dimmers. Some lamps or luminaires have specific dimmer compatibility, and failure to confirm compatibility may result in damage to the luminaire, dimmer, or both.

White-tuning and color-tuning luminaires often have separate power and data, though some (e.g., many tape-light products) do have direct power applied to the LEDs. Control is applied to the luminaire, or an external driver, using standardized signals.

Proprietary control signals are typically used between luminaires and controllers from the same manufacturer. When all the sources in a space are from the same manufacturer, a proprietary protocol may be a simple way to control white or color tuning in a space. When luminaires in a space have different sources, and especially sources with different gamuts, a standardized protocol should be used.

DMX512-controlled luminaires in particular may have options for control through either direct emitter control or indirect control such as chromaticity coordinates. Direct emitter control is generally preferable between the controller and luminaire, as it gives the greatest flexibility and control. As with proprietary controls, chromaticity control can make simple systems with a single luminaire type easy to program. However, when multiple luminaire types are used, it is unlikely

that chromaticity calibration will be the same between luminaires, and color aberrations may occur where luminaires mix. An example of this is a ballroom, where there may be color-tunable cove lights and downlights. Matching the color of the cove lights and the downlights is necessary, but they almost certainly have different gamuts. Trimming the control ranges can help match the different source colors.

User interfaces are different from the control signals between controls and luminaires. There is considerable complexity involved in representing the full color spectrum, due to the number of discrete but interrelated parameters that need to be manipulated to accurately produce an exact color. This complexity often makes direct emitter control preferable for control communication, while a color space, or basic color picker (a user color-selection interface), is generally preferable for the user interface. This approach will allow for simple user control for applications that do not require a high degree of color accuracy, while very precise control is available when needed.

If extremely precise color accuracy is required—for example, if an exact RAL color has to be matched—more care is needed. The luminaire will need to be studied and mocked up so that it illuminates the exact project material, and the control inputs required to provide the desired output are then documented. This information can then be used to specify the final control schema that enables the desired output to be provided in the final installation. An example of CIN and SOO for tunable lighting is shown in **Table 5-14**.

**5.2.11 Circadian.** Circadian lighting is becoming common due to the potential for productivity, wellness, and operational improvements. There are several metrics available to determine the circadian stimulation that might be appropriate or required for a project, but a thorough discussion of the topic is beyond the scope of this document. Luminaires used to provide circadian-effective lighting generally fall into white-tunable or color-tunable categories, and the CIN and SOO example in **Section 5.2.13** may be used by designers familiar with the requirements of circadian entrainment as a base for a complete circadian lighting specification.



**Table 5-14. Tunable Lighting Example**

| Control Intent Narrative                                                                                                                                                                                                                                                                                                                                                                                                                                   | Sequence of Operation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Tunable lighting is adjusted in &lt;room names or designations&gt; to provide lighting color control for occupants. Between &lt;hh:mm&gt; a.m. and &lt;hh:mm&gt; p.m., lighting will be neutral white, with a local control provided to select other colors and white tones. After &lt;hh:mm&gt; p.m., lighting will slowly change to a warm white unless overridden by occupants. When the alarm system is armed, lights will fade out completely.</p> | <ul style="list-style-type: none"> <li>• On occupancy detection between &lt;hh:mm&gt; a.m. and &lt;hh:mm&gt; p.m., tunable luminaires of Type &lt;aa&gt; located in spaces &lt;space names or designations&gt; shall go to a tuned CCT of 4000 K, with light output reduced to 50% of maximum.</li> <li>• On occupancy detection between &lt;hh:mm&gt; p.m. and &lt;hh:mm&gt; a.m., tunable luminaires shall go to a tuned CCT of 2700 K, with light output reduced to 35% of maximum.</li> <li>• On occupant adjustment of the tuned light color, luminaires shall hold the selected color until occupancy detector timeout.</li> <li>• When the alarm system is armed, lights shall fade out completely and remain off unless the alarm is activated.</li> <li>• When the alarm system goes into alarm condition, the tunable lights shall go to full output with a CCT of 4000 K and remain at that output until the alarm condition is cleared, regardless of occupant input on the control.</li> </ul> |

### 5.3 Lighting Control Strategies – By Space Type

The use of a CIN and an SOO to establish contractual documents is typically not exhibited in discrete control strategies as described in **Section 5.2** and its subsections, but rather in layered strategies combined to provide fully functional illumination in specific space applications. The subsections that follow provide examples of CINs and SOOs for specific building spaces.

As with the discrete individual control-strategy examples, these examples should not be copied and used for a real project; instead, they should be adapted and modified to suit the project's budget and specific control requirements.

When specifying a lighting control system to accomplish a specific CIN and SOO, a designer will need to determine what type of lighting control equipment has the features and capability to implement the CIN. To meet base user control and code-compliant implementation, almost any lighting control system will typically suffice. These could be standalone, room-based or networked lighting controls that can deliver the specified SOO. The owner or designer might further prefer a system type based on its cost, complexity, familiarity, or compatibility. Although lighting control systems may communicate

wired or wirelessly between devices, the communication architecture is usually not limited to the CIN and SOO, but is more of an installation, cost, and deployment preference decision.

Where CINs require greater features or a connected system to achieve design intent, it is critical that the designer examine, select, and specify a capable system with all the required features and connectivity. As an example, a CIN might require a lighting system to execute a building-wide demand response strategy. In this case, a networked lighting control system is needed to propagate the demand control signal across the lighting control system to achieve the demand response action. As another example, data collection from the lighting control system, such as electrical energy monitoring and reporting, might part of the CIN. This would require a system that aggregates the energy monitoring and data across the lighting control network. This information might also be transmitted to cloud-based storage via internet connectivity.

**5.3.1 Enclosed Private Office Example.** Enclosed private offices are spaces which should allow good adjustability and control of lighting by the occupant of the space. **Table 5-15** shows an example of a CIN and

**Table 5-15. Example of a CIN and SOO for an Enclosed Private Office**

| Control Intent Narrative                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Sequence of Operation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| <p>General lighting in enclosed private offices automatically turns on to the code-maximum illuminance when occupants enter. Lighting remains on while the room is occupied and may be adjusted by the occupant with a dimmer switch at the entrance. When the room is no longer occupied, all lighting automatically turns off after the maximum occupancy-sensor time delay allowed by the energy code.</p> <p>When the energy code wattage thresholds for the general electric lighting are achieved in daylight areas, the general electric lighting continuously dims to maintain a uniform and consistent illumination when there is sidelighting contribution of daylight. When sufficient daylight is available, the lighting will turn completely off. Daylighting control ramp rates should slowly dim the lighting after appropriate delay, to avoid adjustments that are distractive, and raise lighting output quickly when daylight level decreases due to passing cloud cover. To save energy, lighting should quickly be turned off when the room is only briefly occupied. All configuration, programming, and parameter adjustments and settings are accessible to authorized personnel while standing on the floor.</p> | <ul style="list-style-type: none"> <li>• Upon initial occupant detection, general lighting shall be automatically turned on to &lt;a%&gt; of full power. The dimmer switch shall provide manual control to turn on, increase, decrease, or turn off general room lighting.</li> <li>• All lighting in the room shall be automatically turned off within &lt;b&gt; minutes of all occupants having left the room.</li> <li>• Occupancy sensor mounting and sensitivity shall be set for major-motion detection coverage of the full room and minor-motion detection coverage at the workstation.</li> <li>• Occupancy sensor walk-through mode shall be enabled, automatically turning lighting off within &lt;b&gt; minutes if room occupancy is no longer detected after &lt;c&gt; seconds of initial detection.</li> <li>• Dimmer switches shall allow occupants to temporarily change the lighting output state, but shall not permanently override the occupancy sensor shutoff control.</li> <li>• The daylighting control photosensor shall be set to continuously increase or decrease the electric lighting luminous intensity to maintain the daylight area's target illuminance.</li> <li>• When combined electric and daylight illuminance exceeds &lt;d%&gt; of the target levels for &lt;e&gt; seconds, electric light output shall be reduced by &lt;f%&gt; per second.</li> <li>• When combined electric and daylight illuminance falls below &lt;g%&gt; of the target levels for &lt;e&gt; seconds, electric light shall be increased by &lt;f%&gt; per second.</li> <li>• When the combined electric and daylight illuminance exceeds &lt;h%&gt; of target levels for &lt;e&gt; seconds, the electric lighting shall be automatically turned off.</li> <li>• The interface shall not allow the user to override the electric lighting controls in a way that will provide greater illuminance than allowed by the daylight control photosensor.</li> <li>• The daylight control photosensor shall not require physical adjustment at the sensor. All settings shall be accessible by authorized personnel when standing on the floor of the space for which the sensor has control.</li> </ul> |

SOO providing the occupant dimmable control of an enclosed office space and compliance with an energy code, by employing automatic shutoff using occupancy sensors and daylighting control when the code lighting power threshold has been met or exceeded.

**5.3.2 Open Office.** Open office spaces have multiple users and are often fairly automated in their control.

More recent energy codes require the use of occupancy sensors and limit the maximum size of control-zone areas. The smaller occupancy-sensor zones increase the lighting energy efficiency in these spaces. **Table 5-16** shows an example of a CIN and an SOO for an open-office scenario that provides zone-by-zone occupancy-detection control, reducing lighting when zones are unoccupied, and completely turning off all lighting when



**Table 5-16. Example of a CIN and SOO for an Open Office**

| Control Intent Narrative                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Sequence of Operation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| <p>Dimmer switches that turn on, turn off, and dim general lighting in the entire open office are provided at the main entrances. General lighting is automatically controlled by occupancy sensors in zones of no greater than &lt;x&gt; square &lt;meters or feet&gt;, as limited by the energy code and as shown on the construction documents. When the initial occupant enters the open office, the lighting is turned on, but only in zones where there is detection, and to the last lighting level set by occupants at the manual dimmer switch. All other unoccupied detection zones are automatically turned on to no more than the energy code-allowed “unoccupied” setpoint of &lt;y%&gt; of full lighting power. Lighting in any individual detection zone is reduced to the “unoccupied” setpoint when occupancy in the zone is no longer detected for the energy code-maximum time delay of &lt;z&gt; minutes. When the entire open-office space is no longer occupied, the general lighting and all other room lighting will be automatically turned off after the maximum time delay of &lt;z&gt; minutes allowed by the code.</p> <p>When energy code wattage thresholds for the general electric lighting have been achieved in primary and secondary side-lit daylighting areas, the general lighting will continuously dim to maintain a uniform and consistent task-level illumination when there is a contribution of daylight. Each primary and secondary side-lit area operates independently with its own daylight photosensor and controller. When sufficient daylight is available, the lighting will turn completely off. Daylighting control ramp rates should slowly dim lighting after an appropriate time delay to avoid adjustments that are distractive, and raise lighting output quickly when the amount of daylighting decreases due to passing cloud cover. All configuration, programming, and parameter adjustments and settings are accessible to authorized personnel while standing on the floor. The daylighting controls will not raise light levels above the level set at the dimmer switch or, in unoccupied zones, above the “unoccupied” setpoint of &lt;y%&gt; of full lighting power, whichever is less.</p> | <ul style="list-style-type: none"> <li>• Upon initial occupant detection in a detection zone, general lighting for that zone shall be automatically turned on to the last lighting level set at the dimmer switches. General lighting in the remainder of the open office detection zones shall be automatically turned on to either the “unoccupied” setpoint of &lt;a%&gt; of full lighting power or to the last illumination level set at the dimmer switches, whichever is less.</li> <li>• Dimmer switches shall provide manual control to turn on, increase, decrease, or turn off the general lighting.</li> <li>• Dimmer switches shall allow occupants to temporarily change the lighting output state but shall not permanently override the occupancy-sensor shutoff control.</li> <li>• Occupancy sensor mounting and sensitivity shall be set for major-motion detection coverage of the full room and minor-motion detection coverage at workstations.</li> <li>• Lighting in individual detection zones shall automatically be reduced to the “unoccupied” setpoint of &lt;c%&gt; full lighting power within &lt;d&gt; minutes of all occupants having left the zone.</li> <li>• Within &lt;e&gt; minutes of all occupants having left the last occupied detection zone, lighting in the entire open office area shall be automatically turned off.</li> <li>• The daylighting control photosensor shall be set to continuously increase or decrease the electric lighting luminous intensity to maintain the target illuminance for the daylight area.</li> <li>• When the combined electric and daylight illuminance exceeds &lt;f%&gt; of target levels for &lt;g&gt; seconds, the electric light shall be reduced by &lt;h%&gt; per second.</li> <li>• When the combined electric and daylight illuminance falls below &lt;i%&gt; of target levels for &lt;j&gt; seconds, electric light shall be increased by &lt;k%&gt; per second.</li> <li>• When the combined electric and daylight illuminance exceeds &lt;m%&gt; of target levels for &lt;n&gt; seconds, the electric lighting shall be automatically turned off.</li> <li>• The interface shall not allow the user to override the electric lighting controls in a way that will provide greater illuminance than allowed by the daylight control photosensor.</li> <li>• The daylight control photosensor shall not require physical adjustment at the sensor. All settings shall be accessible by authorized personnel when standing on the floor of the space for which the sensor has control.</li> </ul> |

all zones (i.e., the entire open office space) are unoccupied. Daylighting control is also used when the code lighting-power threshold has been met or exceeded.

**5.3.3 Corridor.** Corridor lighting is typically manually or automatically controlled to maintain the full design illuminance when the corridor is occupied. More recent energy codes require at least a 50% reduction in lighting power when occupancy is not detected in corridors. The example in **Table 5-17** shows a CIN and SOO for a corridor, utilizing on/off control by time switch schedule according to facility operating hours, with reduction control when the corridor is not occupied during these times. It also deploys an after-hours full-occupancy sensor control.

**5.3.4 School or University Classroom.** Classrooms typically have specific lighting and use cases to support the teaching environment. In addition to the general lighting in the classroom, special application lighting is

often present to highlight teaching areas, whiteboards, or speaker locations. It is important that the classroom lighting system also provide controllability to support audio, visual, and other display needs. Classrooms are also good opportunities for integration between lighting and HVAC systems, as both lighting and HVAC have setback requirements when there is no occupancy detected in the classroom. **Section 6.1 HVAC** and/or BMS provides a discussion of integration with the HVAC system and an example diagram. Such a diagram should be included as a detail in both the lighting and the HVAC plans, and referenced in the SOO. (If scene buttons are used in lieu of dimmer switches, see **Section 5.3.8 Conference Room** for an example of scene controls.)

An example CIN and SOO for a classroom found in secondary or higher-education settings in which independent control of general lighting and special-application lighting is provided is shown in **Table 5-18**. The system described provides dimming to give

**Table 5-17. Example of a CIN and SOO for a Corridor**

| Control Intent Narrative                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Sequence of Operation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| <p>Wall switches turning corridor general lighting on or off are provided at each entrance. Using time switch scheduled control, general lighting is automatically turned fully on at the start of the building weekday operating hours, and off at the end of weekday operating hours. During building operating hours, occupancy sensors reduce the corridor lighting to the energy code's maximum-allowed "unoccupied" setpoint within the maximum-allowed time delay after all occupants have left the corridor. After the scheduled facility operating hours and during weekends and holidays, corridor lighting is controlled by occupancy detection to come fully on upon detection, and to be automatically turned off after the energy code maximum-allowed time delay after all occupants have left the corridor. The wall switch may turn lighting on and off irrespective of occupancy detection, but manual switches may not permanently override the occupancy sensor function.</p> | <ul style="list-style-type: none"> <li>• Wall switches shall provide manual control to turn the general corridor lighting fully on and off.</li> <li>• Time-switch control shall turn corridor lighting on to the "unoccupied" setpoint of &lt;a%&gt; of full lighting power each weekday at &lt;hh:mm&gt; a.m. and off at &lt;hh:mm&gt; p.m., according to the building's scheduled operating hours.</li> <li>• When occupancy is detected during scheduled operating hours, corridor lighting shall be increased from the "unoccupied" setpoint of &lt;b%&gt; of full lighting power, to full lighting output. Within &lt;x&gt; minutes of all occupants having left the corridor, the lighting shall be reduced to the "unoccupied" setpoint of &lt;c%&gt; of full lighting output.</li> <li>• After operating hours and during weekends and holidays, the lighting shall be automatically turned on to full lighting power upon detection of occupancy, and off within &lt;d&gt; minutes of all occupants having left the corridor.</li> <li>• Occupancy detection full-on lighting output shall take precedence for the entire occupancy-detection time delay during transitions between time switch scheduled control and occupancy detection-only control.</li> <li>• Wall switches shall allow the occupant to temporarily change the lighting output state but shall not permanently override the occupancy sensor or time switch-scheduled shutoff control events.</li> </ul> |

**Table 5-18. Example of a CIN and SOO for a Classroom**

| Control Intent Narrative                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Sequence of Operation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| <p>Dimmer switches to turn on, turn off, and dim the general lighting are located at the classroom entrances and at the teaching station in the front of the classroom. General lighting is manually turned on by these dimmer switches. Specific application lighting illuminating the teaching station area and white board are controlled independently by dimmer switches located at the teaching station controls.</p> <p>Lighting remains on while the room is occupied and may be adjusted by the occupant by using the dimmer switches. When the room is no longer occupied, all lighting will be automatically turned off after the energy code maximum-allowed time delay. To save energy, lighting should be quickly turned off when the room has been only briefly occupied.</p> <p>When energy code lighting power thresholds have been achieved in primary and secondary side-lit daylighting areas, the general lighting will be continuously dimmed to maintain uniform and consistent task-level illumination when there is contribution of daylight. Each primary and secondary side-lit area operates independently with its own closed-loop daylight photosensor and controller. When sufficient daylight is available, the general lighting will turn all the way off. Daylighting control ramp rates should slowly dim the general lighting after an appropriate delay to avoid adjustments that are distractive, and raise light output quickly when daylighting drops due passing cloud cover. All configuration, programming, and parameter adjustments and settings are accessible to authorized personnel while standing on the floor. The daylighting controls will not raise light levels above the level set at the dimmer switches.</p> <p>The lighting control occupancy-detection system will integrate and inform the HVAC system of classroom occupancy status, providing HVAC “occupied” mode and “unoccupied” standby mode signals on an individual classroom basis.</p> | <ul style="list-style-type: none"> <li>• Independent dimmer switches shall provide manual control to turn on, increase, decrease, or turn off the general lighting, teaching station lighting, and whiteboard lighting. Lighting shall not be automatically turned on.</li> <li>• Upon detection of occupancy, closure of an occupancy sensor dry-contact relay shall signal the HVAC system to go to “occupied” mode.</li> <li>• Within &lt;a&gt; minutes of all occupants having left the classroom, all lighting shall be automatically turned off and a dry contact relay shall open, signaling the HVAC system to go into “unoccupied” standby mode.</li> <li>• Occupancy sensor mounting and sensitivity shall be set for major-motion detection coverage of the full classroom and minor-motion detection coverage at the teaching station, desk, and student seating areas.</li> <li>• Occupancy sensor walk-through mode shall be enabled, automatically turning all lighting off within &lt;b&gt; minutes if occupancy is no longer detected after &lt;c&gt; seconds of initial detection.</li> <li>• Dimmer switches shall allow the occupant to temporarily change the lighting output state but shall not permanently override the occupancy sensor control.</li> <li>• The daylighting control photosensor shall be set to continuously increase or decrease the electric lighting luminous intensity to maintain the daylight area’s target illuminance.</li> <li>• When combined electric and daylight illuminance exceeds &lt;d%&gt; of target levels for &lt;e&gt; seconds, the electric light shall be reduced by &lt;f%&gt; per second.</li> <li>• When combined electric and daylight illuminance falls below &lt;g%&gt; of target levels for &lt;h&gt; seconds, the electric light shall be increased by &lt;i%&gt; per second.</li> <li>• When the combined electric and daylight illuminance exceeds &lt;j%&gt; of target levels for &lt;k&gt; seconds, the electric lighting shall be automatically turned off.</li> <li>• The interface shall not allow the user to override electric lighting in a way that will provide greater illuminance than allowed by the daylight control photosensor.</li> <li>• The daylight control photosensor shall not require physical adjustment at the sensor. All settings shall be accessible by authorized personnel when standing on the floor of the space for which the sensor has control.</li> </ul> |

flexible light level control, utilizes manual-on control, and integrates daylight-responsive control for primary and secondary daylight areas. It also has integration between the lighting occupancy detection and the HVAC system to set back HVAC operation when the space is unoccupied.

**5.3.5 Restroom.** Restrooms are spaces with unpredictable occupancy use patterns and do not typically require much in the way of lighting adjustability by the occupant. In many cases, restrooms may incorporate occupancy detection for control of both lighting and ventilation. For safety reasons, it may be undesirable for public restrooms to have accessible manual switches. **Table 5-19** shows an example of a public restroom CIN and SOO in which the ventilation system is integrated and public access is restricted to manual control of the lighting.

**5.3.6 Breakroom.** Breakrooms spaces are shared with building occupants and often have minimal space ownership and limited control needs. **Table 5-20** shows an example of a CIN and SOO that provide basic auto-on control and compliance with the energy code via automatic shutoff using occupancy sensors. Daylighting control is utilized when the energy code's lighting power threshold has been met or exceeded in the daylight area.

**5.3.7 Multipurpose Room.** Multipurpose room lighting controls typically need significant flexibility due to the various functions performed in the space. Many multipurpose rooms incorporate movable wall partitions or stages (common in secondary education facilities), which may additionally have theatrical lighting. It is important that multipurpose lighting also provide controllability to support audio, visual, and other display needs. Multipurpose rooms are good opportunities for integration of lighting and HVAC systems, as both lighting and HVAC have setback requirements when there is no occupancy detected. **Table 5-21** shows an example CIN and SOO for a multipurpose room found in secondary or higher education settings, providing independent control of general lighting and dimmable stage lighting. It provides dimming to give flexible light level control and utilizes manual-on control. It also provides integration between the lighting system's occupancy detection and the HVAC system, to set back the HVAC operation when the space is unoccupied.

**5.3.8 Conference Room.** Conference rooms have a variety of users and use applications; this requires the flexibility of an intuitive user interface. Conference rooms are also common examples of integration among a variety of systems. For example, because conference rooms are often vacant, both lighting and HVAC systems can be set back when the room is

**Table 5-19. Example of a CIN and SOO for a Public Restroom**

| Control Intent Narrative                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Sequence of Operation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| <p>When occupants enter the public restroom, the lighting will be automatically turned on to full lighting power, and the ventilation will go to "occupied" mode. Lighting and ventilation will remain on while the room is occupied. Lighting will be automatically turned off after the energy code maximum-allowed time delay, and ventilation will be signaled to go to "unoccupied" standby mode.</p> <p>Key-operated wall switches will allow authorized personnel to manually turn on and off the lighting in the restroom, while keeping switches inaccessible to public use, for safety reasons.</p> <p>The wall switch may turn the lighting on and off irrespective of occupancy detection but may not permanently override the occupancy sensor function.</p> | <ul style="list-style-type: none"> <li>• Upon occupant detection, the general lighting shall be automatically turned on to full lighting power, and a dry-contact relay closure shall signal the ventilation to go to "occupied" mode.</li> <li>• All lighting shall be automatically turned off within &lt;x&gt; minutes of all occupants having left the restroom, and a dry-contact relay shall open to signal the ventilation to go to "unoccupied" standby mode.</li> <li>• Occupancy sensor mounting and sensitivity shall be set for major-motion detection coverage of the full room and minor-motion detection coverage in the restroom stalls.</li> <li>• The key-operated wall switch shall allow authorized personnel to temporarily change the lighting output state but shall not permanently override the occupancy-sensor shutoff control.</li> </ul> |

Table 5-20. Example of a CIN and SOO for a Breakroom

| Control Intent Narrative                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Sequence of Operation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| <p>General breakroom lighting will be automatically turned on to the code maximum-allowed illuminance when occupants enter the room. Lighting remains on while the room is occupied and may be turned to full on or off by the occupant at the two manual on/off switches located at the space entrance. Manual on/off switches are available when task-specific lighting exist under cabinets or over countertops. When the room is no longer occupied, the general lighting and all other room lighting will be automatically turned off after the energy code maximum-allowed time delay.</p> <p>When the energy-code power thresholds for general lighting have been achieved in defined daylight areas with a sidelight contribution of daylight, the general lighting in those areas will continuously dim as needed to maintain a uniform and consistent level of illumination. When sufficient daylight is available, the lighting will turn all the way off. Daylighting-control ramp rates should slowly dim the lighting after appropriate delay to avoid adjustments that are distractive, and raise lighting output quickly when daylighting drops due to passing cloud cover. To save energy, lighting should be quickly turned off when the room is only briefly occupied. All configuration, programming, and parameter adjustments and settings are accessible to authorized personnel while standing on the floor.</p> | <ul style="list-style-type: none"> <li>• Upon occupant detection, general lighting shall be automatically turned on to &lt;a%&gt; of full power. The manual wall switches shall provide control to turn the general room lighting on to full or to turn it off.</li> <li>• Under-cabinet lighting shall be manually controlled on and off separately from the general lighting by a manual switch located convenient to the cabinets.</li> <li>• All lighting in the room shall be automatically turned off within &lt;b&gt; minutes of all occupants having left the room.</li> <li>• Occupancy sensor mounting and sensitivity shall be set for major-motion detection coverage of the full room and minor-motion detection coverage at the seating areas.</li> <li>• Occupancy-sensor walk-through mode shall be enabled, automatically turning lighting off within &lt;c&gt; minutes if room occupancy is no longer detected after &lt;d&gt; seconds of initial detection.</li> <li>• Manual switches shall allow occupants to temporarily change the lighting state but shall not permanently override the occupancy-sensor shutoff control.</li> <li>• The daylighting control photosensor shall be set to continuously increase or decrease the electric lighting luminous intensity to maintain the daylight area's target illuminance.</li> <li>• When combined electric and daylight illuminance exceeds &lt;e%&gt; of target levels for &lt;f&gt; seconds, the electric light shall be reduced by &lt;g%&gt; per second.</li> <li>• When combined electric and daylight illuminance falls below &lt;h%&gt; of target levels for &lt;i&gt; seconds, the electric light shall be increased by &lt;j%&gt; per second.</li> <li>• When the combined electric and daylight illuminance exceeds &lt;k%&gt; of target levels for &lt;m&gt; seconds, the electric lighting shall be turned off.</li> <li>• The interface shall not allow the user to override electric lighting in a way that will provide greater illuminance than allowed by the daylight control photosensor.</li> <li>• The daylight control photosensor shall not require physical adjustment at the sensor. All settings shall be accessible by authorized personnel when standing on the floor of the space for which the sensor has control.</li> </ul> |



**Table 5-21. Example of a CIN and SOO for a Multipurpose Room**

| Control Intent Narrative                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Sequence of Operation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| <p>Dimmer switches to turn on, turn off, and dim the general lighting are provided at the room entrances and on the stage. Stage general lighting is controlled independent of the multipurpose room lighting, as is the lighting on each stair entrance to the stage. General lighting is manually turned on by dimmer switches. Stage theatrical lighting is controlled independently by manual dimmer switches located on the stage.</p> <p>General lighting remains on while the room is occupied and may be adjusted by the occupant, using the dimmer switches. When the room is no longer occupied, all lighting will be automatically turned off after the energy code maximum-allowed time delay. To save energy, lighting should be quickly turned off when the room is only briefly occupied.</p> <p>The lighting control occupancy-detection system will integrate and inform the HVAC system of room occupancy status, providing HVAC “occupied” mode and “unoccupied” standby mode signals.</p> | <ul style="list-style-type: none"> <li>Independent dimmer switches shall provide manual control to turn on, increase, decrease, or turn off the general lighting in the multipurpose room, on the stage, and at the stair entrances to the stage. Lighting shall not be automatically turned on.</li> <li>Dimmer switches located on the stage shall provide independent manual control of the theatrical stage lighting.</li> <li>Upon detection of occupancy, closure of an occupancy sensor dry-contact relay shall signal the HVAC system to go to “occupied” mode.</li> <li>Within &lt;x&gt; minutes of all occupants having left the space, all lighting shall be automatically turned off and a dry-contact relay shall open, signaling the HVAC system to go into “unoccupied” standby mode.</li> <li>Occupancy sensor mounting and sensitivity shall be set for major-motion detection coverage of the full multipurpose and stage areas.</li> <li>Occupancy sensor walk-through mode shall be enabled, automatically turning all lighting off within &lt;y&gt; minutes if occupancy is no longer detected after &lt;z&gt; seconds of initial detection.</li> <li>Dimmer switches shall allow the occupant to temporarily change the lighting output state but shall not permanently override the occupancy-sensor shutoff control.</li> </ul> |

unoccupied. **Table 5-22** shows an example CIN and SOO for a conference room in an office building setting. Discrete control of individual lighting zones and room control is provided using scene or preset controls. The buttons and their individual labels should not only be described in the SOO, but also in a detail diagram (see **Figure 7.5** in **Section 7.5.1.2 Completed Enhanced Matrix, Example #1** for an example). Integration between the lighting system’s occupancy detection and the HVAC system is provided to set back HVAC operation when the space is unoccupied.

**5.3.9 Storage Room.** Storage rooms have limited and infrequent occupancy and are therefore best suited for basic on/off control based on occupancy detection. **Table 5-23** shows an example CIN and SOO for a storage room that is occupancy controlled, and provides the user independent manual on/off control.

**5.3.10 Warehouse.** Warehouse lighting is typically controlled to maintain the full design illuminance when a warehouse rack aisle is occupied. Energy codes may require at least a 50% reduction in lighting power when occupancy is not detected in the rack aisles. **Table 5-24** shows an example of a warehouse CIN and SOO utilizing on/off control via time switch according to facility operating hours, with reduction control when the rack aisles are not occupied during these times. It also incorporates daylight-responsive control to reduce electric light output with the contribution of daylight from the skylights.

**5.3.11 Religious Worship Space (Congregation Seating Area, Pulpit, and Choir Area).** Lighting in religious worship spaces is application specific for the functions of religious worship and congregational meetings. These can range from simple community settings, with basic lighting, to large church campuses and facilities that broadcast worship services and have full theatrical

**Table 5-22. Example of a CIN and SOO for a Conference Room**

| Control Intent Narrative                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Sequence of Operation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| <p>The dimming scene controller is used to recall previously recorded scenes for meetings, projector presentations, and video conferencing, as well as all-off. User controls are provided at the room entrance(s) for the following lighting zones:</p> <ul style="list-style-type: none"> <li>a. General illumination over the table</li> <li>b. Downlighting room front</li> <li>c. Downlighting room sides</li> <li>d. Downlighting room back</li> </ul> <p>The lighting in the room shall require manual activation (vacancy mode). The lighting remains on while the room is occupied and may be adjusted using raise and lower overrides. When the room is no longer occupied, all lighting will be automatically turned off after the energy code maximum-allowed time delay.</p> <p>The sensors will integrate and inform the HVAC system of room occupancy status, providing HVAC “occupied” mode and “unoccupied” standby mode signals.</p> | <ul style="list-style-type: none"> <li>• Lighting shall be manually turned on at the user scene-control panel at the conference room entrance(s). Lighting shall not be automatically turned on.</li> <li>• Manual on/off/dim control of individual lighting zones shall be provided at the scene control panel.</li> <li>• Preset scene buttons shall be set for the following lighting scenes:             <ol style="list-style-type: none"> <li>1. General lighting. a: &lt;w%&gt;; b: &lt;x%&gt;; c: &lt;y%&gt;; d: &lt;z%&gt;</li> <li>2. Video conf. a: &lt;w%&gt;; b: &lt;x%&gt;; c: &lt;y%&gt;; d: &lt;z%&gt;</li> <li>3. Projection. a: &lt;w%&gt;; b: &lt;x%&gt;; c: &lt;y%&gt;; d: &lt;z%&gt;</li> <li>4. All off. a: 0%; b: 0%; c: 0%; d: 0%</li> </ol> </li> <li>• Upon detection of occupancy, the lighting control system shall signal the HVAC system to go to “occupied” mode.</li> <li>• Within &lt;a&gt; minutes of all occupants having left the space, all lighting shall be automatically turned off, and the lighting control shall signal the HVAC system to go into “unoccupied” standby mode.</li> <li>• Occupancy sensor mounting and sensitivity shall be set for major-motion and minor-motion detection coverage of the entire conference room.</li> <li>• The system shall allow the occupant to temporarily change the lighting output state but shall not permanently override the occupancy-sensor shutoff control.</li> </ul> |

**Table 5-23. Example of a CIN and SOO for a Storage Room**

| Control Intent Narrative                                                                                                                                                                                                                                                                                                                                                   | Sequence of Operation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| <p>The lights in the storage rooms will be automatically turned on when occupants enter the room. Lighting remains on while the room is occupied and may be turned off by the occupant at the user interface at the door. When the room is no longer occupied, the general lighting will be automatically turned off after the energy code maximum-allowed time delay.</p> | <ul style="list-style-type: none"> <li>• Upon occupant detection, the general lighting shall be automatically turned on to &lt;x%&gt; of full light output. The user interface shall provide manual control to turn the general lighting on or off.</li> <li>• All lighting in the room shall be automatically turned off within &lt;y&gt; minutes of all occupants having left the room.</li> <li>• Occupancy sensor mounting and sensitivity shall be set for major-motion detection coverage of the entire room.</li> </ul> |

staging. **Table 5-25** shows an application with multiple lighting groups utilizing on/off control. Because of the communal nature of these spaces, there are often multiple user groups with various skill levels. This necessitates clear labelling of interface stations so that their functions are clear. In this example, controls are provided for various lighting groups, along with time-based automatic shutoff control for compliance with the prevailing energy codes.

**5.3.12 Retail Floor.** The lighting of a retail floor typically operates in three key modes: store open (sales hours); store closed for maintenance and stocking; and store closed and unoccupied. Retail lighting controls operate to satisfy the lighting needs under these time-based modes. If skylights are incorporated into the space, daylight-responsive controls should be utilized. Override-control interfaces are typically placed in



**Table 5-24. Example of a CIN and SOO for a Warehouse**

| Control Intent Narrative                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Sequence of Operation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| <p>Using time-scheduled control, general lighting is automatically turned on to the “unoccupied” lighting level in each rack aisle at the start of daily operation. At the end of the day, general lighting is turned off. At all times, occupancy detection in a rack aisle turns lights on to the preset “occupied” level. When occupancy is no longer detected by the sensors in a rack aisle, lighting is reduced to the “unoccupied” setpoint level within the maximum-allowed time delay during operational hours, and off during non-operational periods.</p> <p>Daylight-responsive controls modulate the lighting output to maintain a consistent level of illumination in the rack aisles. The illuminance in the rack aisles shall not exceed the “unoccupied” setpoint when there is no activity in the aisles. The daylighting controls should be able to turn the lighting off if the incident illuminance from daylight meets or exceeds the setpoint.</p> <p>The override interface may control the lighting but may not disable the occupancy sensor function.</p> <p>An override interface is provided at each warehouse main entry point.</p> | <ul style="list-style-type: none"> <li>• The time-switch control shall turn on the warehouse rack aisle lighting to the “unoccupied” setpoint of &lt;a%&gt; each weekday at &lt;hh:mm&gt; a.m. and off at &lt;hh:mm&gt; p.m., according to scheduled operating hours.</li> <li>• Override interfaces shall allow for the rack aisle lighting to be adjusted as needed. In each independent rack aisle, when occupancy is detected, lighting shall increase from the “unoccupied” setpoint of &lt;b%&gt; of full lighting power, to the “occupied” setpoint. When occupancy is no longer detected by the sensors in a rack aisle, the lighting shall be reduced to the “unoccupied” setpoint of &lt;c%&gt; within &lt;d&gt; minutes during operational hours, and off during non-operational periods.</li> <li>• Timed events shall not override nor reset occupancy commands during time delays.</li> <li>• The daylighting control photosensor shall be set to continuously increase or decrease the electric lighting luminous intensity to maintain the target illuminance. When combined electric and daylight illuminance exceeds &lt;e%&gt; of target levels for &lt;f&gt; seconds, the electric light output shall be reduced by &lt;g%&gt; per second.</li> <li>• When combined electric and daylight illuminance falls below &lt;h%&gt; of target levels for &lt;i&gt; seconds, the electric light output shall be increased by &lt;j%&gt; per second.</li> <li>• The interface shall not allow the occupant to override the electric lighting controls in a way that will provide greater illuminance than allowed by the daylight control photosensor.</li> <li>• The daylight control photosensor shall not require physical adjustment at the sensor. All settings shall be accessible by authorized personnel when standing on the floor of the space for which the sensor has control.</li> </ul> |

**Table 5-25. Example of a CIN and SOO for a Religious Worship Space**

| Control Intent Narrative                                                                                                                                                                                                                                                                                                                                                                               | Sequence of Operation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>The lighting in the worship space is controlled by the user at a wall station located at each entrance to the worship space. Entrance stations only control the cove and perimeter lights. An interface panel at the pulpit provides control of all lighting groups. Lighting is automatically turned off by a scheduled “sweep.” Five minutes prior to the sweep, a blink warning shall occur.</p> | <ul style="list-style-type: none"> <li>• All interface stations in the worship space shall be labelled according to approved templates.</li> <li>• Cove and perimeter downlighting shall be controlled by a wall station located at each entrance to the worship space.</li> <li>• An interface panel shall be located at the pulpit for control of all lighting zones. A scheduled sweep shall occur Monday through Saturday at &lt;hh:mm&gt; p.m., and Sunday at &lt;hh:mm&gt; p.m. (after hours).</li> <li>• At &lt;x&gt; minutes prior to the scheduled sweep, a warning blink shall occur.</li> </ul> |

remote locations for security reasons. **Table 5-26** shows a CIN and SOO example for a retail store that utilizes time-based controls integrated with the alarm system.

**5.3.13 Exterior Lighting.** Exterior lighting typically is often categorized in two groups: “operational” and “aesthetic.” Each group is handled differently at various times of day, and code jurisdictions may define these groups in slightly different ways. The “operational group” typically includes the building entry, building exit, walkway, pathway, plaza, and parking lot lighting. It is generally lighting for visual acuity and safety. This group is off during daylight hours, comes on generally at dusk, and remains on when it is dark. During periods of limited activity or as dictated by code, the operational group may dim to a lower power level, commonly referred to as a “setback mode.”

The “aesthetic group” includes lighting that highlights the building and grounds for aesthetic effect only. This lighting turns on when it is dark and then turns off during times of limited activity. More information on the application of exterior lighting control can be found in Section 7.6.3 of LP-2-20.<sup>1</sup> **Table 5-27** provides a CIN and SOO example for both common groups of exterior lighting.

**5.3.14 Electrical Equipment Room and Elevator Control Room.** Some codes have requirements that prohibit automatic shutoff (for example, from sensors or timers) in for spaces with mechanical or electrical gear. Therefore, control is generally best handled in the simplest manner possible, e.g., a standalone toggle switch. **Table 5-28** shows a CIN and SOO example for mechanical rooms in a building.

**Table 5-26. Example of a CIN and SOO for a Retail Floor**

| Control Intent Narrative                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Sequence of Operation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| There are three time-based store operating modes with associated lighting presets: store open (sales hours); store closed for maintenance and stocking; and store closed and unoccupied. The lighting levels are primarily controlled via the system time switch. Time switch monitoring may be modified to timing of operating modes. Override controls are located in the employee area. The general lighting is dimmable based on operating mode. Accent lighting from track-mounted luminaires is also controlled based on operating mode. Display-case lighting for high-end merchandise can be locally overridden by employees. The security system and the lighting control system are connected so that the security system may trigger lighting presets, including an alarm or break-in condition. | <ul style="list-style-type: none"> <li>Monday through Saturday at &lt;hh:mm&gt; a.m., and Sunday at &lt;hh:mm&gt; a.m. sales-floor general lighting and all decorative, track, display, cornice, and signage lighting shall recall the associated preset when the store is open for customers.</li> <li>At store closing, Monday through Saturday at &lt;hh:mm&gt; p.m. and Sunday at &lt;hh:mm&gt; p.m., all lighting shall recall the associated preset.</li> <li>When the security system is tripped or signals a break-in alarm, a signal is sent to the lighting control system to turn on all general lighting to full lighting power.</li> <li>The time switch shall provide programmability to adjust scheduling.</li> <li>Override stations located in a secure employee area allow the lighting presets to be recalled and temporarily adjusted without recording new levels. The override stations shall be labeled and have a light indicating which preset is active.</li> </ul> |

**Table 5-27. Example of a CIN and SOO for an Exterior Lighting Area**

| Control Intent Narrative                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Sequence of Operation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>As per code, the exterior lighting is controlled in two independent groups. These groups may consist of several control zones each. Lighting for visual acuity and safety (building entry, building exit, pathways, plaza, parking, and walkway) will be on from dusk to dawn and will be reduced in lighting power by &lt;x%&gt; as required by code. Façade and landscape lighting will be turned on at dusk and turned off &lt;y&gt; hour(s) after business close. This lighting will remain off as required by code. Façade and landscape lighting will be turned on during early morning hours as allowed by code, and should be turned off one hour after sunrise.</p> <p>“Dusk” and “dawn” will be determined by the system astronomical time switch. Photocells may provide override capabilities for providing illuminance above and below established thresholds due to cloud cover and weather.</p> <p>Scheduled control is accomplished by programming. The controller shall have at least 1 week of programming back-up memory during power failure, individual scheduling capability for each day of the week, and at least two independent schedules for holiday use.</p> | <ul style="list-style-type: none"> <li>Building entry, building exit, pathway, plaza, parking, and walkway lighting shall be turned on to 100% lighting power by photocell control when illuminance from daylight decreases to less than &lt;a&gt; &lt;lux or fc&gt;. The light output from each luminaire shall be uniformly reduced by &lt;b%&gt; to the code-specified lighting power at &lt;hh:mm&gt; p.m. The lighting shall be turned on to 100% lighting power at &lt;hh:mm&gt; a.m., remaining fully on until being turned off when illuminance from daylight becomes greater than &lt;c&gt; &lt;lux or fc&gt;.</li> <li>Building façade and landscape lighting shall be turned on to 100% lighting power by photocell control when illuminance from daylight decreases to less than &lt;d&gt; &lt;lux or fc&gt;. This lighting shall be turned off at &lt;hh:mm&gt; p.m. The lighting shall be turned on to 100% lighting power at &lt;hh:mm&gt; a.m., remaining fully on until being turned off when illuminance from daylight becomes greater than &lt;e&gt; &lt;lux or fc&gt;.</li> </ul> |

**Table 5.28. Example of a CIN and SOO for a Building’s Mechanical Rooms**

| Control Intent Narrative                                                                                               | Sequence of Operation                                                                                                                                 |
|------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mechanical and engine rooms should not rely on automatic shutoff methods, but should have a toggle switch in the room. | <ul style="list-style-type: none"> <li>A wall mounted toggle switch shall provide control for all lights in spaces with working machinery.</li> </ul> |

## 6.0 Integrations With Other Systems

Any building system capable of communication with other systems can be interfaced with the lighting control system. Most building owners and operators are not interested in integrating lighting controls with other systems, without a specific reason to do so. However, there are many benefits to be gained by having strategic interfaces between key building systems. These integrations can be leveraged to provide additional cost savings and energy savings; to reduce the number of devices to be purchased, installed, and maintained; to minimize clutter on ceilings; or to provide user interfaces that make the building easier to operate and more comfortable to occupy, and which simplify

common or repetitive tasks. When used properly, these integrations can be leveraged to provide owners and operators with key benefits that are important to them, often while reducing overall project costs.

The subsections that follow provide examples of common types of integration and integration methodologies. Directions on how to choose and specify integrations are not in the scope of this document; however, the information provided here should help to inspire thought as to how integrating systems can provide the building owner and operator with the benefits and value they desire.

### 6.1 HVAC and/or BMS

There are several protocols commonly used to interface between the lighting system and the HVAC and BMS

systems. Interfacing the two systems is most commonly used to turn a room's HVAC off (or decrease it) when the space is unoccupied to save energy. This is called demand-controlled ventilation (DCV). If energy savings are to be realized, it requires coordination between the HVAC and lighting designers through a carefully created sequence of operations, details, plans, specifications, and networking infrastructure.

Current industry practice leaves the choice of interface type to the HVAC engineer; however, the lighting designer and electrical engineer take an active role because the capabilities of the lighting control system are the primary determining factor. Lighting control-system hardware and software capabilities differ among vendors, and details will need to be created for each project. Good practice recommends working with the vendor or manufacturer representative, including reviewing the Protocol Implementation Conformance Statement (PICS, or PIC Statement) to understand the capabilities of the lighting control system and to specify the data point(s) that will be used for the systems to communicate with one another (see also **Sidebar – PICS**, in this section). Coordination with IT and other systems is critical, given that lighting and HVAC integration often relies on IT infrastructure, and the high likelihood of coordination issues around IT network architecture.

The implementation method (hardware or network integration) affects the sequence of operation. If the systems are integrated on the hardware side, integration is dependent on the design of the rest of the lighting

control system. As an example, a line-voltage occupancy sensor with an auxiliary relay can be the simplest method of integration; however, this approach may not comply with current energy codes. Digital or low-voltage occupancy sensors may provide the desired functionality while also complying with energy codes. A good practice is to describe the subcontractor's responsibilities in the detail, including wiring, conduit, terminations, and programming. A better practice is to duplicate the detail in each discipline's drawing section (e.g., mechanical, electrical) because subcontractors might not review other sections when bidding projects or creating scope.

It is important that the HVAC integration consider that the lighting zoning is often more granular than the HVAC zoning; therefore, the HVAC system programming should be carried out with this in mind. Large open offices with several occupancy sensors may require extensive detailing or programming to effectively join the inputs from all the sensors. **Table 6-1** shows an example CIN and SOO for an occupancy-based DCV strategy that is implemented at the local or zone level.

## 6.2 Security

Dry contacts are the most common form of signal used for interfacing with security and life-safety systems, but other protocols could be used, depending on the facility or institutional requirements. Building owners and operators may look for specific terms with which they are familiar when reviewing lighting control intent information. Typically, direction will come from the

**Table 6-1. BMS Example**

| Control Intent Narrative                                                                                                                                                                                       | Sequence of Operation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The lighting control system will share occupancy status with the BMS in each room as indicated on the lighting plans for the purpose of meeting demand-controlled ventilation requirements of the energy code. | <ul style="list-style-type: none"> <li>The lighting control system shall be connected to the BMS via BACnet. Occupancy status changes detected by the lighting control system in the spaces with specified occupancy sensor points shall be sent to the BMS via the BACnet connection.</li> <li>Upon detection of occupancy, the sensor shall send a BACnet command to the HVAC system. Upon detection of vacancy, the sensor shall send a BACnet command.</li> <li>No information is required to be exchanged from the BMS to the lighting control system.</li> </ul> <p>(Also see the HVAC SOO.)</p> |

**SIDEBAR – PICS**

At its heart, a Protocol Implementation Conformance Statement (PICS, or PIC Statement) is a document that facilitates communication between two different control systems.

All electronic devices exchange information through a shared communication protocol (often just referred to as a protocol or control protocol). As an example, DALI and 0-10V (i.e., 10V0) are control protocols commonly used in lighting control systems. BACnet is one protocol used by HVAC and building management systems. Many control systems that internally communicate with different protocols can also communicate externally using BACnet.

Communication with the same protocol is the first step in integrating two systems. The next step is to make sure they are exchanging information in a way that both can understand. Idioms may not always be understood in various parts of the world, even if the same language is used. A computer's ability to infer meaning from syntax is much more limited.

A PIC Statement gives a detailed description not only of the information that a given system can share, but also of the types of data (or object type) supported (e.g., analog value, binary value), networking options, and much more. It is a publicly available document generated by manufacturers for each device they make. The best way to obtain these documents is through an internet search for "PICS" (or occasionally "Technical Bulletin") and the manufacturer and model number of the device. This information is sometimes readily available on the manufacturer's website (again, search there for "PICS" or "Technical Bulletin"). Unfortunately, many distributors are unfamiliar with this kind of document, and requests may generate a poor response. As integration becomes a more common aspect of standard building practice, the usage of PICS and other types of equipment documentation should become more commonplace.

Additional information about PIC Statements can be found at the BACnet official website: <http://www.bacnet.org/DL-Docs/>. (Accessed 2021 Sep 15).

security and/or life-safety vendor, manufacturer, or institutional office. Lighting control systems are more flexible than security systems; therefore, adaptation is made on the lighting control system side as needed to implement a successful interface and control logic outcome. The example shown in **Table 6-2** would be appropriate for a museum application where the lighting control system needs to interface with the security system.

### 6.3 Fire and Life Safety

Because of the nature of the events that trigger these systems, communication from fire and life-safety systems will supersede any other integration or control signal. The initiation of such a signal will always trigger an override condition that brings the lights up to the level determined onsite by the Lighting Designer. Dry contacts are typically used to interface between lighting and fire or life-safety systems, where such an interface is required. The examples shown in **Table 6-3** and **Table 6-4** provide simplified language for fire and power-loss scenarios, respectively.

### 6.4 Audio-Visual (AV) Systems

There are several ways in which AV equipment can be interfaced with lighting controls to simplify operation. The level of complexity required of the integration is one of many factors in determining how to specify this integration. Others include distance the between devices, the number of signals to be shared, the organizational hierarchy of the systems, and the presence or absence of tracking requirements among systems. Care should be taken to indicate whether system communication is unidirectional or bidirectional. Almost any protocol can be used for these connections, from contact closure to full TCP or UDP network connection. It is critical that lighting control hardware be specified that has the physical interfaces required for the protocol that will be used, as well as software functionality to logically deal with sending and receiving these commands. The example in **Table 6-5** shows a simple application with a lighting system subordinate to an AV system without the control of individual lighting control zones.

**Table 6-2. Security Example**

| Control Intent Narrative                                                                                                                                                                                                                                                                                     | Sequence of Operation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| The lighting control system will receive triggers from the security system via contact closure to indicate armed, unarmed, and security-sweep status. The security sweep turns on a specific lighting preset and triggers a 2-hour timer, after which the lighting system will revert to its previous state. | <p>The lighting control system shall receive 3 dry-contact closures from the &lt;name of security system or manufacturer&gt; security system. This interface device is located in &lt;device location&gt;, and this contact shall be connected to &lt;lighting control system device&gt; in &lt;room location&gt;.</p> <ul style="list-style-type: none"> <li>• Security contact closure 1, "Armed," shall trigger the "Night" preset.</li> <li>• Security contact closure 2, "Unarmed," shall trigger the "Day" preset.</li> <li>• Security contact closure 3, "Security Sweep," shall trigger the "Cleanup" preset. When this contact closure is actuated, the lighting system shall begin a 2-hour countdown, after which the lighting system shall revert to its previous preset.</li> </ul> |

**Table 6-3. Fire Example**

| Control Intent Narrative                                                                                                                                                                          | Sequence of Operation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The lighting control system will receive a contact-closure signal when an alarm condition exists, triggering all emergency lighting to turn on to the emergency preset level for building egress. | <ul style="list-style-type: none"> <li>• A change in the contact-closure status shall signify an alarm condition. When this occurs, the lighting control system shall command the emergency lighting preset as a system-wide override. The emergency lighting preset will be recorded onsite by the Lighting Designer and will, at a minimum, include all emergency-designated (EM) luminaires at full light output. No luminaires shall be commanded to turn off in this alarm condition.</li> <li>• Upon resumption of the normal contact condition, luminaires shall return to their normal operative condition.</li> </ul> |

**Table 6-4. Power Loss Example**

| Control Intent Narrative                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Sequence of Operation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| The lighting system will continuously monitor each of the three phases of the power source feeding the luminaires. Upon a voltage deviation beyond <x%>, the lighting system shall command all luminaires to an emergency lighting preset, as well as command all emergency (EM) luminaires to full light output. Given that startup time may exist in normal luminaires and lighting components that may have been powered off during the alarm condition, the EM devices shall continue to command EM levels for the time it takes for said luminaires to reach their previous output. | <ul style="list-style-type: none"> <li>• The lighting control system shall continuously monitor each of the three voltage phases that feed all luminaires. Upon voltage deviation of any phase of power by more than &lt;x%&gt;, the lighting system shall bypass the commanded lighting levels in all rooms and shall force all EM-designated luminaires to full light output and all other architectural lighting levels to the emergency lighting preset. The emergency lighting preset will be recorded onsite by the Lighting Designer and will, at a minimum, include all EM luminaires at full light output. No luminaires shall be commanded to turn off in this alarm condition.</li> <li>• Upon indication that normal power has returned, the lighting system shall immediately command normal power to be restored to all non-emergency lighting devices. The EM devices shall continue to command EM levels for &lt;y&gt; seconds, at which time, the EM devices shall return to normal conditions and control, and all luminaires shall return to the presets and/or power levels they occupied prior to the alarm condition.</li> </ul> |



**Table 6-5. A/V Example**

| Control Intent Narrative                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Sequence of Operation                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The lighting control system will be integrated with the AV system. This integration will allow each system to see and track the state of the other. If a room state is changed via the AV system, the lighting control system will be sent a trigger to recall a preset. When a lighting control button is pressed, that trigger may be sent to the AV system as well. The AV system will be able to query status from the lighting control system (i.e., the preset affecting a given room). Discreet zone lighting levels need not be shared between systems, only presets. | <p>The lighting control presets in the room shall be:</p> <ul style="list-style-type: none"> <li>• &lt;Preset A name&gt;</li> <li>• &lt;Preset B name&gt;</li> <li>• &lt;Preset C name&gt;</li> <li>• &lt;Preset D name&gt;</li> <li>• &lt;Preset E name&gt;</li> </ul> <p>These presets shall be recallable from the AV system via UDP commands. The AV system shall be able to query whether the lighting preset is active in the room via UDP polling, and send an associated UDP response.</p> |

## 6.5 Entertainment

The requirements for live-entertainment lighting control design are many and complicated, and are worthy of an entire document by itself. In entertainment facilities and venues, there may be overlap between the performance lighting and the architectural lighting (the latter could include, for example, the auditorium house lighting and the lobby lighting). It is critical in these overlap cases to ensure that the architectural and entertainment systems interface with and are compatible with one another, and that in all cases the emergency egress lighting requirements are achieved. The designer will need to pay attention to the requirements of the primary programming of the space, as well as any client-requested alternative programs or events, as event spaces typically cater to a wide range of clients and event types. Further discussion is outside the scope of this document.

## 6.6 Dynamic Architectural Lighting

A system that uses control elements that allow for dynamic effects in an architectural environment (sometimes colloquially called “architainment”) should be referred to as an architecturally integrated entertainment system (AIES). These systems often use components from live entertainment control systems. Dynamic lighting, as a technique, is used to attract attention and improve the aesthetic of the illuminated element. In addition to changes in color, intensity, and movement, AIES lighting includes, for example, interactive, human-scale media, exhibits, amusement, and themed rides. An AIES still uses basic control

strategies such as triggers, sequences, and schedules, but is defined by a more complex application of these strategies.

The primary concern when specifying dynamic architectural lighting, besides compatibility, is to ensure that the control protocol utilized can provide the desired functionality. Typically, dynamic architectural lighting will use a less latency-constrained protocol such as ANSI E1.11 – USITT DMX512-A; however, it may also be possible to achieve the desired results using a series of static looks fading from one to another using other protocols. It is important that the designer specify exactly what the lighting system has to do, the schedule by which it will function, and how this schedule is to be maintained, updated, or overridden. Other considerations include ensuring that nighttime light-pollution reduction measures are met. This may include turning off supplemental lighting during curfew hours and ensuring that the lighting complies with all government regulations.

## 6.7 Daylighting Management Systems

As used here, the term *daylight management system* refers to a system that controls physical elements such as shades and fins that move or change to block incident daylight. (Note: The term *daylight management* is defined in ANSI/IES LS-1-20.<sup>4</sup>) These systems are sometimes integrated into the lighting control system, providing the end user with a single illumination control system that regulates both electric light and daylight in a given space. At its simplest, this connection can be a



pair of contact closures to, for example, raise and lower blinds. This integration is effective but does not provide an interface to allow stopping at intermediate (other than open or closed) levels. For that level of control, a more robust type of connection is required, such as RS-232, RS-485, or BACnet. **Table 6-8** shows a CIN and SOO for a window-shade system. **Table 6-9** shows a CIN and SOO for an architectural-shade system.

### 6.8 IoT (Internet of Things) Platforms and Devices

The evolution and application of lighting technology has expanded in recent years beyond lighting design and control. The digitalization of LED systems as well as drivers, sensors, gateways, and ever more powerful on-premises servers and cloud services has enabled many different use cases and value propositions that previously had no relationship to lighting. In some

instances, this interaction with the lighting system uses sensors that are integral to luminaires, or are remotely located but wired to luminaires. In other cases, lighting can be powered by Power over Ethernet wiring, which allows digital control of the lighting devices in addition to enabling integration with multiple other building systems in ways not previously possible, such as cameras, phones, security systems, access systems, lift systems, and A/V technologies.

In general, lighting designers and installers are not leading the charge toward the use of IoT in buildings. It is typically owners and tenants who are most interested in these technologies, because it is they who realize significant value after building occupancy. To navigate the application of IoT and digital technologies through the building specification, procurement, construction,

**Table 6-8. Window Shade Example**

| Control Intent Narrative                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Sequence of Operation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| In rooms <names or designations>, the lighting system will control window shades as specified in CSI Section <appropriate section number under 12 24 00>. The lighting system will provide time switch-based commands to raise and lower the shades. Additionally, an end user can manually raise or lower the shades via the lighting control system's user interfaces. Shades or groups of shades for windows with different facing orientations will be controlled independently. | <ul style="list-style-type: none"> <li>The lighting control system shall command the window shade systems in rooms &lt;names or designations&gt;. The lighting control system shall provide two contact closures for each window shade or shade group in each room. One contact closes to fully raise the window shades, and one contact closes to fully lower the shades.</li> <li>Automatic shade adjustments shall be made: <ul style="list-style-type: none"> <li>At &lt;morning glare start time, hh:mm a.m.&gt;, the shades are lowered.</li> <li>At &lt;morning glare end time, hh:mm a.m.&gt;, the shades are raised.</li> <li>At &lt;afternoon glare start time, hh:mm p.m.&gt;, the shades are lowered.</li> <li>At &lt;afternoon glare end time, hh:mm p.m.&gt;, the shades are raised.</li> </ul> </li> <li>At any time, an occupant of the room may manually command the shades to raise or lower via the &lt;lighting control interface device in the room (specify buttons, what they are called, and if they are the toggle type or two separate buttons)&gt;.</li> <li>The lighting control system shall command the window shades to be only fully raised or fully lowered, depending on their current state at the time of the command.</li> <li>The lighting control system is not responsible for any endpoint adjustment, configuration, or programming of the window shade system.</li> </ul> |

**Table 6-9. Architectural Shading Example**

| Control Intent Narrative                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Sequence of Operation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| <p>In rooms &lt;names or designations; or exterior space name or designation&gt;, the lighting system will control the architectural shading system, as detailed in the architectural drawings.</p> <p>The lighting control system will provide continuously updating closed-loop feedback such that the system commands a fixed illuminance level in the &lt;room or space&gt; via the lighting system's closed-loop photosensors. When illumination levels drop below the target illuminance from sunlight, the architectural shades are &lt;opened, rotated, or other action to let more light in&gt;. When the shades are fully &lt;open, rotated&gt;, electric light is added as needed to reach the target illuminance. When illumination levels increase, the electric lighting is dimmed down to off using the daylighting control function. When the electric lighting is off, illumination levels can be further reduced by &lt;closing, rotating, or other action to let less light in&gt; the architectural shades such that the target illuminance is provided. After sunset and before sunrise, the architectural shades will be left fully &lt;open, rotated&gt;, and the photocells will command only electric-light dimming.</p> | <p>The lighting control system shall be integrated with the architectural shading system so that both the electric lighting and daylight contributions are controlled via a closed-loop photosensor in the lighting control system. This integration shall occur via a &lt;protocol name&gt; connection between the lighting control system and the architectural shading system, and shall be continuously updated to achieve the sequence of operation specified herein.</p> <ol style="list-style-type: none"> <li>1. A closed-loop daylighting control photosensor shall be set to the illuminance target and shall continuously monitor the daylight contribution toward the target and increase or decrease the electric lighting as needed within the space.</li> <li>2. When the daylight level exceeds &lt;a%&gt; of the illuminance target for &lt;b&gt; minutes, the electric lighting shall be completely turned off, and the architectural shading system shall be articulated to control illumination by decreasing the daylight entering the space.</li> <li>3. When the illuminance is below &lt;c%&gt; of the target and the architectural shading system has been fully open for &lt;d&gt; minutes, the lighting control system shall command electric lights to be energized so as to maintain the target illuminance.</li> <li>4. The architectural shading transition speed shall range from the minimum articulation speed of the shading system to &lt;f%&gt; per second. The acceleration rate of the shade opening or closing shall be &lt;g%&gt; per second.</li> <li>5. When the daylight level decreases below &lt;h%&gt; of the target illuminance for &lt;i&gt; minutes, the architectural shading system shall be articulated to its fully open position, and the electric lighting shall be commanded to a level that will provide the target illuminance.</li> <li>6. The interface shall not allow the occupant to override the electric lighting controls in a way that will provide greater illuminance than allowed by the daylight control photosensor.</li> <li>7. All configuration, programming, and parameter adjustments and settings shall be accessible only by authorized personnel. Authorized personnel shall be able to access the programming mode when standing on the floor in the space being controlled.</li> </ol> |

and occupation process, it is important to understand how the technologies can benefit different stakeholders. This makes it important to understand who wants what within the OPR and BOD, so that the CIN and SOO can properly reflect and/or enable requirements that may go well beyond lighting.

Most of the IoT and digital use cases that are desired by owners and tenants are related to indoor location services (ILS). The most common technologies used to enable ILS are wireless sensors, frequently integral to luminaires and installed in a highly dense or “granular” pattern. These sensors can perform the double duties

of providing lighting control while also gathering data about space occupancy, and occupant location over time. Sensors connected to lighting systems are required to be geolocated onto a digital map used to enable location services.

### Sidebar: IoT and Indoor Location Services

Indoor location services are still actively evolving but primarily fall into seven categories: 1) space utilization; 2) asset tracking; 3) wayfinding; 4) space reservation systems; 5) geofencing; 6) contact tracing; and 7) time-of-flight functionality.

These types of systems are still too new to provide useful generalizations. However, with new technologies it can be just as helpful to anticipate risks and know the best questions to ask from the start of the project. Some of the issues and variables associated with IoT and digital technologies that should be considered throughout a project include:

- Sensor locations driven by data collection may not align with luminaire locations most suitable for lighting design goals. Some sensor control systems offer critical flexibility with an option of independent sensors that can be used without luminaires to make sure the data grid is complete, and/or sensors that can be wired to a luminaire but externally mounted.
- Not all sensors can be integrated into all luminaires; it is important to investigate luminaire options in advance, considering the effect on aesthetics as well as the cost for the overall project.
- Advanced technologies offering beyond lighting features and benefits are limited overall. If the OPR specifies a particular beyond-lighting capability, it is not uncommon for only one or two systems to be able to deliver that benefit. In many cases, manufacturers have created specialized offerings for particular building types or applications (e.g., office, retail, healthcare, industrial).
- Wireless communication protocols are highly varied and often have significant relevance to the ability to deliver beyond-lighting benefits. At a minimum the following variables should be considered,

because each has pros and cons directly related to a system's features and benefits: speed of communication; distance of communication; volume of transmitted data; ability to communicate with other technologies, devices, systems, and bandwidth usage; and cybersecurity.

- Benefits such as indoor location services (ILS) require specialized firmware, software, and data analytics in order to make the information actionable in the form of friendly user interfaces and dashboards.
- It is critical that digital systems are capable of being updated to ensure ongoing functionality. Often referred to as "future-proofing," this typically includes updates to firmware in various digital devices necessary to ensure ongoing compatibility of not only the lighting system components but also the related connected systems.
- Methods of configuration and commissioning to enable a CIN and SOO on a digital system are highly variable. The feasibility and simplicity of programming light levels; dimming rates and thresholds; daylight harvesting setpoints; and color and spectral control should be validated in advance to ensure viability.
- Some sensors receive their commands in real time from a different device such as a hub or router, while others have programmed behaviors saved within the sensor. It is important to consider how the lighting system will function if the digital wired or wireless network is temporarily down or disabled. Will it still function to code without the network or hub or server (e.g., auto-shutoff, emergency, daylight harvesting)? Are there large areas that will not function properly due to the loss of a device?
- The software packages that enable functionality and features are sometimes offered by the lighting manufacturer, but in other cases may be provided by non-lighting software companies that are using data from lighting-enabled APIs and microservices to program feature sets.

Ultimately, while there may be additional value associated with IoT and digital systems, the bottom line for lighting professionals is to ensure that the lighting outcomes are achieved, while staying in alignment with the Owner's Project Requirements. Consideration

of these factors will help to mitigate unintended consequences related to the pursuit of beyond-lighting use cases.

### 6.9 Indoor Location Services (ILS) Using Wireless Sensing

In addition to meeting energy codes for auto-shutoff and daylight harvesting, some sensing systems can be used to capture occupancy information for other purposes useful to the occupant, tenant, and/or owner or operator, such as space utilization and space reservation. **Table**

**6-10** shows an example of a CIN and SOO for a digital wireless system that can deliver these benefits. When one system performs for dual purposes, it is important to make sure that the needs of both are met.

## 7.0 The Lighting Control Matrix

SOOs are an excellent way to convey system operation and instructions for programming but

**Table 6-10. Example**

| Control Intent Narrative                                                                                                                                                        | Sequence of Operation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sensors will be used to determine occupant presence for the purpose of auto-shutoff as well as for space utilization analysis and integration with the room-reservation system. | <p>Sensor density shall be sufficient for the analytics of the software to function properly.</p> <p>Space utilization shall be determined using a combination of percentage of space occupied and time duration, with analytic results delivered via electronic dashboard.</p> <p>Where possible, sensors are to be integral to luminaires. Exceptions are expected and required when sensor locations and design intent are not aligned. The sensor system shall be capable of providing externally located sensors wired to luminaires, as well as fixtureless sensors powered independently without relationship to lighting control.</p> <p>Lighting control functionality and domain behaviors required for energy code compliance, aesthetics, and design intent shall remain intact and functional when the wireless lighting network is non-functional. Typically, this requires that lighting behavior profiles are digitally retained within the sensor rather than a remote hub or router location.</p> <p>Room reservations shall be capable of automatic adjustment based on occupancy, allowing the ability to rescind a reservation if occupancy is not detected. Durations shall be programmable by onsite occupants.</p> <p>All aspects of the digital system shall be capable of long-term performance and forward compatibility. This includes over-the-air updates to firmware as necessary for a seamless software functionality, including access to analytics and dashboards.</p> <p>Where more than one device or software manufacturer is to be used, a prototype installation and demonstration are required. This includes testing for interoperability for devices as well as at the cloud level when APIs are utilized.</p> |

can be intimidating and difficult to digest quickly. A sequence in a matrix format can remedy this. The simplified format allows non-experts to understand system operation with minimal effort. That said, it is very easy to write a poor sequence in matrix format by leaving out setpoints and other important information necessary for programming the lighting control system. It is absolutely critical for the matrix to evolve through each design package and after design is complete. These bullets describe how the matrix evolves through each phase:

- *Phase 1, Schematic Design:* Matrix is created; includes lighting control concepts, space types, and components.
- *Phase 2, Design Development:* Design illuminance and other setpoints are added; notes are added with more detail regarding how the system operates or other hardware required.
- *Phase 3, Construction Documents:* The sequences are identified and tied to the plans; space types

are refined to specific rooms; notes are clarified; all setpoints are identified.

- *Phase 4, Installation and Programming:* The matrix is sorted by room number; it includes changes from RFIs, submittals, and/or design modifications.

### 7.1 Phase 1 – Schematic Design

This matrix-style SOO is useful in the Schematic Design (SD) phase of a project. It provides a relatively good idea of the components required to accomplish the lighting control intent for each of the lighting control strategies. The key items to include in this phase are:

- Space type
- Control strategies
- A checkmark or other designation identifying the control strategy each space will use

However, it is lacking a description of the control functions, such as time delay of occupancy sensors, scene settings, and daylight harvesting setpoints.

**Table 7-1. Example of a Phase 1 (SD) Matrix (Showing Components by Space Type)**

| Space Type        | Line-Voltage Switch | On/Off/Raise/Lower Keypad | Scene Select Keypad | Occupancy Sensor (Auto Part On) | Vacancy Sensor | Time Schedule | Astro Time Switch | Daylight Photosensor | Demand Response | Emergency Ltg Control |
|-------------------|---------------------|---------------------------|---------------------|---------------------------------|----------------|---------------|-------------------|----------------------|-----------------|-----------------------|
| Break Rm, Lounge  |                     | •                         |                     |                                 | •              |               |                   | •                    | •               |                       |
| Conference Rms    |                     |                           | •                   |                                 | •              |               |                   | •                    | •               |                       |
| Corridors, Halls  |                     | •                         |                     | •                               |                | •             |                   |                      | •               | •                     |
| Elect, Elev, Mech | •                   |                           |                     |                                 |                |               |                   |                      |                 | •                     |
| Elevator Lobby    |                     | •                         |                     | •                               |                | •             |                   |                      |                 | •                     |
| Janitor, Storage  | •                   |                           |                     |                                 | •              |               |                   |                      |                 |                       |
| Lobby             |                     |                           | •                   |                                 | •              | •             | •                 | •                    | •               | •                     |
| Office <25 sq.m.  |                     | •                         |                     | •                               |                |               |                   | •                    | •               |                       |
| Office >25 sq.m.  |                     | •                         |                     |                                 | •              | •             |                   | •                    | •               | •                     |
| Parking Garage    |                     | •                         |                     | •                               |                |               | •                 | •                    | •               | •                     |
| Restrooms         |                     | •                         |                     | •                               | •              | •             |                   |                      |                 | •                     |
| Retail Sales      |                     |                           | •                   |                                 | •              | •             |                   |                      | •               | •                     |
| Science Lab       |                     | •                         |                     |                                 | •              |               |                   |                      | •               | •                     |
| Stairwells        |                     | •                         |                     | •                               |                |               |                   |                      |                 | •                     |
| Site Lighting     |                     | •                         |                     | •                               |                |               | •                 |                      |                 | •                     |

Furthermore, the Space Types are generic and lack specifics such as room numbers.

After Phase 1, the designer goes to Phase 2, or directly to Phase 3.

## 7.2 Phase 2 – Design Development

This type of matrix-style SOO provides the level of information that is needed for the Design Development (DD) phase. The key items to include in this phase are (new items are italicized):

- Space type
- *Design illuminance*
- Control strategies, with *a reference to additional details for that control sequence*
- A checkmark or other designation identifying the control strategy each space will use
- *Other notes to provide additional control details applicable to the space type*

To the component matrix of Phase 1, a column is added for the Design Illuminance because this value is needed for setting the task-tuning adjustment and for setting the daylight harvesting setpoints. A column is added to indicate where task tuning is desired. Notes N1 through N11 are added to the respective control strategies to provide additional details and setpoints. Notes N12, N13 and N14 are added for specific space types. These notes could include details for how the space operates, directions for scene switches, or information regarding additional hardware required to enact the sequences. Three locations within the matrix indicate that decisions are to be determined (TBD).

An example of a Phase 2 matrix is shown in **Table 7-2**. After Phase 2, the designer goes to Phase 3.

## 7.3 Phase 3 – Construction Documents

This matrix builds on the last one and adds specifics to tie the sequences in the matrix to the drawings. The key items to include in this phase are (new items are italicized):

- Lighting control designation

- *Room numbers* (added to Space Type information)
- Design illuminance
- Control strategies, with a reference to additional details for that control sequence
- A checkmark or other designation identifying the control strategy each space will use
- Other notes to provide additional control details applicable to the space type

A Lighting Control Designation column has been added to the matrix. These designations or identifiers (LC1 through LC15) are intended to be shown on the lighting plans in the respective rooms, and they serve as the link between the plans and the matrix. Also, specific room numbers are included with the Space Types to remove any potential uncertainty as to which rooms on the plans are matched to the Space Type in the matrix. The three TBD locations have been clarified.

An example of a Phase 3 matrix is shown in **Table 7-3**. After Phase 3, the designer goes to Phase 4.

## 7.4 Phase 4 – Installation and Commissioning

This matrix contains the same information as shown in the previous matrix, but now it is sorted by specific room number. This matrix is useful during the startup installation and during commissioning.

The key items to include in this phase are (new items are italicized):

- *Updates based on RFIs, submittals, or design changes*
- Room numbers (*separated from Space Type and sorted*)
- Lighting control designation
- Design illuminance
- Control strategies, with a reference to additional details for that control sequence
- A checkmark or other designation identifying the control strategy each space will use
- Other notes to provide additional control details applicable to the space type

An example of a Phase 3 matrix is shown in **Table 7-4**.

**Table 7-2. Example of a Phase 2 (DD) Matrix (Functionality by Space Type)**

| Space Type           | Design Illuminance (flux) | N1 – Line-Voltage Switch | N2 – On/Off/Raise/Lower Keypad | N3 – Scene Select Keypad | N4 – Task Tuning | N5 – Occupancy Sensor (Auto Part On) | N6 – Vacancy Sensor | N7 – Time Schedule | N8 – Astro Time Switch | N9 – Daylight Harvesting | N10 – Demand Response | N11 – Emergency Ltg Control | Other Notes |
|----------------------|---------------------------|--------------------------|--------------------------------|--------------------------|------------------|--------------------------------------|---------------------|--------------------|------------------------|--------------------------|-----------------------|-----------------------------|-------------|
| Break Rm, Lounge     | 250                       |                          | •                              |                          | •                |                                      | •                   |                    | •                      | •                        | •                     |                             |             |
| Conference Rms       | 350                       |                          |                                | •                        | •                |                                      | •                   |                    | •                      | •                        | •                     |                             |             |
| Corridors, Halls     | 200                       |                          | •                              |                          | •                | •                                    |                     | •                  | •                      |                          | •                     | •                           |             |
| Elect, Elev, Mech Rm | 200                       | •                        |                                |                          |                  |                                      |                     |                    |                        |                          |                       | •                           |             |
| Elevator Lobby       | 200                       |                          | •                              |                          | •                | •                                    |                     | •                  |                        |                          |                       | •                           | N12         |
| Janitor, Storage     | 150                       | •                        |                                |                          |                  |                                      |                     |                    |                        |                          |                       |                             |             |
| Lobby                | 250                       |                          |                                | •                        | •                |                                      | •                   | •                  | •                      | •                        | •                     | •                           |             |
| Office <25 sq.m.     | 300                       |                          | •                              |                          | •                | •                                    |                     |                    | •                      | •                        |                       |                             |             |
| Office >25 sq.m.     | 300                       |                          | •                              |                          | •                |                                      |                     | •                  | •                      | •                        | •                     | •                           |             |
| Parking Garage       | 150                       |                          | •                              |                          |                  | •                                    |                     |                    | •                      | •                        | •                     | •                           |             |
| Restrooms            | 250                       |                          | TBD                            | TBD                      | •                | •                                    | •                   | •                  | •                      |                          |                       | •                           |             |
| Retail Sales         | 500                       |                          | •                              | •                        | •                |                                      |                     | •                  | •                      |                          | •                     | •                           |             |
| Science Lab          | 500                       |                          | •                              |                          | •                |                                      | •                   |                    |                        |                          | •                     | •                           |             |
| Stairwells           | 100                       |                          | TBD                            |                          |                  | •                                    |                     |                    |                        |                          |                       | •                           | N13         |
| Site Lighting        | 30                        |                          | •                              |                          |                  | •                                    |                     |                    |                        |                          |                       | •                           | N14         |

*Table notes:*

- N1 Manual on/off device located within the area it controls.
- N2 Controls that allow the light output to be adjusted up and down.
- N3 8-buttons: On, Off, Raise, Lower, and 4 additional scene buttons.
- N4 Adjust the maximum allowed illuminance to be no more than 10% above the design illuminance as measured at the task. Record the new maximum setpoint.
- N5 Auto-on to 50% of maximum light output. Manual raise/lower. Manual-off and Auto-off.
- N6 During normal hours, Auto-off after 15 minutes of vacancy, using a 5-second fade. During non-normal hours, Auto-off after 5 minutes of vacancy, using a 5-second fade.
- N7 Lighting controls are enabled during normal hours of operation (7 a.m. to 7 p.m. weekdays). During non-normal hours, lighting is turned off automatically following a flash warning. May be manually overridden to on, or automatically via an occupancy sensor.
- N8 At 30 minutes after civil dusk, interior lighting shall be reduced by 25% over 10 minutes. At 30 minutes after civil dawn, interior lighting shall return to its normal levels over 10 minutes.
- N9 When illuminance from daylight is more than 150% of the design illuminance, lighting power shall be reduced by 65%. When illuminance from daylight is more than 200% of the design illuminance, lighting power shall fade to off over 60 seconds.

- N10 During a demand-response event, the lighting power shall be reduced by 15% and shall not be allowed to exceed that value during the event. After the demand-response event, the lighting shall return to the level preceding the event.
- N11 If the general-lighting luminaires are used to provide the code-required emergency illumination, then means shall be provided (e.g., via a UL924 relay) to cause the designated emergency luminaires to go to full light output upon loss of the normal lighting power to the area.
- N12 Elevator lobbies shall provide a minimum of 100 lux 24/7 at the door sill of the elevators. The sills may have greater illuminance to match adjacent corridors during normal hours.
- N13 Stairwells shall have a minimum of 100 lux while the stairwell is occupied and shall remain at this lighting level for 10 minutes after vacancy. During vacancy, the illuminance shall not be less than 10 lux.
- N14 Site Lighting: All lighting shall be tied to an astronomical time switch. Occupancy sensors shall be used only where required by code.



**Table 7-3. Example of a Phase 3 (CD) Matrix (Functionality by Space Type With Room Numbers)**

| Lighting Control Designation | Space Type and Room Numbers        | Design Illuminance (lux) | N1 – Line-Voltage Switch | N2 – On/Off/Raise/Lower Keypad | N3 – Scene Select Keypad | N4 – Task Tuning | N5 – Occupancy Sensor (Auto Part On) | N6 – Vacancy Sensor | N7 – Time Schedule | N8 – Astro Time Switch | N9 – Daylight Harvesting | N10 – Demand Response | N11 – Emergency Ltg Control | Other Notes |
|------------------------------|------------------------------------|--------------------------|--------------------------|--------------------------------|--------------------------|------------------|--------------------------------------|---------------------|--------------------|------------------------|--------------------------|-----------------------|-----------------------------|-------------|
| LC1                          | Break Rm, Lounge 108, 111          | 250                      |                          | •                              |                          | •                |                                      | •                   |                    | •                      | •                        | •                     |                             |             |
| LC2                          | Conference Rms 103, 106            | 350                      |                          |                                | •                        | •                |                                      | •                   |                    | •                      | •                        | •                     |                             |             |
| LC3                          | Corridors, Halls 105, 116          | 200                      |                          | •                              |                          | •                | •                                    |                     | •                  | •                      |                          | •                     | •                           |             |
| LC4                          | Elect, Elev, Mech Rm 107, 117, 121 | 200                      | •                        |                                |                          |                  |                                      |                     |                    |                        |                          |                       | •                           |             |
| LC5                          | Elevator Lobby 109                 | 200                      |                          | •                              |                          | •                | •                                    |                     | •                  |                        |                          |                       | •                           | N12         |
| LC6                          | Janitor, Storage 110, 112          | 150                      | •                        |                                |                          |                  |                                      |                     |                    |                        |                          |                       |                             |             |
| LC7                          | Lobby 101                          | 250                      |                          |                                | •                        | •                |                                      | •                   | •                  | •                      | •                        | •                     | •                           |             |
| LC8                          | Office <250SF 102, 115             | 300                      |                          | •                              |                          | •                | •                                    |                     |                    | •                      | •                        |                       |                             |             |
| LC9                          | Office >250SF 113                  | 300                      |                          | •                              |                          | •                |                                      |                     | •                  | •                      | •                        | •                     | •                           |             |
| LC10                         | Parking Garage 001                 | 150                      |                          | •                              |                          |                  | •                                    |                     |                    | •                      | •                        | •                     | •                           |             |
| LC11                         | Restrooms 118, 119                 | 250                      |                          | •                              |                          | •                | •                                    | •                   | •                  | •                      |                          |                       | •                           |             |
| LC12                         | Retail Sales 104                   | 500                      |                          | •                              | •                        | •                |                                      |                     | •                  | •                      |                          | •                     | •                           |             |
| LC13                         | Science Lab 120                    | 500                      |                          | •                              |                          | •                |                                      | •                   |                    |                        |                          | •                     | •                           |             |
| LC14                         | Stairwells 114                     | 100                      |                          | •                              |                          |                  | •                                    |                     |                    |                        |                          |                       | •                           | N13         |
| LC15                         | Site Lighting 300                  | 30                       |                          | •                              |                          |                  | •                                    | •                   |                    |                        |                          |                       | •                           | N14         |

*Table notes:*

- N1 Manual on/off device located within the area it controls.
- N2 Controls that allow the lighting level to be adjusted up and down.
- N3 8-buttons: On, Off, Raise, Lower, and 4 additional scene buttons.
- N4 Adjust the maximum allowed illuminance to be no more than 10% above the design illuminance as measured at the task. Record the new maximum setpoint.
- N5 Auto-on to 50% of maximum light output. Manual raise/lower. Manual-off and Auto-off.
- N6 During normal hours, Auto-off after 15 minutes of vacancy using a 5-second fade. During non-normal hours, Auto-off after 5 minutes of vacancy using a 5-second fade.
- N7 Lighting controls are enabled during normal hours of operation (7 a.m. to 7 p.m. weekdays). During non-normal hours, lighting is turned off automatically following a flash warning. May be manually overridden to on, or automatically via occupancy sensor.
- N8 At 30 minutes after civil dusk, interior lighting shall be reduced by 25% over 10 minutes. At 30 minutes after civil dawn, interior lighting shall return to its normal levels over 10 minutes.
- N9 When illuminance from daylight is more than 150% of design illuminance, lighting power shall be reduced by 65%. When daylight illuminance is more than 200% of design illuminance, lighting power shall fade to off over 60 seconds.

- N10 During a demand-response event, the lighting power shall be reduced by 15% and shall not be allowed to exceed that value during the event. After the demand-response event, the lighting shall return to the level preceding the event.
- N11 If the general-lighting luminaires are used to provide the code-required emergency illumination, then means shall be provided (e.g., via a UL924 relay) to cause the designated emergency luminaires to go to full light output upon loss of the normal lighting power to the area.
- N12 Elevator lobbies shall provide a minimum of 100 lux 24/7 at the door sill of the elevators. The sills may have greater illuminance to match adjacent corridors during normal hours.
- N13 Stairwells shall have a minimum of 100 lux while the stairwell is occupied and shall remain at this lighting level for 10 minutes after vacancy. During vacancy, the illuminance shall not be less than 10 lux.
- N14 Site Lighting: All lighting shall be tied to an astronomical time switch. Occupancy sensors shall be used only where required by code. Where occupancy sensors are required, the lights shall remain on for a duration of 20 minutes after last occupancy.

**Table 7-4. Example of a Phase 4 (Installation and Commissioning) Matrix (Functionality by Room Number)**

| Room Number | Space Type         | Control Designation | Design Illuminance (lux) | N1 – Line-Voltage Switch | N2 – On/Off/Raise/Lower Keypad | N3 – Scene Select Keypad | N4 – Task Tuning | N5 – Occupancy Sensor (Auto Part On) | N6 – Vacancy Sensor | N7 – Time Schedule | N8 – Astro Time Switch | N9 – Daylight Harvesting | N10 – Demand Response | N11 – Emergency Ltg Control | Other Notes |
|-------------|--------------------|---------------------|--------------------------|--------------------------|--------------------------------|--------------------------|------------------|--------------------------------------|---------------------|--------------------|------------------------|--------------------------|-----------------------|-----------------------------|-------------|
| 001         | Parking Garage     | LC10                | 150                      |                          | •                              |                          |                  | •                                    |                     |                    | •                      | •                        | •                     | •                           |             |
| 101         | Lobby              | LC7                 | 25                       |                          |                                | •                        | •                |                                      | •                   | •                  | •                      | •                        | •                     | •                           |             |
| 102         | Office-1 <25 sq.m. | LC8                 | 300                      |                          | •                              |                          | •                | •                                    |                     |                    | •                      | •                        |                       |                             |             |
| 103         | Conference Rm 1    | LC2                 | 350                      |                          |                                | •                        | •                |                                      | •                   |                    | •                      | •                        | •                     |                             |             |
| 104         | Retail Sales       | LC12                | 500                      |                          | •                              | •                        | •                |                                      |                     | •                  | •                      |                          | •                     | •                           |             |
| 105         | Corridor           | LC3                 | 200                      |                          | •                              |                          | •                | •                                    |                     | •                  | •                      |                          | •                     | •                           |             |
| 106         | Conference Rm 2    | LC2                 | 350                      |                          |                                | •                        | •                |                                      | •                   |                    | •                      | •                        | •                     |                             |             |
| 107         | Elev Machine Rm    | LC4                 | 200                      | •                        |                                |                          |                  |                                      |                     |                    |                        |                          |                       | •                           |             |
| 108         | Lounge             | LC1                 | 250                      |                          | •                              |                          | •                |                                      | •                   |                    | •                      | •                        | •                     |                             |             |
| 109         | Elevator Lobby     | LC5                 | 100                      |                          |                                |                          |                  |                                      |                     |                    |                        |                          |                       |                             | N12         |
| 110         | Janitor            | LC6                 | 150                      | •                        |                                |                          |                  |                                      |                     |                    |                        |                          |                       |                             |             |
| 111         | Break Rm           | LC1                 | 250                      |                          | •                              |                          | •                |                                      | •                   |                    | •                      | •                        | •                     |                             |             |
| 112         | Storage            | LC6                 | 150                      | •                        |                                |                          |                  |                                      |                     |                    |                        |                          |                       |                             |             |
| 113         | Office >25 sq.m.   | LC9                 | 300                      |                          | •                              |                          | •                |                                      |                     | •                  | •                      | •                        | •                     | •                           |             |
| 114         | Stairwells         | LC14                | 100                      |                          |                                |                          |                  | •                                    |                     |                    |                        |                          |                       | •                           | N13         |
| 115         | Office-2 <25 sq.m. | LC8                 | 300                      |                          | •                              |                          | •                | •                                    |                     |                    | •                      | •                        |                       |                             |             |
| 116         | Hall               | LC3                 | 200                      |                          | •                              |                          | •                | •                                    |                     | •                  | •                      |                          | •                     | •                           |             |
| 117         | Mech Rm            | LC4                 | 200                      | •                        |                                |                          |                  |                                      |                     |                    |                        |                          |                       | •                           |             |
| 118         | Restrooms – Men    | LC11                | 250                      |                          |                                |                          | •                | •                                    | •                   | •                  | •                      |                          |                       | •                           |             |
| 119         | Restrooms – Women  | LC11                | 250                      |                          |                                |                          | •                | •                                    | •                   | •                  | •                      |                          |                       | •                           |             |
| 120         | Science Lab        | LC13                | 500                      |                          | •                              |                          | •                |                                      | •                   |                    |                        | •                        | •                     | •                           |             |
| 121         | Elect Rm           | LC4                 | 200                      | •                        |                                |                          |                  |                                      |                     |                    |                        |                          |                       | •                           |             |
| 300         | Site Lighting      | LC15                | 30                       |                          | •                              |                          |                  | •                                    | •                   |                    |                        |                          |                       | •                           | N14         |

Table notes:

- N 1 Manual on/off device located within the area it controls.
- N 2 Controls that allow the lighting level to be adjusted up and down.
- N 3 8-buttons: On, Off, Raise, Lower, and 4 additional scene buttons.
- N 4 Adjust the maximum allowed illuminance to be no more than 10% above the design illuminance as measured at the task. Record the new maximum setpoint.
- N 5 Auto-on to 50% of maximum light output. Manual raise/lower. Manual-off and Auto-off.
- N 6 During normal hours, Auto-off after 15 minutes of vacancy using a 5-second fade. During non-normal hours, Auto-off after 5 minutes of vacancy using a 5-second fade.
- N 7 Lighting controls are enabled during normal hours of operation (7 a.m. to 7 p.m. weekdays). During non-normal hours, lighting is turned off automatically following a flash warning. May be manually overridden to on, or automatically via occupancy sensor.
- N 8 At 30 minutes after civil dusk, interior lighting shall be reduced by 25% over 10 minutes. At 30 minutes after civil dawn, interior lighting shall return to its normal levels over 10 minutes.

- N 9 When daylight illuminance from daylight is >150% of design illuminance, reduce lighting power by 65%. When illuminance is more than 200% of design illuminance, lighting power shall fade to off over 60 seconds.
- N 10 During a demand-response event, the lighting power shall be reduced by 15% and shall not be allowed to exceed that value during the event. After the demand-response event, the lighting shall return to the level preceding the event.
- N 11 If the general-lighting luminaires are used to provide the code-required emergency illumination, then means shall be provided (e.g., via a UL924 relay) to cause the designated emergency luminaires to go to full light output upon loss of the normal lighting power to the area.
- N 12 Elevator lobbies shall provide a minimum of 100 lux 24/7 at the door sill of the elevators. The sills may have greater illuminance to match adjacent corridors during normal hours.
- N 13 Stairwells shall have a minimum of 100 lux while the stairwell is occupied and shall remain at this lighting level for 10 minutes after vacancy. During vacancy, the illuminance shall not be less than 10 lux.
- N 14 Site Lighting: All lighting shall be tied to an astronomical time switch. Occupancy sensors shall be used only where required by code. Where occupancy sensors are required, the lights shall remain on for a duration of 20 minutes after last occupancy.

## 7.5 Enhanced Lighting Control Matrix Examples

The examples provide in the following subsections show an enhanced matrix. The first example project is a university building containing an atrium and classrooms. A preliminary matrix and completed matrix were produced for the project. The second example project is a hotel, including exterior lighting and various amenities.

**7.5.1 Enhanced Matrix – Example #1.** This is an example of a project in which a lighting controls specifier was engaged later in the project to provide a sequence of operations. A preliminary matrix was used by the design team and Owner to confirm the functionality for each space. The completed matrix was submitted to an electrical contractor and a lighting-controls manufacturer's field engineer before submittal review. The submittal was reviewed to ensure that the equipment could enact the sequence and that the submittal faithfully captured the requirements of the SOO.

**7.5.1.1 Preliminary Enhanced Matrix, Example #1.** Since this matrix was created later in the project, room names and locations of rooms were already settled by the time this matrix was created. However, control functionalities had to be agreed to first, such as which space would have multiple scenes, or which space would automatically change scenes based on time of day.

**Figure 7-1** shows brief description of proposed control functionality at the top row, and room names and numbers were listed on the first column. Specific setpoints were mostly to be determined later.

For some rooms, multiple scenes were desired for different space uses, to enhance the appearance of the space and to increase energy savings. **Figures 7-2a** and **7-2b** show names of scenes listed for each room, but output for each zone was not yet determined.

**7.5.1.2 Completed Enhanced Matrix, Example #1.** The completed enhanced matrix is shown in **Figure 7-3**. At this point, all setpoints have been finalized. In addition, a Scene Schedule was completed, with light output for each zone (**Figures 7-4a** through **7-4c**), and keypad types and diagrams (**Figure 7-5**). Construction-level control documents provide information needed for a fully informed startup installation. It is important

to mention that although setpoints were provided in the sequence and programmed during the startup installation, they may need to be adjusted. The design team and the Owner's representative should review the setpoints and scenes in the building with a lighting control manufacturer's field engineer and fine-tune them. Best practice would be to include a note in the matrix or narrative that all setpoints will be adjusted in the field. This will allow the controls contractor to budget and allow time for this activity.

**7.5.2 Preliminary Enhanced Matrix – Example #2.** An example is shown in **Figure 7-6** of an SOO supplied on a project where a lighting designer is engaged longer than in Example #1 (see **Section 7.5.1**). The lighting designer would be contracted for the duration of the design and would be onsite during construction for fine-tuning the lighting control system. Therefore, this style of SOO provides details on what kinds of controls will be utilized in specific areas, and provides a base level of code-required setpoints, but does not go so far as to define illumination levels. On this type of project, the lighting designer will provide final setpoints onsite in an aiming and level-setting session with direct input from the architect, interior designer, owner, and/or end user.

## List of Abbreviations

ANSI: American National Standards Institute  
 API: application programming interface  
 BMS: Building management system  
 BOD: Basis of Design  
 CCT: correlated color temperature  
 CIN: Control Intent Narrative  
 CSI: Construction Specifications Institute  
 DCV: demand-controlled ventilation  
 HVAC: heating, ventilating, and air conditioning  
 ILS: indoor location services  
 OPR: Owner's Project Requirements  
 PICS: Protocol Implementation Conformance Statement  
 PIR: passive-infrared (sensor)  
 SOO: Sequence of Operations  
 TCP: Transmission Control Protocol  
 UDP: User Datagram Protocol

## Lighting Practice: Documenting Control Intent Narratives and Sequences of Operations

|                                                                                                         |                    |                  |                                    |                                         |                                          |                                         |                         |
|---------------------------------------------------------------------------------------------------------|--------------------|------------------|------------------------------------|-----------------------------------------|------------------------------------------|-----------------------------------------|-------------------------|
| Partition sensor integration                                                                            | INTEGRATION        |                  |                                    |                                         |                                          |                                         | •                       |
| AV integration                                                                                          |                    |                  |                                    |                                         |                                          | •                                       | •                       |
| Emergency light fixtures: on when losing normal power, otherwise controlled with non-emergency fixtures | EMERGENCY CONTROL  |                  | •                                  | •                                       | •                                        |                                         | •                       |
| On call for Demand Response, dim 15%, 10sec fade                                                        | DEMAND RESPONSE    |                  |                                    | •                                       | •                                        | •                                       | •                       |
| Ramp up rate 20%/sec and ramp down rate 2%/sec                                                          | DAYLIGHTING        |                  | •                                  | •                                       | •                                        |                                         | •                       |
| Daylight fc value for lighting fixtures dimmed to 35%                                                   |                    | TBD              | TBD                                | TBD                                     |                                          |                                         | TBD                     |
| Fade rate for Occ Sensor                                                                                | OCCUPANCY SENSOR   |                  | 2 sec                              |                                         | 2 sec                                    | 2 sec                                   | 2 sec                   |
| 15min delay, light all off when vacant; walk thru is ON                                                 |                    |                  |                                    |                                         | •                                        | •                                       | •                       |
| 15min delay, auto dim when vacant, Dim setting (Note 1)                                                 |                    | TBD              |                                    | TBD                                     |                                          |                                         |                         |
| Dual Tech Occ Sensor, auto on                                                                           |                    | TBD              |                                    | TBD                                     |                                          |                                         |                         |
| 30min delay, auto dim when vacant, Bright Setting (Note 1)                                              |                    | TBD              |                                    | TBD                                     |                                          |                                         |                         |
| Dual Tech Occ Sensor, auto on, Bright Setting (Note 1)                                                  |                    | TBD              |                                    | TBD                                     |                                          |                                         |                         |
| Dual Tech Occ Sensor, auto on to Scene 2 (Medium ambient)                                               |                    |                  |                                    |                                         | •                                        | •                                       | •                       |
|                                                                                                         |                    |                  |                                    |                                         |                                          |                                         |                         |
| Fade rate for automatic scenes by time clock                                                            | SCHEDULED CONTROL  |                  | 10 sec                             | 10 sec                                  | 10 sec                                   |                                         |                         |
| Scheduled scenes change based on astronomical time switch                                               |                    |                  | •                                  | •                                       | •                                        |                                         |                         |
| Turn Day scene on 6am                                                                                   |                    |                  | •                                  | •                                       | •                                        |                                         |                         |
| Sweep all off at 3am except lights controlled by Occ Sensor                                             |                    |                  | •                                  | •                                       | •                                        |                                         |                         |
| Fade rate for push button                                                                               | MANUAL CONTROLS    |                  | 3 sec                              | 3 sec                                   | 3 sec                                    | 2 sec                                   | 2 sec                   |
| Override scheduled scene setting max 2hr (push button)                                                  |                    |                  | •                                  | •                                       | •                                        |                                         |                         |
| Continuous dimming adjustment (hold button)                                                             |                    |                  | •                                  | •                                       | •                                        | •                                       | •                       |
| Select scene (push button)                                                                              |                    |                  | •                                  | •                                       | •                                        | •                                       | •                       |
|                                                                                                         | SPACE / ROOM TYPES | Entry hall 1000A | Lower Atrium 1st Floor 1000A, 1202 | Lower Atrium 2nd Floor 2200, 2202, 2204 | Classroom/seminar with camera 1211, 1207 | Classrooms 1203, 1205, 1213, 1215, 1217 | Lecture Hall 1102, 1104 |

Note 1: 6am to one hour after Sunset: Bright setting. One hour after Sunset to 6am: Dim setting.

Note 2: Building primarily used 7am-7pm M-F. Will be Users present outside of the prime working hours. Users have access to the building 24/7 and are not restricted to 7-7, M-F.

Note 3: No off season.

**Figure 7-1. Preliminary enhanced-matrix example: Sequence of Operation.** (Image courtesy of Yukiko Yoshida, Auerbach Glasow)

| PRELIMINARY LIGHTING CONTROL SCENES                              |                                      |                                                  |                 |             |
|------------------------------------------------------------------|--------------------------------------|--------------------------------------------------|-----------------|-------------|
| Room Type and Room Number                                        | Keypad Type                          | Scene                                            | Label on Button | Photosensor |
| Entry Hall 1000A                                                 | No keypad required                   | Automatic by astronomical time schedule * Note 1 |                 |             |
|                                                                  |                                      | 1. Day                                           |                 | Enable      |
|                                                                  |                                      | 2. Twilight                                      |                 | Disable     |
|                                                                  |                                      | 3. Evening                                       |                 | Disable     |
|                                                                  |                                      | 4. Late evening                                  |                 | Disable     |
|                                                                  |                                      | OFF                                              |                 |             |
| Entry Hall 1000A                                                 | (1) 5 Button Keypad for 1,2,3 & OFF  | Manual override                                  |                 |             |
|                                                                  |                                      | 1. Dim Ambient                                   | DIM             | Disable     |
|                                                                  |                                      | 2. Medium Ambient                                | MED             | Disable     |
|                                                                  |                                      | 3. Bright Ambient                                | BRT             | Enable      |
|                                                                  |                                      | OFF                                              | OFF             |             |
| Lower Atrium 1st Floor & 2nd Floor 1000A, 1202, 2200, 2202, 2204 | No keypad required                   | Automatic by astronomical time schedule          |                 |             |
|                                                                  |                                      | 1. Day                                           |                 | Enable      |
|                                                                  |                                      | 2. Twilight                                      |                 | Disable     |
|                                                                  |                                      | 3. Evening                                       |                 | Disable     |
|                                                                  |                                      | 4. Late evening                                  |                 | Disable     |
|                                                                  |                                      | OFF                                              |                 |             |
| Lower Atrium 1st Floor 1000A, 1202                               | (1) 5 Button Keypad for 1,2,3 & OFF. | Manual override                                  |                 |             |
|                                                                  |                                      | 1. Dim Ambient                                   | DIM             | Disable     |
|                                                                  |                                      | 2. Medium Ambient                                | MED             | Disable     |
|                                                                  |                                      | 3. Bright Ambient                                | BRT             | Enable      |
|                                                                  |                                      | OFF                                              | OFF             |             |
| Lower Atrium 2nd Floor 2200, 2202, 2204                          | (1) 5 Button Keypad for 1,2,3 & OFF. | Manual override                                  |                 |             |
|                                                                  |                                      | 1. Dim Ambient                                   | DIM             | Disable     |
|                                                                  |                                      | 2. Medium Ambient                                | MED             | Disable     |
|                                                                  |                                      | 3. Bright Ambient                                | BRT             | Enable      |
|                                                                  |                                      | OFF                                              | OFF             |             |

**Figure 7-2a. Preliminary enhanced-matrix example: Scene Schedule, page 1.** (Image courtesy of Yukiko Yoshida, Auerbach Glasow)

| Room Type and Room Number                    | Keypad Type                                                                                                                         | Scene                                                        | Label on Button  | Photosensor |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|------------------|-------------|
| Classroom/ Seminar with camera<br>1207, 1211 | (1) 8 Button Keypad at door for scene 1,2,3, 4 & OFF.<br>5 and 6 accessed via AV touch panel.                                       | Manual override                                              |                  |             |
|                                              |                                                                                                                                     | 1. Dim Ambient                                               | DIM              | NA          |
|                                              |                                                                                                                                     | 2. Bright at front and Medium Ambient                        | MED              | NA          |
|                                              |                                                                                                                                     | 3. Dim at projection and Medium Ambient                      | PRJ              | NA          |
|                                              |                                                                                                                                     | 4. Bright Ambient for cleaning                               | BRT              | NA          |
|                                              |                                                                                                                                     | 5. Front at Presenter's left bright for Course Capture       | Presenter Left   | NA          |
|                                              |                                                                                                                                     | 6. Front at Presenter's right bright for Course Capture      | Presenter Right  | NA          |
| Classrooms<br>1203, 1205, 1213, 1217, 1215   | (2) 8 Button Keypads for scenes 1,2,3,4, 5 & OFF.                                                                                   | OFF                                                          | OFF              |             |
|                                              |                                                                                                                                     | Manual override                                              |                  |             |
|                                              |                                                                                                                                     | 1. Dim Ambient                                               | DIM              | NA          |
|                                              |                                                                                                                                     | 2. Bright at front and Medium Ambient                        | MED              | NA          |
|                                              |                                                                                                                                     | 3. Dim at projection and Medium Ambient                      | PRJ              | NA          |
|                                              |                                                                                                                                     | 4. Bright at lectern and Medium Ambient                      | LEC              | NA          |
|                                              |                                                                                                                                     | 5. Bright Ambient for cleaning                               | CLN/BRT          | NA          |
| Lecture Hall<br>1102                         | (2) 8 Button Keypads at front and back for scene 1,2,3,4, 5 & OFF.<br>6,7, and 8 will be accessed through AV system AV touch panel. | OFF                                                          | OFF              |             |
|                                              |                                                                                                                                     | Manual override                                              |                  |             |
|                                              |                                                                                                                                     | 1. Dim Ambient                                               | DIM              | NA          |
|                                              |                                                                                                                                     | 2. Bright at front and Medium Ambient                        | MED              | NA          |
|                                              |                                                                                                                                     | 3. Bright at lectern and Medium Ambient                      | LEC              | NA          |
|                                              |                                                                                                                                     | 4. Dim at front for projection and Medium Ambient            | PRJ              | NA          |
|                                              |                                                                                                                                     | 5. Bright Ambient for cleaning                               | CLN/BRT          | NA          |
|                                              |                                                                                                                                     | 6. Videotaping: Very Bright at front area for Course Capture | Video whiteboard | NA          |
|                                              |                                                                                                                                     | 7. Videotaping: Very Bright at lectern for Course Capture    | Video lectern    | NA          |
|                                              |                                                                                                                                     | 8. Videotaping: Dim at projection for Course Capture         | Projection       | NA          |
|                                              |                                                                                                                                     | OFF                                                          | OFF              |             |

Figure 7-2b. Preliminary enhanced-matrix example: Scene Schedule, page 2. (Image courtesy of Yukiko Yoshida, Auerbach Glasow)

| Room Type and Room Number                     | Keypad Type                                                                  | Scene                                                        | Label on Button  | Photosensor |
|-----------------------------------------------|------------------------------------------------------------------------------|--------------------------------------------------------------|------------------|-------------|
| <b>Lecture Hall 1104</b>                      | (1) 8 Button Keypad at front & back for scenes 1,2,3,4,5 & OFF.              | Manual override                                              |                  |             |
|                                               |                                                                              | 1. Dim Ambient                                               | DIM              | Disable     |
|                                               |                                                                              | 2. Bright at front and Medium Ambient                        | MED              | Disable     |
|                                               |                                                                              | 3. Bright at lectern and Medium Ambient                      | LEC              | Disable     |
|                                               |                                                                              | 4. Dim at front for projection and Medium Ambient            | PRJ              | Disable     |
|                                               | 6,7, and 8 accessed via AV system AV touch panel.                            | 5. Bright Ambient for cleaning                               | CLN/BRT          | Enable      |
|                                               |                                                                              | 6. Videotaping: Very Bright at front area for Course Capture | Video whiteboard | Disable     |
|                                               |                                                                              | 7. Videotaping: Very Bright at lectern for Course Capture    | Video lectern    | Disable     |
|                                               |                                                                              | 8. Videotaping: Dim at projection for Course Capture         | Projection       | Disable     |
|                                               |                                                                              | OFF                                                          | OFF              |             |
| <b>Lecture Halls combined 1102 &amp; 1104</b> | The keypads in each room will be used. 9 and 10 accessed via AV touch panel. | Manual override                                              |                  |             |
|                                               |                                                                              | 1. Dim Ambient                                               | DIM              | Disable     |
|                                               |                                                                              | 2. Bright at front and Medium Ambient                        | MED              | Disable     |
|                                               |                                                                              | 3. Bright at lecterns and Medium Ambient                     | LEC              | Disable     |
|                                               |                                                                              | 4. Dim at front for projection and Medium Ambient            | PRJ              | Disable     |
|                                               |                                                                              | 5. Bright Ambient for cleaning and Science Fair              | BRT              | Enable      |
|                                               |                                                                              | 9. Panel discussion and Medium Ambient                       | Discussion panel | Disable     |
|                                               |                                                                              | 10. Bright at one Lectern and Medium Ambient                 | Center lectern   | Disable     |
|                                               |                                                                              | OFF                                                          | OFF              |             |
|                                               |                                                                              |                                                              |                  |             |

Note 1:

Scene schedule for astronomical time switch:  
 "Day" Scene starts: 6am  
 "Twilight" Scene starts: One hour before Sunset  
 "Evening" Scene starts: One hour after Sunset  
 "Late evening" starts: 9:40pm

Note 2:

Sunrise and Sunset in Berkeley CA  
 Date Sunrise Sunset  
 March 20 7:12am 7:21pm  
 June 21 5:47am 8:35pm  
 September 22 6:57am 7:05pm  
 December 21 7:21am 4:53pm

**Figure 7-2c. Preliminary enhanced-matrix example: Scene Schedule, page 3.** (Image courtesy of Yukiko Yoshida, Auerbach Glasow)



## Lighting Practice: Documenting Control Intent Narratives and Sequences of Operations

|                                                                                                         |                    |                  |                                    |                                         |                                          |                                         |                         |
|---------------------------------------------------------------------------------------------------------|--------------------|------------------|------------------------------------|-----------------------------------------|------------------------------------------|-----------------------------------------|-------------------------|
| Partition sensor integration                                                                            | INTEGRATION        |                  |                                    |                                         |                                          |                                         |                         |
| AV integration                                                                                          |                    |                  |                                    |                                         | •                                        |                                         | •                       |
| Emergency light fixtures: on when losing normal power, otherwise controlled with non-emergency fixtures | EMERGENCY CONTROL  | •                | •                                  | •                                       |                                          |                                         | •                       |
| On call for Demand Response, dim 15%, 10sec fade                                                        | DEMAND RESPONSE    |                  | •                                  | •                                       | •                                        | •                                       | •                       |
| Ramp up rate 20%/sec and ramp down rate 2%/sec                                                          | DAYLIGHTING        | •                | •                                  | •                                       |                                          |                                         | •                       |
| Daylight fc value for lighting fixtures dimmed to 35%                                                   |                    | 30fc             | 45fc                               | 45fc                                    |                                          |                                         | B=60fc                  |
| Fade rate for Occ Sensor                                                                                | OCCUPANCY SENSOR   | 2 sec            |                                    | 2 sec                                   | 2 sec                                    | 2 sec                                   | 2 sec                   |
| 15min delay, light all off when vacant; walk thru is ON                                                 |                    |                  |                                    |                                         | •                                        | •                                       | •                       |
| 15min delay, auto dim when vacant, Dim setting (Note 1)                                                 |                    | 5%               |                                    | 5%                                      |                                          |                                         |                         |
| Dual Tech Occ Sensor, auto on, Dim setting (Note 1)                                                     |                    | 25%              |                                    | 25%                                     |                                          |                                         |                         |
| 30min delay, auto dim when vacant, Bright Setting (Note 1)                                              |                    | 25%              |                                    | 25%                                     |                                          |                                         |                         |
| Dual Tech Occ Sensor, auto on, Bright Setting (Note 1)                                                  |                    | 100%             |                                    | 100%                                    |                                          |                                         |                         |
| Dual Tech Occ Sensor, auto on to Scene 2 (Medium ambient)                                               |                    |                  |                                    |                                         | •                                        | •                                       | •                       |
| Fade rate for automatic scenes by time clock                                                            | SCHEDULED CONTROL  | 10 sec           | 10 sec                             | 10 sec                                  |                                          |                                         |                         |
| Scheduled scenes change based on astronomical time switch                                               |                    | •                | •                                  | •                                       |                                          |                                         |                         |
| Turn Day scene on 6am                                                                                   |                    | •                | •                                  | •                                       |                                          |                                         |                         |
| Sweep all off at 3am except lights controlled by Occ Sensor                                             |                    | •                | •                                  | •                                       |                                          |                                         |                         |
| Fade rate for push button                                                                               | MANUAL CONTROLS    | 3 sec            | 3 sec                              | 3 sec                                   | 2 sec                                    | 2 sec                                   | 2 sec                   |
| Override scheduled scene setting max 2hr (push button)                                                  |                    | •                | •                                  | •                                       |                                          |                                         |                         |
| Continuous dimming adjustment (hold button)                                                             |                    | •                | •                                  | •                                       | •                                        | •                                       | •                       |
| Select scene (push button)                                                                              |                    | •                | •                                  | •                                       | •                                        | •                                       | •                       |
|                                                                                                         | SPACE / ROOM TYPES | Entry hall 1000A | Lower Atrium 1st Floor 1000A, 1202 | Lower Atrium 2nd Floor 2200, 2202, 2204 | Classroom/seminar with camera 1211, 1207 | Classrooms 1203, 1205, 1213, 1215, 1217 | Lecture Hall 1102, 1104 |

Note 1: 6am to one hour after Sunset: Bright setting. One hour after Sunset to 6am: Dim setting.

Note 2: Building primarily used 7am-7pm M-F. Will be Users present outside of the prime working hours. Users have access to the building 24/7 and are not restricted to 7-7, M-F.

Note 3: No off season.

**Figure 7-3. Completed enhanced-matrix example: Sequence of Operation.** (Image courtesy of Yukiko Yoshida, Auerbach Glasow)

| LIGHTING CONTROL SCENES CD PHASE                                 |                                           |                                                 |                                 |      |     |      |      |      |      |      |      |      |      |      |     |
|------------------------------------------------------------------|-------------------------------------------|-------------------------------------------------|---------------------------------|------|-----|------|------|------|------|------|------|------|------|------|-----|
| Room Type and Number                                             | Keypad Type                               | Scene                                           | Control Group/Zone & Output (%) |      |     |      |      |      |      |      |      |      |      |      |     |
| Entry Hall 1000A                                                 | No keypad required                        | Automatic by astronomical time schedule * Note1 | A1                              | A2   | A3  |      |      |      |      |      |      |      |      |      |     |
|                                                                  |                                           | 1. Day                                          | 0%                              | 100% |     |      |      |      |      |      |      |      |      |      |     |
|                                                                  |                                           | 2. Twilight                                     | 80%                             | 80%  |     |      |      |      |      |      |      |      |      |      |     |
|                                                                  |                                           | 3. Evening                                      | 20%                             | 25%  |     |      |      |      |      |      |      |      |      |      |     |
|                                                                  |                                           | 4. Late evening                                 | 10%                             | 25%  |     |      |      |      |      |      |      |      |      |      |     |
|                                                                  |                                           | OFF                                             | 0%                              | 0%   |     |      |      |      |      |      |      |      |      |      |     |
| Entry Hall 1000A                                                 | KP-1 (1) 5 Button Keypad for 1,2,3 & OFF  | Manual override                                 | A1                              | A2   | A3  |      |      |      |      |      |      |      |      |      |     |
|                                                                  |                                           | 1. Dim Ambient                                  | 10%                             | 10%  |     |      |      |      |      |      |      |      |      |      |     |
|                                                                  |                                           | 2. Medium Ambient                               | 50%                             | 50%  |     |      |      |      |      |      |      |      |      |      |     |
|                                                                  |                                           | 3. Bright Ambient                               | 100%                            | 100% |     |      |      |      |      |      |      |      |      |      |     |
|                                                                  |                                           | OFF                                             | 0%                              | 0%   |     |      |      |      |      |      |      |      |      |      |     |
| Lower Atrium 1st floor & 2nd Floor 1000A, 1202, 2200, 2202, 2204 | No keypad required                        | Automatic by astronomical time schedule         | A4                              | A5   | A6  | A7   | A8   | B1   | B2   | B3   | B4   | B5   | B6   | B7   | B8  |
|                                                                  |                                           | 1. Day                                          | 100%                            | 100% | 40% | 40%  | 100% | 100% | 100% | 100% | 100% | 100% | 100% | 100% | 40% |
|                                                                  |                                           | 2. Twilight                                     | 80%                             | 80%  | 60% | 60%  | 100% | 100% | 100% | 100% | 100% | 100% | 80%  | 80%  | 60% |
|                                                                  |                                           | 3. Evening                                      | 40%                             | 60%  | 25% | 40%  | 100% | 25%  | 25%  | 70%  | 40%  | 50%  | 40%  | 80%  | 25% |
|                                                                  |                                           | 4. Late evening                                 | 0%                              | 10%  | 10% | 10%  | 100% | 25%  | 25%  | 20%  | 0%   | 10%  | 0%   | 10%  | 10% |
|                                                                  |                                           | OFF                                             | 0%                              | 0%   | 0%  | 0%   | 0%   | 0%   | 0%   | 0%   | 0%   | 0%   | 0%   | 0%   | 0%  |
| Lower Atrium 1st Floor 1000A, 1202                               | KP-1 (1) 5 Button Keypad for 1,2,3 & OFF. | Manual override                                 | A4                              | A5   | A6  | A7   | A8   |      |      |      |      |      |      |      |     |
|                                                                  |                                           | 1. Dim Ambient                                  | 10%                             | 20%  | 10% | 20%  | 100% |      |      |      |      |      |      |      |     |
|                                                                  |                                           | 2. Medium Ambient                               | 40%                             | 60%  | 40% | 50%  | 100% |      |      |      |      |      |      |      |     |
|                                                                  |                                           | 3. Bright Ambient                               | 100%                            | 100% | 80% | 100% | 100% |      |      |      |      |      |      |      |     |

Figure 7-4a. Completed enhanced-matrix example: Scene Schedule, page 1. (Image courtesy of Yukiko Yoshida, Auerbach Glasow)

| Room Type and Number                               | Keypad Type                                                                                                         | Scene                                                   | Control Group/Zone & Output (%) |      |      |      |      |                                         |      |     |  |  |
|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|---------------------------------|------|------|------|------|-----------------------------------------|------|-----|--|--|
|                                                    |                                                                                                                     |                                                         | B1                              | B2   | B3   | B4   | B5   | B6                                      | B7   | B8  |  |  |
| Lower Atrium 2nd Floor<br>2200, 2202, 2204         | KP-1<br>(1) 5 Button Keypad for 1, 2, 3 & OFF.                                                                      | Manual override                                         |                                 |      |      |      |      |                                         |      |     |  |  |
|                                                    |                                                                                                                     | 1. Dim Ambient                                          | 30%                             | 10%  | 30%  | 10%  | 20%  | 10%                                     | 20%  | 10% |  |  |
|                                                    |                                                                                                                     | 2. Medium Ambient                                       | 70%                             | 60%  | 70%  | 60%  | 60%  | 40%                                     | 80%  | 40% |  |  |
|                                                    |                                                                                                                     | 3. Bright Ambient                                       | 100%                            | 100% | 100% | 100% | 100% | 100%                                    | 100% | 80% |  |  |
|                                                    |                                                                                                                     | OFF                                                     | 0%                              | 0%   | 0%   | 0%   | 0%   | 0%                                      | 0%   | 0%  |  |  |
| Classroom/<br>Seminar<br>with camera<br>1207, 1211 | KP-3<br>(1) 8 Button Keypad at door for scene 1, 2, 3, 4 & OFF.<br>5 and 6 will be accessed through AV touch panel. | Manual override                                         | C1                              | C2   | C3   | C4   |      |                                         |      |     |  |  |
|                                                    |                                                                                                                     | 1. Dim Ambient                                          | 0%                              | 25%  | 10%  | 10%  |      |                                         |      |     |  |  |
|                                                    |                                                                                                                     | 2. Bright at front and Medium Ambient                   | 0%                              | 100% | 60%  | 60%  |      |                                         |      |     |  |  |
|                                                    |                                                                                                                     | 3. Dim at projection and Medium Ambient                 | 0%                              | 100% | 10%  | 10%  |      |                                         |      |     |  |  |
|                                                    |                                                                                                                     | 4. Bright Ambient for cleaning                          | 50%                             | 100% | 90%  | 90%  |      |                                         |      |     |  |  |
| Classrooms<br>1203, 1205, 1213, 1217, 1215         | KP-2<br>(2) 8 Button Keypads for scenes 1, 2, 3, 4, 5 & OFF.                                                        | 5. Front at Presenter's left bright for Course Capture  | 0%                              | 100% | 10%  | 100% |      |                                         |      |     |  |  |
|                                                    |                                                                                                                     | 6. Front at Presenter's right bright for Course Capture | 0%                              | 100% | 100% | 10%  |      |                                         |      |     |  |  |
|                                                    |                                                                                                                     | OFF                                                     | 0%                              | 0%   | 0%   | 0%   |      |                                         |      |     |  |  |
|                                                    |                                                                                                                     | Manual override                                         | C1                              | C2   | C3   | C4   | C5   |                                         |      |     |  |  |
|                                                    |                                                                                                                     | 1. Dim Ambient                                          | 0%                              | 25%  | 10%  | 10%  | 10%  |                                         |      |     |  |  |
|                                                    |                                                                                                                     | 2. Bright at front and Medium Ambient                   | 0%                              | 100% | 100% | 100% | 100% |                                         |      |     |  |  |
|                                                    |                                                                                                                     | 3. Dim at projection and Medium Ambient                 | 0%                              | 60%  | 0%   | 0%   | 0%   |                                         |      |     |  |  |
|                                                    |                                                                                                                     | 4. Bright at lectern and Medium Ambient                 | 0%                              | 60%  | 100% | 0%   | 0%   | Check which side the lectern is located |      |     |  |  |
|                                                    |                                                                                                                     | 5. Bright Ambient for cleaning                          | 50%                             | 100% | 100% | 100% | 100% |                                         |      |     |  |  |
|                                                    |                                                                                                                     | OFF                                                     | 0%                              | 0%   | 0%   | 0%   | 0%   |                                         |      |     |  |  |

Figure 7-4b. Completed enhanced-matrix example: Scene Schedule, page 2. (Image courtesy of Yukiko Yoshida, Auerbach Glasow)

| Room Type and Number | Keypad Type                                                                                                                                     | Scene                                                        | Control Group/Zone & Output (%) |  |  |  |  |  |  |  |  |  |  |    |    |      |      |      |  |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|---------------------------------|--|--|--|--|--|--|--|--|--|--|----|----|------|------|------|--|
| Lecture Hall 1102    | KP-2<br>(2) 8 Button Keypads at front and back for scene 1,2,3,4 ,5 & OFF.<br><br>6,7, and 8 will be accessed through AV system AV touch panel. | Manual override                                              |                                 |  |  |  |  |  |  |  |  |  |  |    |    |      |      |      |  |
|                      |                                                                                                                                                 | 1. Dim Ambient                                               |                                 |  |  |  |  |  |  |  |  |  |  |    |    | C10  | C11  | C12  |  |
|                      |                                                                                                                                                 | 2. Bright at front and Medium Ambient                        |                                 |  |  |  |  |  |  |  |  |  |  |    |    | 0%   | 0%   | 0%   |  |
|                      |                                                                                                                                                 | 3. Bright at lectern and Medium Ambient                      |                                 |  |  |  |  |  |  |  |  |  |  |    |    | 25%  | 25%  | 25%  |  |
|                      |                                                                                                                                                 | 4. Dim at front for projection and Medium Ambient            |                                 |  |  |  |  |  |  |  |  |  |  |    |    | 0%   | 0%   | 0%   |  |
|                      |                                                                                                                                                 | 5. Bright Ambient for cleaning                               |                                 |  |  |  |  |  |  |  |  |  |  |    |    | 80%  | 80%  | 80%  |  |
|                      |                                                                                                                                                 | 6. Videotaping: Very Bright at front area for Course Capture |                                 |  |  |  |  |  |  |  |  |  |  |    |    | 100% | 100% | 100% |  |
|                      |                                                                                                                                                 | 7. Videotaping: Very Bright at lectern for Course Capture    |                                 |  |  |  |  |  |  |  |  |  |  |    |    | 0%   | 0%   | 100% |  |
|                      |                                                                                                                                                 | 8. Videotaping: Dim at projection for Course Capture         |                                 |  |  |  |  |  |  |  |  |  |  |    |    | 0%   | 0%   | 0%   |  |
| OFF                  |                                                                                                                                                 |                                                              |                                 |  |  |  |  |  |  |  |  |  |  | 0% | 0% | 0%   |      |      |  |
| Lecture Hall 1104    | KP-2<br>(1) 8 Button Keypad at front & back for scenes 1,2,3,4,5 & OFF.<br><br>6,7, and 8 will be accessed through AV system AV touch panel.    | Manual override                                              |                                 |  |  |  |  |  |  |  |  |  |  |    |    |      |      |      |  |
|                      |                                                                                                                                                 | 1. Dim Ambient                                               |                                 |  |  |  |  |  |  |  |  |  |  |    |    |      |      |      |  |
|                      |                                                                                                                                                 | 2. Bright at front and Medium Ambient                        |                                 |  |  |  |  |  |  |  |  |  |  |    |    |      |      |      |  |
|                      |                                                                                                                                                 | 3. Bright at lectern and Medium Ambient                      |                                 |  |  |  |  |  |  |  |  |  |  |    |    |      |      |      |  |
|                      |                                                                                                                                                 | 4. Dim at front for projection and Medium Ambient            |                                 |  |  |  |  |  |  |  |  |  |  |    |    |      |      |      |  |
|                      |                                                                                                                                                 | 5. Bright Ambient for cleaning                               |                                 |  |  |  |  |  |  |  |  |  |  |    |    |      |      |      |  |
|                      |                                                                                                                                                 | 6. Videotaping: Very Bright at front area for Course Capture |                                 |  |  |  |  |  |  |  |  |  |  |    |    |      |      |      |  |
|                      |                                                                                                                                                 | 7. Videotaping: Very Bright at lectern for Course Capture    |                                 |  |  |  |  |  |  |  |  |  |  |    |    |      |      |      |  |
|                      |                                                                                                                                                 | 8. Videotaping: Dim at projection for Course Capture         |                                 |  |  |  |  |  |  |  |  |  |  |    |    |      |      |      |  |
| OFF                  |                                                                                                                                                 |                                                              |                                 |  |  |  |  |  |  |  |  |  |  |    |    |      |      |      |  |

Figure 7-4c. Completed enhanced-matrix example: Scene Schedule, page 3. (Image courtesy of Yukiko Yoshida, Auerbach Glasow)

| Room Type and Number               | Keypad Type                                                                              | Scene                                             | Control Group/Zone & Output (%) |      |     |      |     |      |     |      |      |      |      |     |
|------------------------------------|------------------------------------------------------------------------------------------|---------------------------------------------------|---------------------------------|------|-----|------|-----|------|-----|------|------|------|------|-----|
|                                    |                                                                                          |                                                   | C1                              | C2   | C3  | C4   | C5  | C6   | C7  | C8   | C9   | C10  | C11  | C12 |
| Lecture Halls combined 1102 & 1104 | The keypads in each room will be used. 9 and 10 will be accessed through AV touch panel. | Manual override                                   | 0%                              | 50%  | 0%  | 50%  | 0%  | 50%  | 0%  | 0%   | 0%   | 0%   | 0%   | 0%  |
|                                    |                                                                                          | 1. Dim Ambient                                    | 0%                              | 100% | 0%  | 100% | 0%  | 100% | 25% | 25%  | 25%  | 25%  | 25%  | 25% |
|                                    |                                                                                          | 2. Bright at front and Medium Ambient             | 0%                              | 100% | 0%  | 100% | 0%  | 100% | 0%  | 0%   | 80%  | 0%   | 0%   | 80% |
|                                    |                                                                                          | 3. Bright at lecterns and Medium Ambient          | 0%                              | 60%  | 0%  | 60%  | 0%  | 60%  | 0%  | 0%   | 0%   | 0%   | 0%   | 0%  |
|                                    |                                                                                          | 4. Dim at front for projection and Medium Ambient | 80%                             | 100% | 80% | 100% | 80% | 100% | 80% | 80%  | 80%  | 80%  | 80%  | 80% |
|                                    |                                                                                          | 5. Bright Ambient for cleaning and Science Fair   | 0%                              | 100% | 0%  | 100% | 0%  | 100% | 2%  | 100% | 100% | 100% | 100% | 0%  |
|                                    |                                                                                          | 9. Panel discussion and Medium Ambient            | 0%                              | 60%  | 0%  | 60%  | 0%  | 60%  | 0%  | 0%   | 80%  | 80%  | 0%   | 0%  |
|                                    |                                                                                          | 10. Bright at one Lectern and Medium Ambient      | 0%                              | 0%   | 0%  | 0%   | 0%  | 0%   | 0%  | 0%   | 0%   | 0%   | 0%   | 0%  |
|                                    |                                                                                          | OFF                                               | 0%                              | 0%   | 0%  | 0%   | 0%  | 0%   | 0%  | 0%   | 0%   | 0%   | 0%   | 0%  |
|                                    |                                                                                          |                                                   |                                 |      |     |      |     |      |     |      |      |      |      |     |

Note1: Scene schedule for astronomical time switch:  
 "Day" Scene starts: 6am  
 "Twilight" Scene starts: One hour before Sunset  
 "Evening" Scene starts: One hour after Sunset  
 "Late evening" starts: 9:40pm

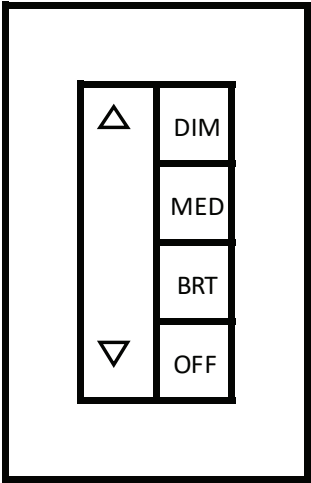
Note2: Sunrise and Sunset in Berkeley CA  
 Date Sunrise Sunset  
 March 20 7:12am 7:21pm  
 June 21 5:47am 8:35pm  
 September 22 6:57am 7:05pm  
 December 21 7:21am 4:53pm

Note3: Control group/zones with red text are controlled by photosensor

Note4: Control group/zones with gray shade are controlled by occupancy sensor

Figure 7-2d. Preliminary enhanced-matrix example: Scene Schedule, page 4. (Image courtesy of Yukiko Yoshida, Auerbach Glasow)

CONTROL KEY PADS CD phase  
Day of issue: XXXX

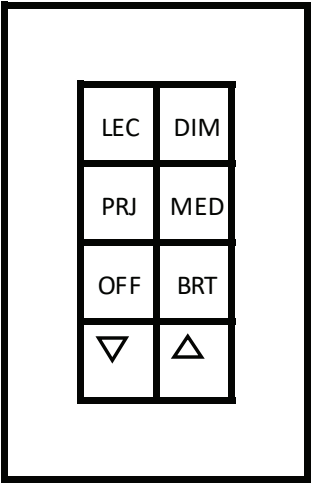


KP-1

Entry Hall

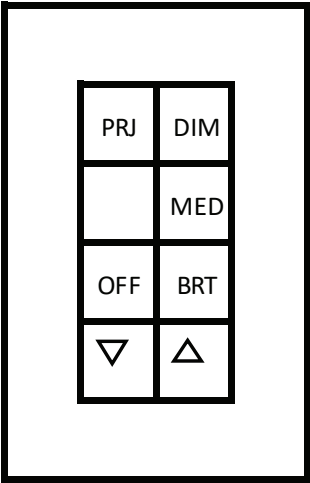
Lower Atrium - 1FL

Lower Atrium - 2FL



KP-2

Classrooms  
& Lecture Halls



KP-3

Classrooms/Seminar  
w/camera

Figure 7-5. Completed enhanced-matrix example: Control key pads. (Image courtesy of Yukiko Yoshida, Auerbach Glasow)

| Sample Hotel - Sequence of Operations                                                                                                                                                                                                  |                  |                         |                            |                                                                                                                                                                                                                                                                                                                                                                                   |                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-------------------------|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| Day                                                                                                                                                                                                                                    | Public Hours     | Staff Hours             | "Day" Schedule             | "Evening" Schedule                                                                                                                                                                                                                                                                                                                                                                | "Night" Schedule |
| Monday - Friday                                                                                                                                                                                                                        | 24 hrs           | 24 hrs                  | 5:00am to Sunset-60min     | Sunset-60min to 1:00am                                                                                                                                                                                                                                                                                                                                                            | 1:00am-5:00am    |
| Saturday                                                                                                                                                                                                                               | 24 hrs           | 24 hrs                  | 5:00am to Sunset-60min     | Sunset-60min to 1:00am                                                                                                                                                                                                                                                                                                                                                            | 1:00am-5:00am    |
| Sunday                                                                                                                                                                                                                                 | 24 hrs           | 24 hrs                  | 5:00am to Sunset-60min     | Sunset-60min to 1:00am                                                                                                                                                                                                                                                                                                                                                            | 1:00am-5:00am    |
| Vacancy Timeout                                                                                                                                                                                                                        | Override Timeout | Sunset Source           | BAS Connection             | User Level Setting                                                                                                                                                                                                                                                                                                                                                                | Sweep Time       |
| 15 minutes                                                                                                                                                                                                                             | 2 hours          | Astronomical Time Clock | Add Alternate              | Via PC and Tablet UI                                                                                                                                                                                                                                                                                                                                                              | 1:00am           |
| Space                                                                                                                                                                                                                                  | System           | Interfaces              | Sequence of Operation      |                                                                                                                                                                                                                                                                                                                                                                                   |                  |
| Exterior Site                                                                                                                                                                                                                          | Central Control  | Astronomical Time Clock | Day:<br>Evening:<br>Night: | Lights Shall be Off<br>Lights Shall be On at Defined Levels<br>Lights Shall be On at Defined Levels - reduced output                                                                                                                                                                                                                                                              |                  |
| Vestibule<br>Lobby<br>Bar Lounge<br>Reception<br>Gallery & Entry<br>Prefunction<br>Parlors<br>Wine Room<br>FOH Corridors<br><br>100, 101, 102, 103, 118,<br>125, 126, 127, 128, 129,<br>B2-100, B2-101, B2-102,<br>M-100, M-101, M-107 | Central Control  | Astronomical Time Clock | Day:<br>Evening:<br>Night: | Lights Shall be On at Defined Levels - high level of output<br>Lights Shall be On at Defined Levels - reduced level out<br>output. Tuned to evening level of illumination with adequate<br>levels for normal operation.<br>Lights Shall be On at Defined Levels - reduced level out<br>output. Tuned to night level of illumination with adequate<br>levels for normal operation. |                  |

Figure 7-6a. Sequence of Operations controls matrix, page 1. (Image courtesy of Casey Diers / Morlights)



| Space                               | System          | Interfaces                                                                                | Sequence of Operation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------------------|-----------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Banquet Room<br>111-1<br>121<br>122 | Central Control | Keypad<br>Occupancy Sensor<br>Photocell<br>Astronomical Time Clock<br>Room Combine Sensor | <p>Day <input type="checkbox"/>: Keypad<br/> <input type="checkbox"/>: Vacancy Timeout<br/> Evening <input type="checkbox"/>: Keypad<br/> <input type="checkbox"/>: Vacancy Timeout<br/> Night <input type="checkbox"/>: Keypad<br/> <input type="checkbox"/>: Vacancy Timeout</p> <p>All:<br/> - When top keypad button "on" is triggered, that shall start a timer for a vacancy timeout. All other presets shall override vacancy timeout and photocells for specialty programming.<br/> - Any rooms left in an override state at the end of day shall reset at sweep time.<br/> - Room combine sensor shall be utilized. When rooms are combined, keypads adjust both rooms. When rooms are separated, keypads adjust local room only.</p> |
| Dining Room<br>107<br>109           | Central Control | Keypad<br>Occupancy Sensor<br>Photocell<br>Astronomical Time Clock                        | <p>Day:<br/> Evening:<br/> - Lights Shall be On at Defined Levels - high level of output<br/> - Lights Shall be On at Defined Levels - reduced level out output. Tuned to evening level of illumination with adequate levels for normal operation.<br/> Night:<br/> - Lights Shall be On at Defined Levels - reduced output for hours room is not occupied<br/> - Room shall employ daylight harvesting by reducing electric light levels proportionally to the amount of daylight entering via exterior windows.</p>                                                                                                                                                                                                                          |

Figure 7-6b. Sequence of Operations controls matrix, page 2. (Image courtesy of Casey Diers / Morlights)

| Space                                                                         | System          | Interfaces                                                         | Sequence of Operation                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------------------------------------------------------------|-----------------|--------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Meeting Room<br>M-108<br>M-109                                                | Central Control | Keypad<br>Occupancy Sensor<br>Photocell<br>Astronomical Time Clock | Day <input type="checkbox"/> : Keypad<br><input type="checkbox"/> : Vacancy Timeout / Keypad<br>Evening <input type="checkbox"/> : Keypad<br><input type="checkbox"/> : Vacancy Timeout / Keypad<br>Night <input type="checkbox"/> : Keypad<br><input type="checkbox"/> : Vacancy Timeout / Keypad<br>-<br>- Room shall employ daylight harvesting by reducing electric light levels proportionally to the amount of daylight entering via exterior windows.<br>- |
| Fitness Center<br>B2-106                                                      | Central Control | Astronomical Time Clock                                            | Day <input type="checkbox"/> : Occupancy Sensor / Keypad<br><input type="checkbox"/> : Vacancy Timeout<br>Evening <input type="checkbox"/> : Occupancy Sensor / Keypad<br><input type="checkbox"/> : Vacancy Timeout<br>Night <input type="checkbox"/> : Occupancy Sensor / Keypad<br><input type="checkbox"/> : Vacancy Timeout                                                                                                                                  |
| FOH Toilets<br>105<br>119<br>120<br>130<br>B2-103<br>B2-104<br>M-102<br>M-103 | Central Control | Button Station<br>Occupancy Sensor                                 | Day <input type="checkbox"/> : Occupancy Sensor / Keypad<br><input type="checkbox"/> : Vacancy Timeout<br>Evening <input type="checkbox"/> : Occupancy Sensor / Keypad<br><input type="checkbox"/> : Vacancy Timeout<br>Night <input type="checkbox"/> : Occupancy Sensor / Keypad<br><input type="checkbox"/> : Vacancy Timeout                                                                                                                                  |
| FOH Support & Storage<br>104                                                  | Central Control | Keypad<br>Occupancy Sensor                                         | Day <input type="checkbox"/> : Occupancy Sensor / Keypad<br><input type="checkbox"/> : Vacancy Timeout<br>Evening <input type="checkbox"/> : Occupancy Sensor / Keypad<br><input type="checkbox"/> : Vacancy Timeout<br>Night <input type="checkbox"/> : Occupancy Sensor / Keypad<br><input type="checkbox"/> : Vacancy Timeout                                                                                                                                  |

Figure 7-6c. Sequence of Operations controls matrix, page 3. (Image courtesy of Casey Diers / Morlights)

| Space                                                                                                                                                    | System                        | Interfaces                         | Sequence of Operation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Guestroom Corridor<br>269<br>369<br>469<br>569                                                                                                           | No Control                    | No Control                         | On At All Times                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Guestroom                                                                                                                                                | Local Switching Master Switch | Local Switching Master Switch      | See Engineering Documents<br>Local switches per plans. All circuits in each room to be fed from room master relay with occupancy sensing capabilities, as specified in engineering documents.                                                                                                                                                                                                                                                                                                                       |
| Stairs<br>ST1<br>ST2<br>ST3<br>ST4                                                                                                                       | Local Control                 | Integral Occupancy Sensor          | <div> Day <input type="checkbox"/>: Occupancy Sensor<br/> <input type="checkbox"/>: Vacancy Timeout<br/> Evening <input type="checkbox"/>: Occupancy Sensor<br/> <input type="checkbox"/>: Vacancy Timeout<br/> Night <input type="checkbox"/>: Occupancy Sensor<br/> <input type="checkbox"/>: Vacancy Timeout </div> <div> - Every other light has an integrated occupancy sensor. Lights with sensors dim down to 10% when unoccupied. Lights without sensors shall remain on at all times.<br/> -<br/> - </div> |
| Back of House<br>Offices<br>Storage<br>Back of House Toilet<br>Cafeteria<br>Housekeeping<br>Administrative<br>Food and Bev Storage<br>Engineering Office | Local Control                 | Button Station<br>Occupancy Sensor | <div> Day <input type="checkbox"/>: Button Station<br/> <input type="checkbox"/>: Vacancy Timeout<br/> Evening <input type="checkbox"/>: Button Station<br/> <input type="checkbox"/>: Vacancy Timeout<br/> Night <input type="checkbox"/>: Button Station<br/> <input type="checkbox"/>: Vacancy Timeout </div>                                                                                                                                                                                                    |
| Back of House Corridor                                                                                                                                   | Local Control                 | Occupancy Sensor                   | <div> Day <input type="checkbox"/>: Occupancy Sensor<br/> <input type="checkbox"/>: Vacancy Timeout<br/> Evening <input type="checkbox"/>: Occupancy Sensor<br/> <input type="checkbox"/>: Vacancy Timeout<br/> Night <input type="checkbox"/>: Occupancy Sensor<br/> <input type="checkbox"/>: Vacancy Timeout </div>                                                                                                                                                                                              |

Figure 7-6d. Sequence of Operations controls matrix, page 4. (Image courtesy of Casey Diers / Morlights)

| Space                                          | System          | Interfaces                             | Sequence of Operation                                                                                                                                                                                                                                                                                                                                             |
|------------------------------------------------|-----------------|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Parking Garage                                 | Local Control   | Integral Occupancy Sensor              | Day <input type="checkbox"/> : Occupancy Sensor<br><input type="checkbox"/> : Vacancy Timeout<br>Evening <input type="checkbox"/> : Occupancy Sensor<br><input type="checkbox"/> : Vacancy Timeout<br>Night <input type="checkbox"/> : Occupancy Sensor<br><input type="checkbox"/> : Vacancy Timeout<br><br>Lights with sensors dim down to 50% when unoccupied. |
| Kitchen<br>111                                 | Local Control   | Local Toggle Switches                  | Toggle switches only for life safety.                                                                                                                                                                                                                                                                                                                             |
| Electrical, Plumbing and Fire Protection Rooms | Local Control   | Timer Switches                         | Timer switches only for life safety.                                                                                                                                                                                                                                                                                                                              |
| Lobby<br>Vestibule<br>A100<br>A101<br>R109     | Central Control | BAS Contact<br>Astronomical Time Clock | Day: Lights Shall be On at Defined Levels - high level of output<br>Evening: Lights Shall be On at Defined Levels - reduced level of output. Tuned to evening level of illumination with adequate levels for normal operation.<br>Night: Lights Shall be On at Defined Levels - reduced output                                                                    |

Figure 7-6e. Sequence of Operations controls matrix, page 5. (Image courtesy of Casey Diers / Morlights)

| Space                                   | System          | Interfaces                                                                                                        | Sequence of Operation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------------|-----------------|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Program Rooms<br>A113<br>A114           | Central Control | Keypad<br>Occupancy Sensor<br>Photocell<br>Astronomical Time Clock<br>Room Combine Sensor<br>AV System Connection | <p>Day <input type="checkbox"/>: Keypad<br/> <input type="checkbox"/>: Vacancy Timeout<br/> Evening <input type="checkbox"/>: Keypad<br/> <input type="checkbox"/>: Vacancy Timeout<br/> Night <input type="checkbox"/>: Keypad<br/> <input type="checkbox"/>: Vacancy Timeout</p> <p>All:<br/> - Lighting control system and AV system to share room combine triggers. When rooms are combined, keypads should trigger presets in both rooms. When separated, keypads should trigger presets only in local room.<br/> - Room shall employ daylight harvesting by reducing electric light levels proportionally to the amount of daylight entering via exterior windows.<br/> - When top keypad button "on" is triggered, that shall start a timer for a vacancy timeout. All other presets shall override vacancy and photocell timeout for specialty programming. Any rooms left in an override state at the end of day shall reset at sweep time.</p> |
| Imaginarium<br>A115<br>A115.A<br>A115.B | Central Control | Keypad<br>Occupancy Sensor<br>Photocell<br>Astronomical Time Clock<br>Room Combine Sensor<br>AV System Connection | <p>Day <input type="checkbox"/>: Keypad<br/> <input type="checkbox"/>: Vacancy Timeout<br/> Evening <input type="checkbox"/>: Keypad<br/> <input type="checkbox"/>: Vacancy Timeout<br/> Night <input type="checkbox"/>: Keypad<br/> <input type="checkbox"/>: Vacancy Timeout</p> <p>All:<br/> - Room shall employ daylight harvesting by reducing electric light levels proportionally to the amount of daylight entering via exterior windows.<br/> - When top keypad button "on" is triggered, that shall start a timer for a vacancy timeout. All other presets shall override vacancy timeout and photocells for specialty programming.<br/> - Any rooms left in an override state at the end of day shall reset at sweep time.</p>                                                                                                                                                                                                                |

Figure 7-6f. Sequence of Operations controls matrix, page 6. (Image courtesy of Casey Diers / Morlighs)

| Space                                                                                                                                                                           | System        | Interfaces                         | Sequence of Operation                                                                                                                                                                                                                                                                                                            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Toilets<br>R106<br>R122.A<br>R122.B                                                                                                                                             | Local Control | Button Station<br>Occupancy Sensor | Day <input type="checkbox"/> : Occupancy Sensor / Keypad<br><input type="checkbox"/> : Vacancy Timeout<br>Evening <input type="checkbox"/> : Occupancy Sensor / Keypad<br><input type="checkbox"/> : Vacancy Timeout<br>Night <input type="checkbox"/> : Occupancy Sensor / Keypad<br><input type="checkbox"/> : Vacancy Timeout |
| Back of House<br>Offices<br>Storage<br><br>R114.A<br>R116<br>R120<br>R123<br>R130<br>A104<br>A104.A<br>A105<br>A106<br>A107<br>A108<br>A110<br>A115.C<br>A115.D<br>X153<br>X157 | Local Control | Button Station<br>Occupancy Sensor | Day <input type="checkbox"/> : Keypad<br><input type="checkbox"/> : Vacancy Timeout<br>Evening <input type="checkbox"/> : Keypad<br><input type="checkbox"/> : Vacancy Timeout<br>Night <input type="checkbox"/> : Keypad<br><input type="checkbox"/> : Vacancy Timeout                                                          |
| Back of House Corridor                                                                                                                                                          | Local Control | Occupancy Sensor                   | Day <input type="checkbox"/> : Occupancy Sensor<br><input type="checkbox"/> : Vacancy Timeout<br>Evening <input type="checkbox"/> : Occupancy Sensor<br><input type="checkbox"/> : Vacancy Timeout<br>Night <input type="checkbox"/> : Occupancy Sensor<br><input type="checkbox"/> : Vacancy Timeout                            |

Figure 7-6g. Sequence of Operations controls matrix, page 7. (Image courtesy of Casey Diers / Morlights)

## Glossary

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This glossary includes an abridged list of relevant terms. Other lighting and controls terms may be found in *ANSI/IES LS-1-20, Lighting Science: Nomenclature and Definitions for Illuminating Engineering*.<sup>4</sup> For terms not found in this glossary or in ANSI/IES LS-1-21, additional resources include *ANSI C137.0-2017, Lighting Systems – Lighting Systems Terms and Definitions*<sup>5</sup> and *NEMA LSD 64-2019, Lighting Controls Terminology*.<sup>6</sup>

**absence.** (See *unoccupied*.)

**acceptance test.** A formal action, taken by a person with appropriate authority (which may or may not be contractually defined) to declare that some aspect of the project meets defined requirements, thus permitting subsequent activities to proceed.

**blink warning.** A lighting control sequence where the controls will turn off the lighting for a brief moment (usually for one second or less), notifying space occupants that the lighting will soon be automatically turned off unless manually overridden.

**commissioning.** The process of verifying and documenting that a building and all its systems and assemblies are planned, designed, installed, tested, operated, and maintained to meet the owner's project requirements.

**component-based controls.** Lighting-control devices that operate as self-contained units to carry out a complete control sequence without the need for communication with other control devices or systems. Also called *line-voltage controls* or *standalone controls*.

**contact closure interface.** (See *dry contact*.)

**Control Intent Narrative (CIN).** A description of how a function or system is expected to work, but without the detail one would need to program it. Essentially, it is the "what" a system is intending to accomplish.

**curfew.** As used in outdoor lighting: A period (typically between 11:00 p.m. and 7:00 a.m.) defined by a standard

or statute when the outdoor lighting is required to be reduced or turned off to reduce light trespass and pollution.

**daily scheduling.** (See *scheduling*.)

**design intent.** Written specification for the lighting and controls system providing the design criteria, control strategies, and other system requirements.

**demand response.** A control strategy for selectively reducing the load of a system on a temporary basis to reduce energy usage. A building manager or system may utilize demand response to avoid peak pricing or to avoid a condition where electricity demand exceeds supply. In lighting, a lighting control system would reduce or turn off the output of designated lighting in a demand-response event. Also called *load shedding*.

**dry contact.** An electromagnetic relay interface between two components or systems that provides on-state or off-state signal output from one component or system to the other. The electromagnetic relay-interface output is electrically isolated from the component or /system receiving the input, thereby providing no voltage or current, as this is provided from the input component or system. Also called a *contact closure interface*.

**dusk-to-dawn lighting.** Exterior lighting that turns on at or around dusk and off at or around dawn, remaining on all through the night.

**entertainment lighting.** Lighting specific to an entertainment application, such as lighting directed at a stage and/or performers.

**functional test.** Testing of an installed lighting control system whereby the devices and system are tested against the functional requirements and specifications to ensure that the requirements are properly satisfied by individual-device and system operation.

**integral combination sensor.** A control device with more than one function, comprising photocell, motion sensing, dimming, humidity, and/or audio control.

**line-voltage controls.** (See *component-based controls*.)



**low-end trim.** A lighting control strategy that sets the minimum available light output for each unit of control.

**occupancy sensing.** A lighting control strategy that regulates the operation of lighting or other equipment based on detecting the presence or absence of people within a space.

**occupied.** The state of a room or space when one or more occupants are present. The term *presence* is sometimes used instead.

**outdoor motion sensor.** An occupancy sensor rated for outdoor use. It typically is pole or luminaire mounted.

**Owner's Project Requirements (OPR).** A written document that details the functional requirements of a project and the expectations of how it will be used and operated. These include project goals, measurable performance criteria, cost considerations, benchmarks, success criteria, and supporting information.

**part-night outdoor lighting control.** A lighting control strategy that regulates the operation of a portion of outdoor lighting power by time schedules based on hours of operation and frequency of occupants in the outdoor area. Example: Dimming parking lot lighting between the hours of midnight and 6 a.m. when adjoining facilities are closed or have minimal operations.

**presence.** (See *occupied*.)

**room-based controls.** A schema of lighting control-system structure that has most or all lighting elements and controls within the same room. In this system, the control logic exists local to devices or a "room controller" located at or near the luminaires, keypads, and/or sensors. It may be connected to a central processor or building automation system, although this is not a necessary requirement.

**scheduling.** A control strategy that controls lighting, equipment, or systems based on time of day or astronomical event. For example, scheduling building lighting to be automatically turned off at 6 p.m. or at sunset.

**Sequence of Operation (SOO).** A detailed guide on how

to achieve the required control intent. This information can be used as a directive in the programming phase and should include specific information one can program to, including setpoints, levels, and other specific functional parameters. It is the prescriptive "how" a system accomplishes the description provided in the Control Intent Narrative.

**standalone controls.** (See *component-based controls*.)

**start-up.** The part of the commissioning process when the control equipment is configured onsite for a project.

**sweep (as in *lighting sweep* or *sweep-off*).** A scheduled control event where an entire area or building is commanded to a specific level or preset, usually off. Typically used after closing to ensure that lights accidentally left on do not stay on overnight. (This term originates from lighting controls in the 1970s and 1980s, where a product, called a "Motor-Master," would mechanically "sweep" across a set of electrical contacts to turn lighting on and off in sequence.)

**switch leg.** That portion of a lighting branch circuit between a controlling device (e.g., switch, relay, dimmer) and the luminaire.

**theatrical lighting.** (See *entertainment lighting*.)

**warm dim.** A property of an LED luminaire or lamp whereby the CCT shifts so that it appears "warmer" as luminous intensity is reduced. This property enables LEDs to mimic the dimming behavior of incandescent lamps but does not allow the user to adjust CCT separately from luminous intensity. It is, therefore, a property of the luminaire, not of the control system.

**vacant.** (See *unoccupied*.)

**unoccupied.** The state of a room or space when no occupant is present. The terms *vacant* or *absence* are sometimes used instead.

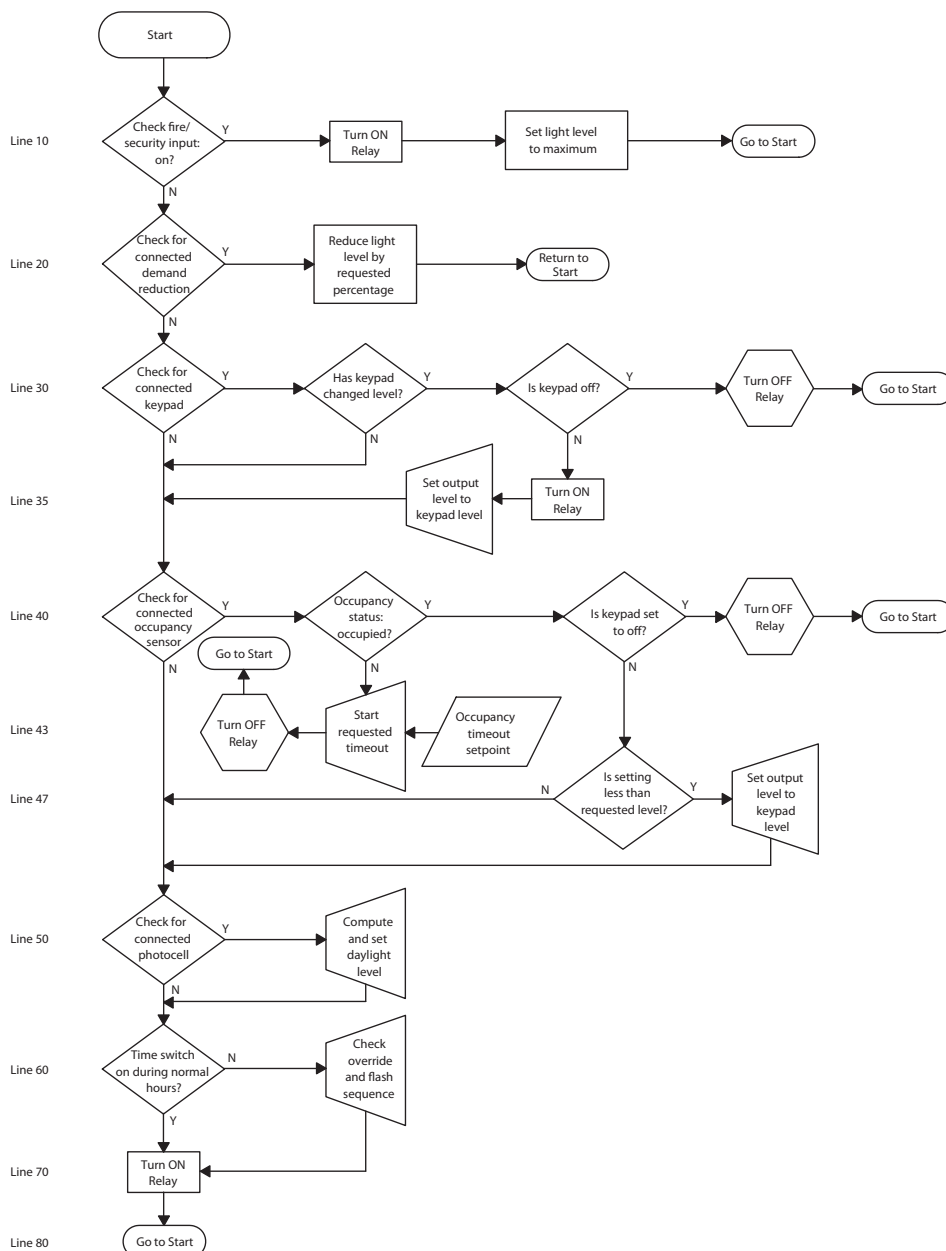
**verification.** The process by which specific documents, components, equipment, assemblies, systems, and interfaces among systems are confirmed to comply with the criteria described in the Owner's Project Requirements (OPR).

## Annex A – Lighting Control Sequence of Operation

The manufacturers of lighting control products provide systems that are able to meet the required Sequence of Operations (SOO) specified on projects. This necessitates that the hardware and software of the lighting controller be capable of accommodating a wide variety of sequences. On any single project, there may be several different sequences required to satisfy the

specified or code-required operation. To ensure that what has been specified can be provided, it is important that the system controller be easily programmable for multiple scenarios. This means the sequence of operation of the controller is more complex than any particular project SOO. This level of complexity is not something the lighting designer or specifying engineer needs to understand or deal with.

An example of a flow diagram of the software logic for a lighting controller is shown in **Figure A-1**. This example



**Figure A-1. Flow diagram for a control system sequence of operations.** (Courtesy of Jim Yorgey)

is simplified to provide a sense of the decisions that need to be made. An important element is recognizing what devices are actually connected to the controller. For example, a daylight sensor might not be required in the particular space. Therefore, that portion of the logic flow would be ignored. This particular logic flow also ensures that the lowest requested illuminance is maintained. Additional logic elements, not shown, can provide for time switch, occupancy sensor, and daylight sensor settings and adjustments.

## Annex B – Control Diagrams

Control diagrams provide a graphical representation of the underlying system that supports the SOO. The connection between the SOO and the lighting control system hardware is not obvious. It is crucial to include diagram to ensure that the required equipment and wiring will be installed. Furthermore, it is expensive to install networking equipment and the wiring the runs between floors and back to the central server. This work is often performed under short deadlines for design or bidding. Vendors and contractors often omit this step unless a diagram is included. Diagrams allow them to quickly understand the hardware and include the required equipment and wiring, in addition to

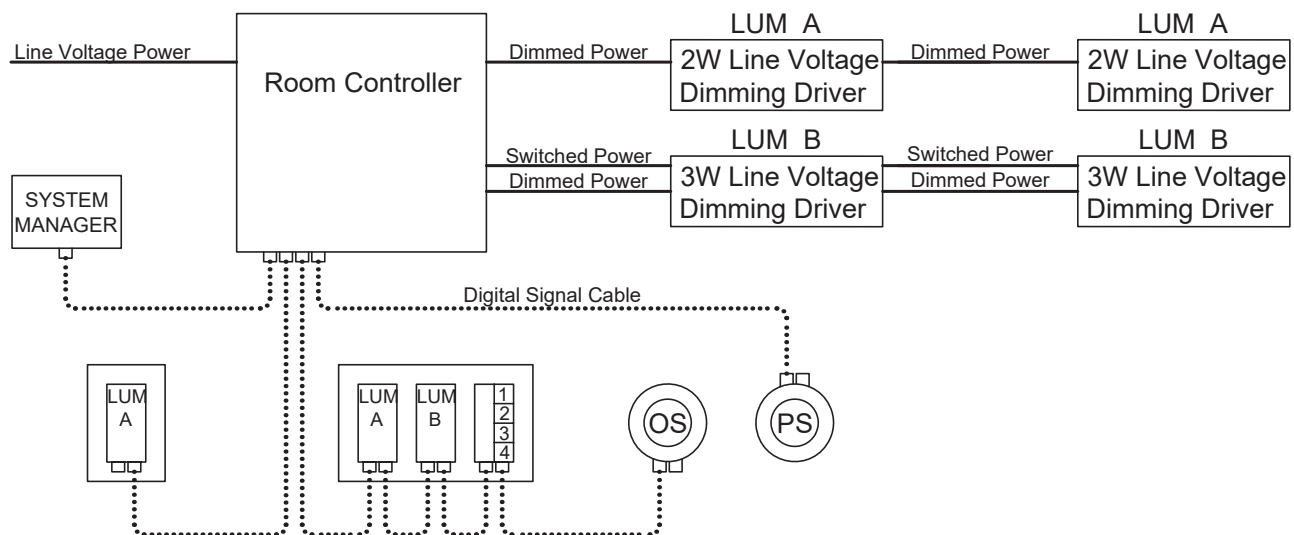
ensuring that there are graphically explicit, contractually enforceable requirements. A diagram will reduce the potential for misunderstandings and help to ensure that the requirements of the SOO are enacted with minimal need for change orders.

Several examples of wiring diagrams for lighting control systems are provided in this annex.

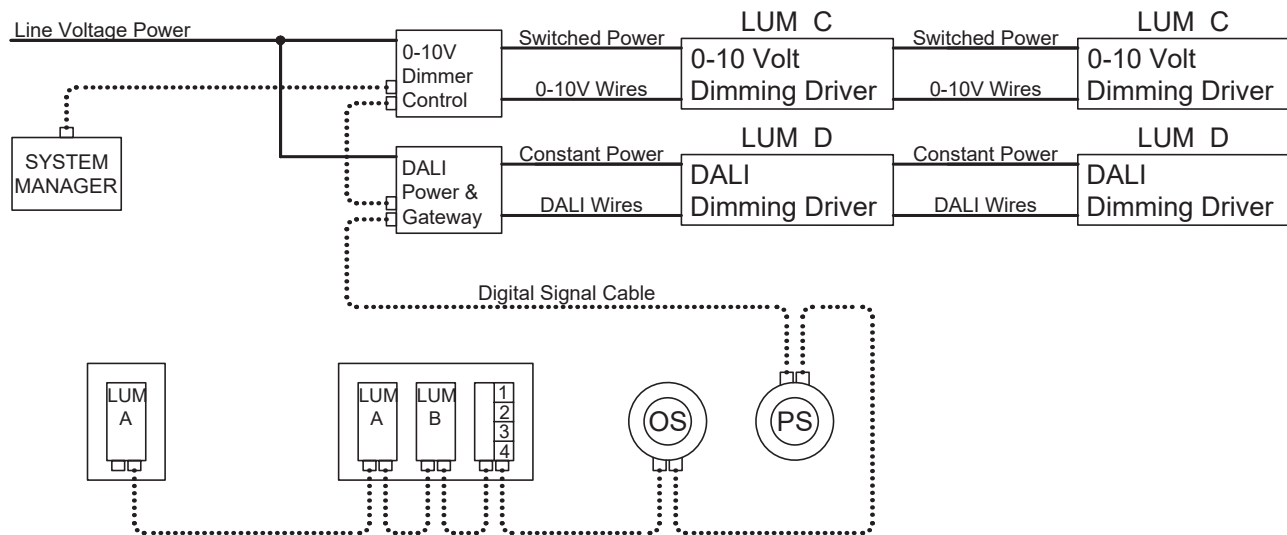
### B.1 Conference Rooms

SOOs for conference rooms with scene controls need to be reinforced by diagrams, including details for scene diagrams. Examples of wiring diagrams for a small conference room are illustrated in **Figures B-1a and B-1b**. Each diagram shows a keypad station for on/off/raise/lower control of the luminaires designated “LUM A”; a second keypad station for on/off/raise/lower control of the LUM A luminaires, on/off/raise/lower control of the LUM B luminaires, and a 4-button scene-select keypad; an occupancy sensor; and a photosensor. Both diagrams have the same sequence of operation. It is important that the SOO be in harmony with the scene keypad provided. Specific details should include the labels for the keypad (see **Figure 7-5 in Section 7.5.1.2 Completed Enhanced Matrix #1** for an example).

Each diagram also shows a System Manager, which would be required if the SOO for the conference room called for an automatic demand response as sent from



**Figure B-1a. Example of a wiring diagram for a small conference room using a room controller. (“2W” = 2-wire; “3W” = 3-wire.)** (Graphic courtesy of Richard N. Miller and Javid Butler)



**Figure B-1b. Example of a wiring diagram for a small conference room using zone controllers.** (Graphic courtesy of Richard N. Miller and Javid Butler)

the utility company, or if the SOO called for a response from a master time schedule. If the SOO did not require a response to such global commands, then the System Manager would not be required for this application.

The room-controller diagram shown in **Figure B-1a** depicts a single dual-channel universal line-voltage controller that allows for the individual control of the LUM A luminaires and the LUM B luminaires. In this diagram, the LUM A luminaires have two-wire (2W) phase-cut dimming drivers, while the LUM B luminaires have three-wire (3W) phase-cut dimming drivers. Also shown is the digital signal cable installed in a free-form topology, as is allowed by that product line.

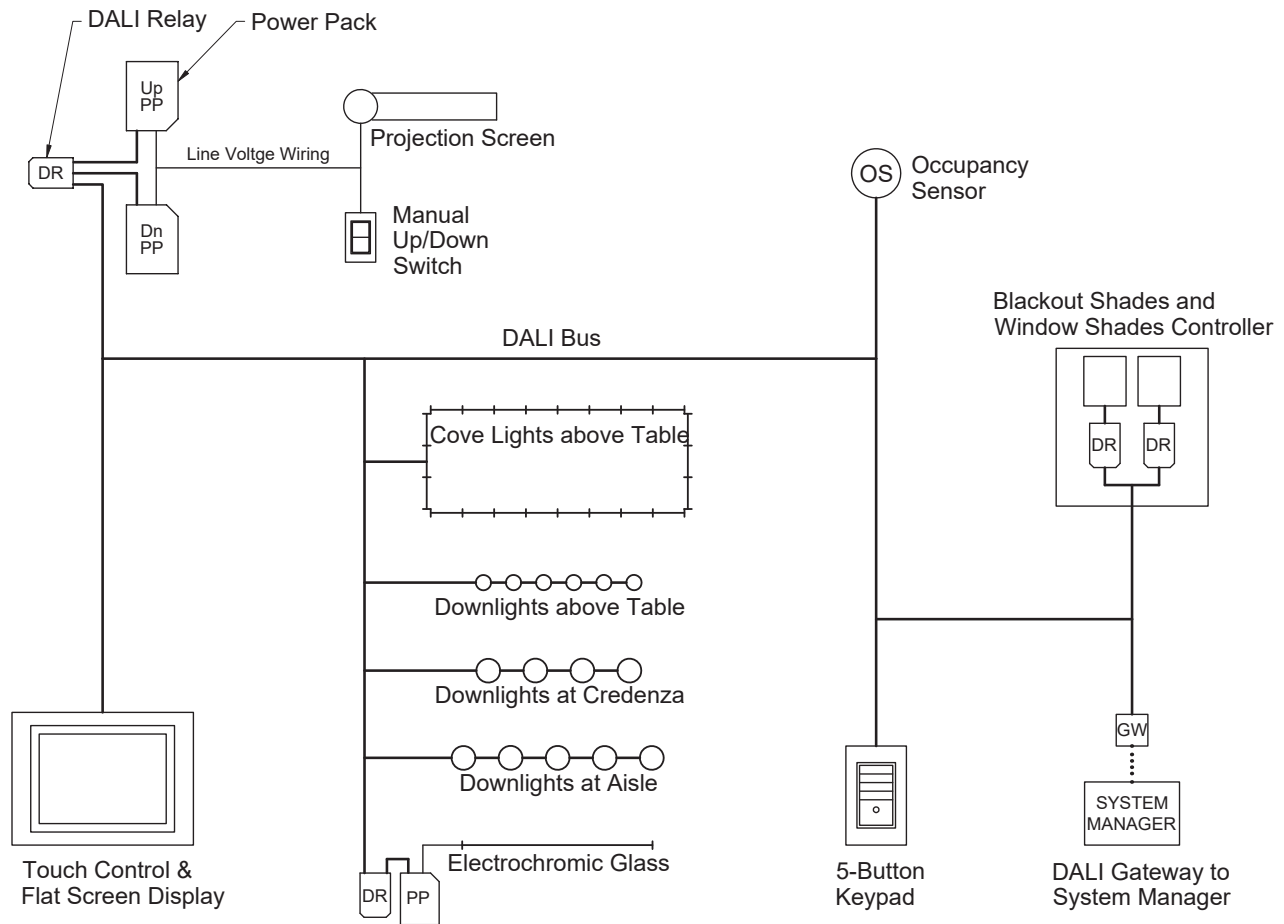
The zone-controller diagram shown in **Figure B-1b** depicts two single-channel controllers for the individual control of the LUM A luminaires and the LUM B luminaires. In this diagram, the LUM A luminaires have 0-10V dimming drivers, while the LUM B luminaires have DALI dimming drivers. The controllers are required to be compatible with their respective luminaires. Also shown is the digital signal cable installed in a strict daisy-chain topology, as is required by that product line.

## B.2 Corporate Boardroom

An illustration of the extent of a single DALI lighting-control bus (sometimes referred to as a loop) in a corporate board room is shown in **Figure B-2**. The

rectangle in the center of the diagram represents a quantity of 22 4-ft strip lights in a cove above the conference table. Below that are shown 6 MR16 downlights above the table, 4 downlights above a credenza, 5 downlights in the aisle, and a DALI relay (DR) and power pack relay (PP) for the electrochromic glass window wall (used to provide visual privacy when needed). Along the right side of the diagram are shown an occupancy sensor (OS), 2 DALI relays for the control of the blackout shades and the window shades, a 5-button scene-select keypad (one of the scenes is All Off), and a gateway between the DALI bus in the board room and the corporate lighting controls System Manager. At the lower left side of the diagram is shown an interactive video display and touchscreen controller, and at the top left is shown a DALI relay controlling two power pack relays for the up/down control of the projection screen. The bold line represents the DALI bus, which physically consists of a pair of #18 AWG conductors. In summary, there are 41 controlled points (DALI luminaires and DALI relays), and three controlling points (keypad, occupancy sensor, and touchscreen).

It is important that additional details for the scene-select keypad and touchscreen controller also be provided, and that the SOO references this equipment and its details and provides instructions for programming the features of the equipment.



**Figure B-2. Example of DALI lighting control for a corporate boardroom.** (Image courtesy of Richard N. Miller)

### B.3 Multistory Building Wireless Lighting Control System

The extent of wiring required in a representative wireless lighting control system is illustrated in **Figure B-3**. Of all the cases for adding a network diagram, this example is most illustrative of its importance. Although the system in **Figure B-3** is wireless, the integration with the other systems for demand response and back-end integration of the HVAC control is not wireless, because the system has to distribute these commands to the local level. (Wireless commands from a central server to equipment across an entire floor and to upper floors are unreliable.)

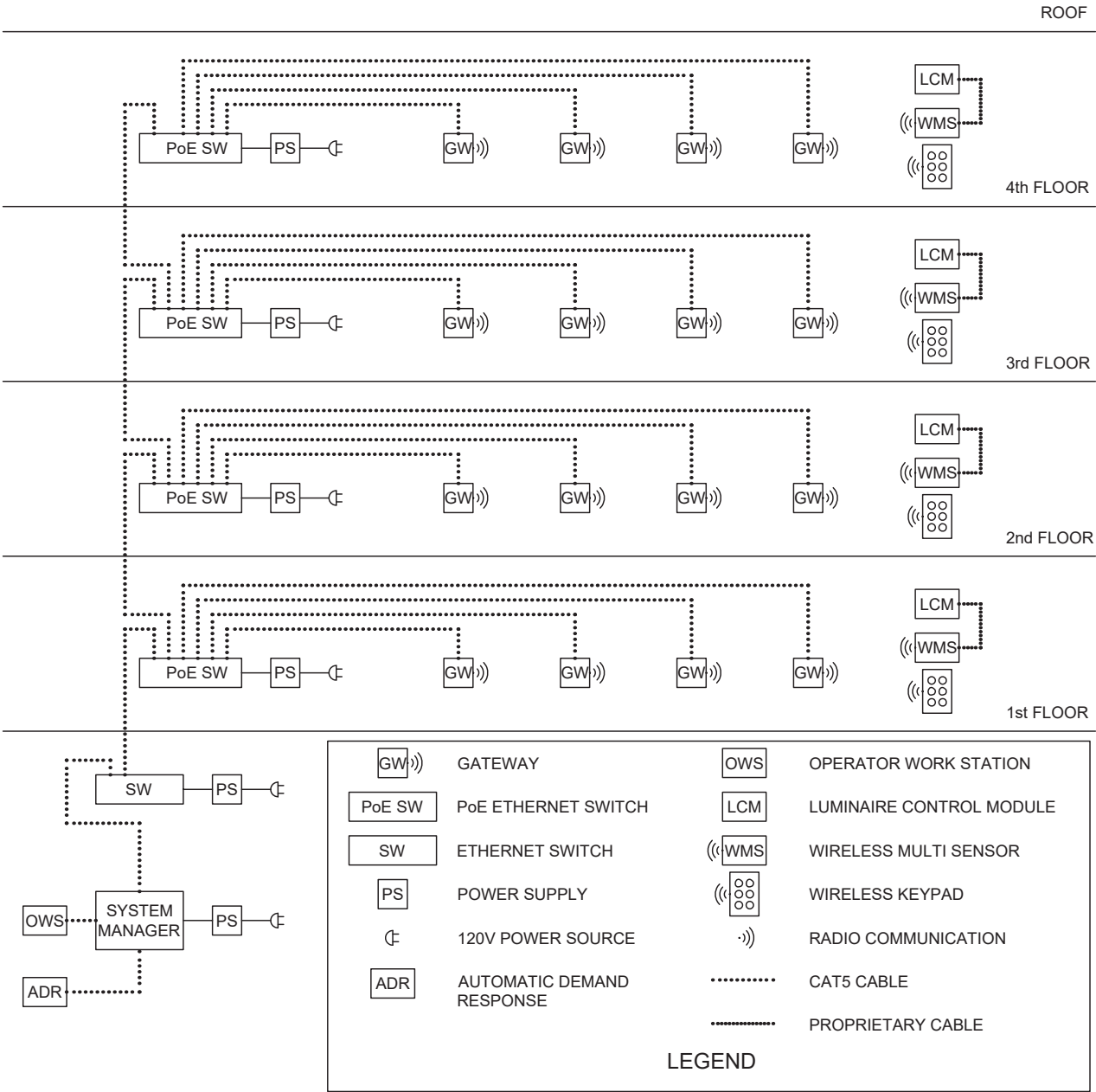
It is crucial to include a network diagram; vendors and contractors often fail to realize that this strategy requires a networked topology based on the SOO alone. Furthermore, it might not be included automatically because it adds significant cost to the project. Areas not

connected to the central server will not be able to enact sequences that require networked commands.

The gateways provide two-way communication between the sensor units and the system manager. The CAT5 cables provide the power to operate the gateways, where the power is obtained from the Power over Ethernet (PoE) network switches.

The wireless keypads send on/off/raise/lower and scene-select commands to the luminaires via the sensor units.

The wireless multi-sensor devices have PIR occupancy sensors, photosensors for daylight harvesting functionality, and ambient-temperature sensors. They have a two-way radio for communicating with the system manager via the gateways. The same radio also receives commands from the wireless keypads. The sensor units communicate with the luminaire control modules via a



**Figure B-3. Example of a schematic for a wireless lighting control system.** (Image courtesy of Richard N. Miller)

multi-wire cable. The sensor unit contains the appropriate sequence of operations for its particular application.

The luminaire control module provides the on/off and dimming protocol signal to the LED drivers. The luminaire control module receives its commands from the associated sensor unit.

In the lower left part of the diagram is shown the “head-end” equipment, consisting of the system manager

and an operator’s workstation. The workstation has a flat screen display, a keyboard, and a mouse. The system manager contains an energy usage log; access to individual luminaires for high-end and low-end trim adjustments, load shedding, and automatic demand response; luminaire outage reports; and other functions and features as selected by the design team.

**B.4 Auditorium**

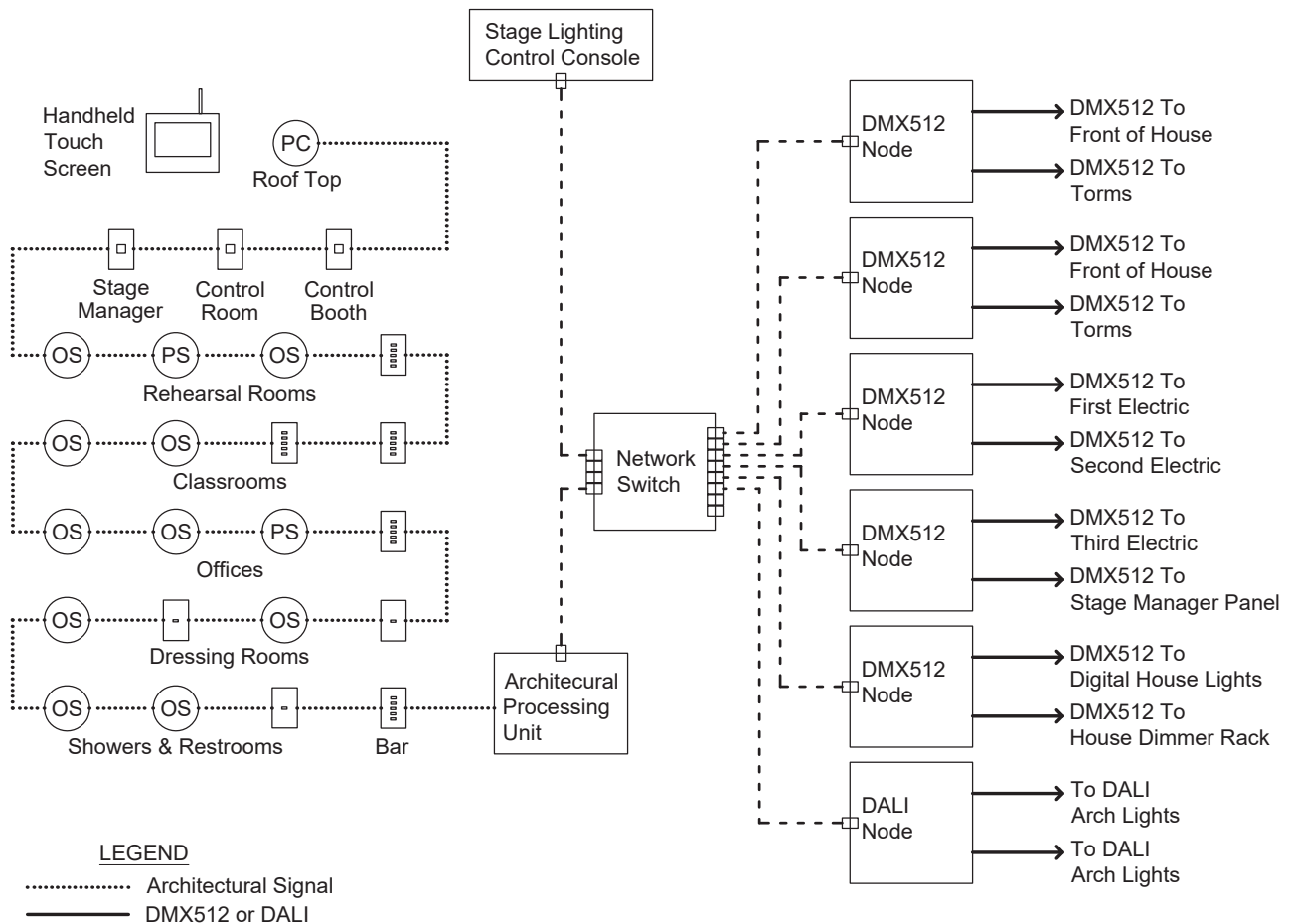
A schematic diagram of the lighting control system for

an auditorium with an architectural processor for the houselights is illustrated in **Figure B-4**. In the center of the diagram is the network switch for data such as Streaming ACN or ArtNet, as well as for the architectural lighting. In the center top of the diagram is the Stage Lighting Control Console.

As with the detail for the networked lighting control system described in **Section B.3**, the control hardware required to enact the sequence of operation is not obvious based on the SOO document alone. The network switch is an essential component to fulfilling the SOO, and is also an expensive component to procure, install, and configure. Failure to include a diagram with this equipment will result in its being “forgotten,” and will cause schedule delays and possibly even change orders.

Nodes to convert the network data to DMX512 format are shown on the right side of the diagram, providing DMX512 control for typical stage lighting locations as well as for the architectural lighting system. It is possible to have a combination of digitally controlled luminaires and line-voltage dimming luminaires, though digitally controlled luminaires offer advantages, such as remote device management and better dimming control, compared with line voltage luminaires.

The left side of the diagram depicts the architectural-lighting input devices, such as keypads, occupancy sensors, and photosensors, all of which communicate to the architectural processing unit via an RS485 digital protocol, which requires a strict daisy-chain topology.



**Figure B-4. Example of a schematic diagram for normal power in an auditorium.** (Image courtesy of Casey Diers, Javid Butler, and Richard N. Miller)



The keypads should also have their own detail drawings to show the button labels, and these details should reflect the SOO.

Emergency lighting is typically required for a theater, but due to varying code and installation requirements, it is not shown on this diagram. The local authority having jurisdiction should be consulted for emergency-lighting requirements on any project.

## Annex C – Sequence of Operation (SOO) Questionnaire

*This annex is not part of ANSI/IES LP-16-22, Lighting Practice: Documenting Control Intent Narratives and Sequences of Operations. It is provided for informational purposes only.*

Lighting control systems are unique for every project. The example SOOs provided in this document may not cover all scenarios where controls are used. In such cases, the designer will need to create one. The questionnaire outlined in this annex is intended to help with that. It can also be used to as a starting point for a dialog with clients to understand their preferences.

### C.1 Instructions

**Instructions:** To start, reference the OPR, your BOD, and lighting control drawings. You will need these to provide the general operating scenarios, project

requirements (e.g., from codes, regulations, owner), and setpoints for a custom SOO.

Create a sequence for each space and use type (e.g., classroom, office, exterior, campus). If you are using the matrix format (see example matrices in **Tables 7-1** through **7-4** and template matrix in **Section C.3**), create a sequence designation and add it to the notes section of the matrix. Finally, write the narrative by responding to the prompt. Do not respond to the questions with a single word (i.e., yes or no); answering with a single word does not aid in the creation of an enforceable sequence. Instead, answer the question with a narrative statement as it will appear on the drawings.

Setpoints transform a CIN into an SOO. They also make the requirements in an SOO enforceable. That said, the best sequence often needs to be adjusted because of site conditions. It is always recommended to add “(adjustable)” or “(adj.)” after most setpoints. (In fact, this is standard practice for HVAC sequences of operation.) For example: The interior designer changed the paint scheme and the formerly black walls are now eggshell-white. The top trim or on level can then be adjusted to accommodate the different reflectivity of the finishes. However, just because setpoints can be adjusted does not mean that specification of adjustability should be applied to all. Emergency light levels are just one example of setpoints that should not be adjustable.

The following example shows what a shutoff time-switch control sequence for an open office would look like when complete:

**Shutoff:** Occupany Sensor

- How will the lights turn on with occupancy? Auto-on 100%, Auto-on 50% (adj.), manual-on, or not at all?
- How long will it take for the lights to turn off? What determines how the lights turn off?
- Which luminaires will be controlled by which occupancy sensors? Will it work with other occupancy sensors in the same space?
- Does the occupancy sensor or lighting control system interface with other systems?

**Sequence Designation:** OS-3

General lighting: Auto-on to 50% of maximum light output and 20-minute delay (adjustable) to turn the lights off when occupancy is not detected. All lights will come on if any occupancy sensor in the space is tripped.  
Fade rate up/down: 1.5 seconds (adjustable). Trigger on PIR; retrigger on PIR or ultrasonic.

Occupancy sensors for the space will integrate with the VAV boxes to shut off HVAC when the space is not occupied. See sequence INT-1 and mechanical SOOs. See lighting control drawings, mechanical schedules, and mechanical ductwork plans for zoning. See details for wiring diagrams. See specifications for other integration requirements.



**C.2 Sequence of Operation Questionnaire****Space name or designation:** \_\_\_\_\_**Local Control and Multi-Level** \_\_\_\_\_**Sequence Designation:** \_\_\_\_\_

- Local on/off?
- Dimming?
- For which luminaires?
- Dimming protocol? Up/down speed?

**Shutoff: Time-Switch Control** \_\_\_\_\_**Sequence Designation:** \_\_\_\_\_

- Will this zone be controlled by the time switch? If there is more than one type of lighting, call out which one(s).
- What are the hours of operation?
- Holidays?
- Sweep? Interval? Blink warning?  
How long between blink and sweep?
- Janitor override? How long?

**Shutoff: Occupancy-Sensor** \_\_\_\_\_**Sequence Designation:** \_\_\_\_\_

- How will the lights turn on? With occupancy? Auto-on 100%, Auto-on 50% (adj.), manual-on, or none?
- How long will it take for the lights to be turned off? What determines how the lights turn off?
- Which luminaires will be controlled by which occupancy sensors? Will it work with other occupancy sensors in the same space?
- Does the occupancy sensor or lighting control system interface with other systems?

**Daylight Harvesting** \_\_\_\_\_**Sequence Designation:** \_\_\_\_\_

- Open or closed loop?
- Target lighting level?
- Minimum dimming level?
- Which luminaires controlled?
- Ramp up/down rate?
- Is there a delay for how long the system should wait to change the luminaire's light output?

**Demand Response** \_\_\_\_\_**Sequence Designation:** \_\_\_\_\_

- Which zones will dim?
- How much will they dim?
- Ramp down/up rate?
- Will decorative lighting shutoff?

**Emergency** \_\_\_\_\_**Sequence Designation:** \_\_\_\_\_

- Will lights come on in an emergency?
- Which ones?
- Will they be on or off at night?

**Priority** \_\_\_\_\_**Sequence Designation:** \_\_\_\_\_

- Are there scenes where daylighting (or other) controls will be disabled?
- Could a control be overridden, and if so, for how long?

**Scenes** \_\_\_\_\_**Sequence Designation:** \_\_\_\_\_

- Scene/button labels?
- What will the buttons do?
- Which lights will be on? At what level?
- Are scenes interlocked with other systems? (e.g., window shades)

**Integration** \_\_\_\_\_**Sequence Designation:** \_\_\_\_\_

- Are there other systems that will accept/receive outputs/inputs from the lighting control system?
- What will happen with those signals?

**Exterior** \_\_\_\_\_**Sequence Designation:** \_\_\_\_\_

- What will turn on the lights? Motion sensor? Photocell? Time switch?
- Will there be a curfew?
- Will there be different light levels? What will trigger the change?

**Other (e.g., Signs, Outlets)** \_\_\_\_\_**Sequence Designation:** \_\_\_\_\_

- How will outlets be controlled?
- Are there any other control sequences not mentioned above?
- Will spaces with multiple luminaires but no scene controls have special labels for the switches?

### C.3 Matrix SOO Template

| Space,<br>Function, or<br>Zone | Local Control | Shutoff | Daylighting | Demand<br>Response | Emergency | Priority | Scene Control | Integration | Exterior | Other |
|--------------------------------|---------------|---------|-------------|--------------------|-----------|----------|---------------|-------------|----------|-------|
|                                |               |         |             |                    |           |          |               |             |          |       |
|                                |               |         |             |                    |           |          |               |             |          |       |

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5. American National Standards Institute. ANSI C137.0-2017, Lighting Systems - Lighting Systems Terms and Definitions. Washington, DC: ANSI; 2017.
6. National Electrical Manufacturers Association. NEMA LSD 64-2019, Lighting Controls Terminology. Rosslyn, Va: NEMA; 2019.

## **Process for Change to an ANSI/IES Standard under Continuous Maintenance**

This standard is maintained under continuous maintenance procedures, for which IES has an established and documented program for regular publication of addenda or revisions, including procedures for timely, documented, consensus action on requests for change to any part of the standard. Committee consideration will be given to proposed changes by June 30 of any given year for proposed changes received by the IES Director of Standards no later than December 31 of the previous year.

### **Submittal Format**

Proposed changes must be submitted to the IES Director of Standards in the announced published format. However, changes may be accepted in an earlier published format, if the differences are immaterial to the proposed change submittal. If the Director of Standards concludes that a current form must be utilized, the proposer may be given up to 20 additional days to resubmit the proposed changes in the current format.

Specific changes in the text or values are required and must be substantiated. Any change proposals that do not meet these requirements will be returned to the proposer. Supplemental background documents to support changes submitted may be included.

### **Submission to the Committee Chair**

The Director of Standards shall forward proposed changes received on appropriate forms to the committee chair for assigning to committee members (responders) to develop responses to submitters of proposed changes.

### **Review and Clarification**

Responders shall review proposals and should contact the proposer if necessary for clarification.

### **Response Recommendation**

Designated responders shall draft a recommended committee response, including any recommended changes to the standard. The 'responders' recommended responses shall be submitted to the committee chair in electronic form usable by Society Staff, including any recommended change to the standard in response to proposals received.

Options for Committee response are limited to:

- a) Proposed change accepted for public review without modification
- b) Proposed change accepted for public review with modification
- c) Proposed change accepted for further study
- d) Proposed change rejected

The responders shall provide reasons for any recommendation other than option (a) above.

The designated responders shall not recommend option (c) unless the further study can be completed by October 1 of that year, and providing the Committee can then vote for option (a), (b), or (d) no later than November 15 of that year.

### **Editing**

The Committee chair or his or her designee shall edit the draft responses and circulate the edited drafts to the committee for review.

## Form for Proposing Change to an ANSI/IES Standard under Continuous Maintenance

*NOTE: Use a separate form for each comment. Submit to the Director of Standards, IES, 120 Wall Street, 17<sup>th</sup> Floor, New York, NY 10005-4001. Email: [standards@ies.org](mailto:standards@ies.org). Fax: 212-248-5017.*

1. Submitter: \_\_\_\_\_  
Affiliation: \_\_\_\_\_  
Address: \_\_\_\_\_  
City: \_\_\_\_\_ State: \_\_\_\_\_ Zip: \_\_\_\_\_ Country: \_\_\_\_\_  
Telephone: \_\_\_\_\_  
Fax: \_\_\_\_\_  
E-mail: \_\_\_\_\_

I hereby grant the Illuminating Engineering Society (IES) the non-exclusive royalty rights, including non-exclusive rights in copyright, in my proposals. I understand that I acquire no rights in publication of the standard in which my proposals in this, or other analogous, form are used. I hereby attest that I have the authority and am empowered to grant this copyright release.

Submitter's signature: \_\_\_\_\_ Date: \_\_\_\_\_

2. Title of publications and year published \_\_\_\_\_

3. Clause (section), sub-clause or paragraph number; and page number: \_\_\_\_\_

4. My proposal (check one):

- ☐ Change to read as follows
- ☐ Delete and substitute as follows
- ☐ Add new text as follows
- ☐ Delete without substitution

*Use underscore to show material to be added (added) and strikethrough for material to be deleted (~~deleted~~). Use additional pages if needed.*

5. Proposed change:

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6. Reason and substantiation:

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Select as applicable:

- ☐ Additional pages are attached. Number of additional pages: \_\_\_\_\_
- ☐ Attachments or referenced materials cited in this proposal accompany this proposed change.

*Please verify that all attachments and references are relevant, current, and clearly labeled to avoid processing and review delays. Please list your attachments here:*









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